

Adopted Levels, Gammas

$Q(\beta^-) = -5966.2$ 7; $S(n) = 12644.76$ 5; $S(p) = 6370.81$ 4; $Q(\alpha) = -6997.90$ 4 [2021Wa16](#)
 $S(2n) = 24152.8$ 4, $S(2p) = 17254.1$ 11 ([2021Wa16](#)).

Isotope discovery ([2012Th10](#)): positive-ray mass spectrograph at Cambridge, UK ([1919As01](#)).

 ^{35}Cl LevelsCross Reference (XREF) Flags

A	^{35}S β^- decay (87.61 d)	K	^{31}P ($^6\text{Li}, d$)	U	^{35}Cl ($p, p'\gamma$)
B	^{35}Ar ε decay (1.7755 s)	L	^{32}S (α, p)	V	^{36}Ar (n, d)
C	^{36}K εp decay (341 ms)	M	^{32}S ($\alpha, p\gamma$)	W	^{36}Ar ($d, ^3\text{He}$)
D	^1H ($^{34}\text{S}, ^{35}\text{Cl}\gamma$)	N	^{33}S (α, d)	X	^{36}Ar ($\text{pol } d, ^3\text{He}$)
E	^2H ($^{34}\text{S}, n\gamma$)	O	^{34}S (p, γ)	Y	^{37}Cl (p, t)
F	^{12}C ($^{28}\text{Si}, \alpha p\gamma$)	P	^{34}S (p, p), ($p, p'\gamma$):resonances	Z	^{39}K (γ, α)
G	^{16}O ($^{24}\text{Mg}, \alpha p\gamma$)	Q	^{34}S (d, n)	Others:	
H	^{24}Mg ($^{16}\text{O}, \alpha p\gamma$)	R	^{34}S ($^3\text{He}, d$)	AA	^{40}Ca ($\mu^-, \nu\alpha n\gamma$)
I	^{27}Al ($^{14}\text{N}, \alpha p n\gamma$)	S	^{35}Cl (γ, γ')	AB	Coulomb excitation
J	^{31}P (α, p), (α, n):resonances	T	^{35}Cl ($n, n'\gamma$)		

<u>E(level)[†]</u>	<u>J^π</u>	<u>T_{1/2} or Γ^{&}</u>	<u>XREF</u>	<u>Comments</u>
0.0	3/2 ⁺	stable	ABC EFGHI KLM O QRSTU VWXYZ	XREF: Others: AA , AB $\mu = +0.821870$ 2 (2013Ja17 , 2019StZV) $Q = -0.0817$ 8 (2008Py02 , 2021StZZ) J ^π : L=2 from 0 ⁺ in (d,n), (n,d), ($^3\text{He}, d$), ($d, ^3\text{He}$), and (pol d, ^3He) and L-1/2 transfer from analyzing power. Spin=3/2 from microwave spectroscopy and atomic beam methods (1948To10 , 1949Da14). μ: Nuclear Magnetic Resonance (2013Ja17). Q: Atomic beam magnetic resonance (2008Py02). Others: -0.08249 2 (1972St38), -0.076 5 (1986El09), -0.817 8 (1993Su36), 0.0819 11 (2000Ha64), 0.0850 11 (2004Al08). Evaluated rms nuclear charge radius $R = 3.365$ fm 19 (2013An02).
1219.38 7	1/2 ⁺	0.119 ps 14	B E H KLM O QRSTU VWXY	XREF: Others: AA , AB B(E2) [†] =0.00076 7 B(E2) [†] : from Coulomb excitation. E(level): weighted average of 1219.2 2 from ^{35}Ar ε decay, 1219.39 7 from (p,γ), and 1219.42 10 from (p,p'γ), 1219.0 10 from ($^{16}\text{O}, \alpha p\gamma$), and 1219.4 6 from (α,pγ). J ^π : L=0 from 0 ⁺ in (d,n), (n,d), ($^3\text{He}, d$), ($d, ^3\text{He}$), and (pol d, ^3He). T _{1/2} : from lifetime $\tau = 0.172$ ps 20: weighted average of 0.29 ps 4 from ($^{34}\text{S}, n\gamma$) (1973Wa10 , DSAM), 0.21 ps +10−8 from (α,pγ) (1969In04 , DSAM), 80 fs 40 (1971Wi13 , DSAM), 175 fs 20 (1972Hu11 , DSAM), 150 fs 20 (1973Fa07 , DSAM), 145 fs 30 (1976Me12 , DSAM) from (p,γ), 0.13 ps +3−2 from (γ,γ') (1966Ho02 , resonance fluorescence), 200 fs 80 from (n,n'γ) (1989Ge09 , DSAM), 0.215 ps 75 (1972Va06 , DSAM), 0.21 ps +6−4 (1972Br45 , DSAM), 0.27 ps +19−10 (1969Du08 , DSAM) from (p,p'γ), 0.16 ps 5 (1969Ha17 , DSAM), and 0.230 ps 25 (1977Sc36 , DSAM) from Coulomb excitation.
1763.15 4	5/2 ⁺	0.36 ps 3	B EFGHI KLM O QRSTU VWXYZ	XREF: Others: AA , AB

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Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

<u>E(level)[†]</u>	<u>J^π[‡]</u>	<u>T_{1/2} or Γ^{&}</u>	<u>XREF</u>	<u>Comments</u>
				T=1/2 B(E2)↑=0.0106 7 XREF: X(1785)Y(1750) B(E2)↑: from Coulomb excitation. E(level): weighted average of 1763.1 2 from ^{35}Ar ε decay, 1763.0 6 from ($^{28}\text{Si},\alpha\text{py}$), 1762.95 17 from ($^{24}\text{Mg},\alpha\text{py}$), 1763.3 2 from ($^{16}\text{O},\alpha\text{py}$), 1763.19 10 from ($^{14}\text{N},\alpha\text{pny}$), 1763.14 4 from (p,$\gamma$), 1763.26 10 from (p,p'$\gamma$). J^π: L=2 from 0⁺ in (d,n); spin=5/2 from pγ(θ) or γ(θ) in (α,py), (p,γ), and (p,p'γ). T_{1/2}: lifetime τ=0.52 ps 5: weighted average of 0.54 ps 7 from ($^{34}\text{S},\text{n}\gamma$) (1973Wa10, DSAM), 0.55 ps 15 from (α,py) (1969In04, DSAM), 0.50 ps 10 (1972Hu11, DSAM), 0.40 ps 14 (1973Fa07, DSAM), 0.53 ps 9 (1976Me12, DSAM) from (p,γ), 0.60 ps 21 from (γ,γ') (1962Bo17, resonance fluorescence), 0.205 ps 75 (1972Va06, DSAM), 0.49 ps +9-6 (1972Br45, DSAM), 1.15 ps +45-63 (1969Du08, DSAM) from (p,p'γ), 0.57 ps 7 (1969Ha17, DSAM), and 0.63 ps 5 (1977Sc36, DSAM) from Coulomb excitation.
2645.67 8	7/2 ⁺	0.159 ps 21	B FGHI LM O Qr TU y	XREF: Others: AA, AB XREF: r(2645)y(2650) E(level): weighted average of 2645.81 18 from ($^{24}\text{Mg},\alpha\text{py}$), 2645.9 3 from ($^{16}\text{O},\alpha\text{py}$), 2645.45 12 from ($^{14}\text{N},\alpha\text{pny}$), 2645.70 8 from (p,$\gamma$), and 2645.74 17 from (p,p'γ). Others: 2645.1 10 from ^{35}Ar ε decay, 2645.8 6 from ($^{28}\text{Si},\alpha\text{py}$), and 2645.8 8 from ($\alpha,\text{py}$). J^π: spin=7/2 from p$\gamma$($\theta$) in ($\alpha,\text{py}$) and γ(θ) in (p,γ); parity from 882γ, M1+E2 to 5/2⁺, 1763 level in (α,py). T_{1/2}: lifetime τ=0.229 ps 30: weighted average of 0.30 ps 9 from (α,py) (1969In04, DSAM), 0.255 ps 65 (1972Hu11, DSAM), 0.200 ps 30 (1976Me12, DSAM) in (p,γ), 0.27 ps 9 in (n,n'γ) (1989Ge09, DSAM), 0.185 ps 50 (1972Va06, DSAM), 0.26 ps 6 (1972Br45, DSAM), 0.35 ps +10-7 (1969Du08, DSAM) in (p,p'γ), and 0.26 ps 9 in Coulomb excitation (1977Sc36, DSAM). Other: τ<1.3 ps in ($^{28}\text{Si},\alpha\text{py}$) (2007Ks01, DSAM).
2694.00 20	3/2 ⁺	32 fs 6	B LM O QRSTU Wxy	XREF: Others: AA XREF: L(2693?)X(2676)y(2650) E(level): weighted average of 2693.8 2 from ^{35}Ar ε decay, 2693.6 13 from (α,py), 2693.9 3 from (p,γ), 2694.50 28 from (p,p'γ). J^π: spin=3/2 from γ(θ) in (p,p'γ); L=2 from 0⁺ in ($^3\text{He},\text{d}$) and (d,^3He). T_{1/2}: lifetime τ=46 fs 9: unweighted average of 70 fs 30 in (α,py) from 1972Br33 with DSAM, 21 fs 3 in (α,py) from 1972Hu11 with DSAM, 19 fs 4 in (α,py) from 1973Fa07 with DSAM, 20 fs 4 in (p,γ) from 1976Me12 with DSAM, 65 fs 20 in (p,γ) from 1972Va06 with DSAM, 62 fs 16 in (p,p'γ) from 1969Du08 with DSAM, and 62 fs 8 in (n,n'γ) from 1989Ge09 with DSAM. Others: 115 fs +95-59 in

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Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

<u>E(level)[†]</u>	<u>J^π[‡]</u>	<u>T_{1/2} or Γ^{&}</u>	<u>XREF</u>	<u>Comments</u>
3002.64 11	5/2 ⁺	22 fs 6	B F HI LM O QRSTU WXY	(p,p'γ) from 1969Du08 (using 931γ) is considered less reliable by the authors, and 11 fs 7 in (p,p'γ) from 1971Ca40 with DSAM is considered an outlier. E(level): weighted average of 3002.3 3 from ^{35}Ar ε decay, 3003.1 8 from ($^{28}\text{Si},\alpha\text{p}\gamma$), 3002.0 10 from ($^{16}\text{O},\alpha\text{p}\gamma$), 3001.98 22 from ($^{14}\text{N},\alpha\text{p}\text{n}\gamma$), 3002.6 15 from ($\alpha,\text{p}\gamma$), 3002.75 8 from (p,γ), 3002.6 5 from (p,p'γ). J ^π : L=2 from 0 ⁺ in ($^3\text{He},\text{d}$), ($\text{d},^3\text{He}$), and (pol d, ^3He) with L+1/2 transfer from analyzing power. Discrepancy: L=3 from 0 ⁺ in (d,n) and the angular distribution fit seems good. T _{1/2} : lifetime τ=32 fs 8: unweighted average of 48 fs 9 in (α,pγ) from 1972Br33 with DSAM, 22 fs 3 in (α,pγ) from 1972Hu11 with DSAM, 14 fs 2 in (α,pγ) from 1973Fa07 with DSAM, 16 fs 4 in (p,γ) from 1976Me12 with DSAM, 72 fs 12 in (n,n'γ) from 1989Ge09 with DSAM, 18 fs +26-15 in (p,p'γ) from 1972Va06 with DSAM, and 31 fs 13 in (p,p'γ) from 1969Du08 with DSAM.
3162.87 9	7/2 ⁻	29.2 ps 12	FGHI KLM O QR TU WX	T=1/2 E(level): weighted average of 3163.1 6 from ($^{28}\text{Si},\alpha\text{p}\gamma$), 3163.43 19 from ($^{24}\text{Mg},\alpha\text{p}\gamma$), 3162.9 3 from ($^{16}\text{O},\alpha\text{p}\gamma$), 3162.64 9 from ($^{14}\text{N},\alpha\text{p}\text{n}\gamma$), 3163.1 11 from (α,pγ), 3162.99 7 from (p,γ), 3162.75 10 from (p,p'γ). J ^π : spin=7/2 from pγ(θ) in (α,pγ) and γ(θ) in (p,γ) and (p,p'γ); L=3 from 0 ⁺ in (d,n), ($^3\text{He},\text{d}$), ($\text{d},^3\text{He}$), and (pol d, ^3He). T _{1/2} : lifetime τ=42.1 ps 17: weighted average of 42 ps 2 (1974Va13, RDM), 41.7 ps 17 (1976Ke02, RDM), 45 ps 6 (1991Ja11, RDM) from $^{24}\text{Mg}(\text{}^{16}\text{O},\alpha\text{p}\gamma)$, 41.8 ps 18 from $^{27}\text{Al}(\text{}^{14}\text{N},\alpha\text{p}\text{n}\gamma)$ (1976Wa11, RDM), 60 ps 7 from (α,pγ) (1969In04, DSAM), 37 ps 4 from (p,γ), (1971Ba23, delayed coincidence), and 46 ps 6 from (p,p'γ) (1974Hu09, RDM). Others: τ=140 ps 40 (1968Az02, delayed coincidence; obtained using NaI(Tl) detectors with broad time resolution) from (p,γ).
3918.47 12	3/2 ⁺	4.9 fs 14	B M O RS u x	XREF: R(3908)u(3930)x(3948) E(level): weighted average of 3918.4 5 from ^{35}Ar ε decay, 3915 2 from (α,pγ), 3918.48 10 from (p,γ). J ^π : spin=3/2 determined by the feeding 5162γ(θ) from 5/2 ⁺ , 9081 level in (p,γ); parity=+ determined by 3918γ, M1+E2 to 3/2 ⁺ ground state. L=2 from 0 ⁺ in (pol d, ^3He). T _{1/2} : lifetime τ=7 fs 2 from (p,γ), (1972Hu11, DSAM). Others: <22 fs from (α,pγ) (1972Br33, DSAM).
3942.9 2	9/2 ⁺	0.18 ps 3	F HI LM O	E(level): weighted average of 3942.0 7 from ($^{28}\text{Si},\alpha\text{p}\gamma$), 3943.4 10 from ($^{16}\text{O},\alpha\text{p}\gamma$), 3942.3 10 from ($^{14}\text{N},\alpha\text{p}\text{n}\gamma$), 3943.4 12 from (α,pγ), and 3942.9 2 from (p,γ). J ^π : spin=9/2 from pγ(θ) in (α,pγ); 2180.1γ E2 to 5/2 ⁺ . T _{1/2} : lifetime τ=0.26 ps 5: weighted average of 0.33 ps 5 in (α,pγ) (1972Br33, DSAM), 0.39 ps 11 (1972Hu11,

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Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

<u>E(level)[†]</u>	<u>J^π[‡]</u>	<u>T_{1/2} or Γ^{&}</u>	<u>XREF</u>	<u>Comments</u>
3967.6 4	1/2 ⁺	11.8 fs 28	B M O R	DSAM), 0.145 ps 50 (1973Fa07, DSAM), and 0.29 ps 5 (1976Me12, DSAM) from (p,γ). Other: τ<0.5 ps in (²⁸ Si,αpγ) (2007Ks01, DSAM). E(level): weighted average of 3967.7 4 from ³⁵ Ar ε decay, 3968 2 from (α,pγ), and 3967.3 6 from (p,γ). J ^π : L=0 from 0 ⁺ in (³ He,d). T _{1/2} : lifetime τ=17 fs 4: weighted average of 13 fs 4 (1972Hu11, DSAM), 20 fs 5 (1973Fa07, DSAM), and 20 fs 4 (1976Me12, DSAM) in (p,γ). Other: τ<54 fs from (α,pγ) (1972Br33, DSAM).
4059.1 4	(3/2) ⁻	13.9 fs 21	LM O QR	E(level): weighted average of 4056.9 27 from (α,p), 4058 3 from (α,pγ), 4059.2 4 from (p,γ). J ^π : L=1 from 0 ⁺ in (³ He,d); (3/2 ⁻) from FRESKO-DWBA analysis of measured σ(θ) of (α,p) (2019Se09). T _{1/2} : lifetime τ=20 fs 3: weighted average of 31 fs 13 from (α,pγ) (1972Br33, DSAM), 22 fs 3 (1972Hu11, DSAM), 20 fs 4 (1973Fa07, DSAM), and 18 fs 3 (1976Me12, DSAM) from (p,γ).
4112.4 16	7/2 ⁺	49 fs 11	LM O Q	E(level): weighted average of 4111.1 29 from (α,p), 4108 2 from (α,pγ), and 4113.7 10 from (p,γ). J ^π : 4113.4γ E2(+M3) to 3/2 ⁺ . T _{1/2} : lifetime τ=70 fs 16 from (α,pγ) (1972Br33, DSAM).
4173.43 23	(5/2 ⁺)	35 fs 6	M O q	T=1/2 XREF: q(4170) E(level): weighted average of 4171 2 from (α,pγ) and 4173.45 19 from (p,γ). J ^π : based on primary transitions from proton resonances in (p,γ): 3827.3γ from 8001.0, (7/2 ⁺), 4456.4γ from 8630.1, 7/2 ⁻ level, 4243.0γ from 8416.7, 1/2 ⁺ level. Note that primary 3099.1γ from 7272.7, 1/2 ⁻ resonance disfavors 5/2 ⁺ assignment, but this γ branch is relatively weak. T _{1/2} : lifetime τ=51 fs 8: weighted average of 68 fs 24 in (α,pγ) (1972Br33, DSAM), 105 fs +35-30 (1971Wi13, DSAM), 55 fs 6 (1972Hu11, DSAM), and 34 fs 9 (1976Me12, DSAM) in (p,γ).
4177.9 2	3/2 ⁻	27.0 fs 28	L O qR	XREF: q(4170)R(4167) J ^π : spin=3/2 from γ(θ) in (p,γ); L=1 from 0 ⁺ in (³ He,d). T _{1/2} : lifetime τ=39 fs 4: weighted average of 32 fs 5 (1972Hu11, DSAM), 42 fs 4 (1973Fa07, DSAM), and 47 fs 10 (1976Me12, DSAM) from (p,γ).
4347.76 7	9/2 ⁻	1.0 ps 6	FGHI LM O	E(level): weighted average of 4347.8 6 from (²⁸ Si,αpγ), 4348.29 24 from (²⁴ Mg,αpγ), 4347.8 3 from (¹⁶ O,αpγ), 4347.43 17 from (¹⁴ N,αpγ), 4347.8 12 from (α,pγ), and 4347.76 6 from (p,γ). J ^π : spin from γ(θ) in (p,γ); parity from 1184.6γ, M1+E2 to 7/2 ⁻ , 3163 level in (p,γ) (1974Lo17). T _{1/2} : lifetime τ=1.5 ps 8: unweighted average of 1.31 ps +30-23 from (²⁸ Si,αpγ) (2007Ks01, DSAM), 2.9 ps +14-7 (1972Br33, DSAM), and 0.33 ps 6 (1976Me12, DSAM) from (p,γ).
4624.3 3	3/2,5/2	40 fs 17	B LM O	XREF: B(?) E(level): other: 4625.3 28 from (α,p); 4618 2 from

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Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

<u>E(level)[†]</u>	<u>J^π</u>	<u>T_{1/2} or Γ^{&}</u>	<u>XREF</u>	<u>Comments</u>	
				(α,py) is discrepant. J ^π : from pγ(θ) in (α,py). T _{1/2} : lifetime τ=58 fs 24 from (α,py) in 1972Br33 with DSAM. E(level): others: 4770.4 26 from (α,p), 4770 15 from (α,py). J ^π : from primary transitions in (p,γ): 4059.2γ from 8829.3, 1/2 ⁻ level, 2778.9γ from 7548.9, 7/2 ⁻ level, 3231.0γ from 8001.0, (7/2 ⁺) level. Other: 7/2,9/2 from pγ(θ) for a group at E=4770 15 in (α,py) is inconsistent and disfavored by γ rays to 1/2 ⁺ and 3/2 ⁺ . T _{1/2} : lifetime τ=110 fs 29: weighted average of 270 fs 95 (1972Hu11 , DSAM) and 105 fs 17 (1976Me12 , DSAM) in (p,γ).	
4769.86 23	(5/2 ⁻)	76 fs 20	LM 0		
4839.10 11	(1/2 ⁺ ,3/2)	9.7 fs 35	0	x	XREF: x(4839) J ^π : from primary γ transitions in (p,γ): 2339.4γ from 7178.6, 1/2 ⁽⁺⁾ level, 2355.4γ from 7194.6, 1/2 ⁻ level, 2867.2γ from 7706.4, 5/2 ⁺ level. T _{1/2} : lifetime τ=14 fs 5 from (p,γ) (1976Me12 , DSAM).
4854.4 19	(1/2,3/2)	4.9 fs 14	L 0	x	XREF: x(4839) E(level): other: 4859.2 28 from (α,p). J ^π : from primary γ transitions in (p,γ): 2340.1γ from 7194.6, 1/2 ⁻ level, 3183.9γ from 8038.4, 1/2 ⁺ ,3/2 ⁺ level; decay 3634.8 γ to 1219.38, 1/2 ⁺ . T _{1/2} : lifetime τ=7 fs 2 from (p,γ) (1976Me12 , DSAM).
4880.96 15	7/2 ⁽⁺⁾	5.6 fs 14	LM 0		E(level): others: 4881.8 29 from (α,p), 4880.8 21 from (α,py). J ^π : spin=7/2 from pγ(θ) in (α,py); 3417.1 primary γ from 8298.2, 3/2 ⁺ resonance in (p,γ). T _{1/2} : lifetime τ=8 fs 2: weighted average of 7 fs 2 (1972Hu11 , DSAM), 7 fs 3 (1973Fa07 , DSAM), and 10 fs 3 (1976Me12 , DSAM) from (p,γ). Other: τ=0.28 ps 6 from (α,py) in 1972Br33 with DSAM.
5009.1 18	(1/2,3/2)	7.6 fs 21	LM 0	R	E(level): weighted average of 5006.9 27 from (α,p) and 5010.1 18 from (p,γ). Others: 5015 20 from (α,py) and 5010 15 from (³ He,d). J ^π : primary 2185.4γ from 7194.6, 1/2 ⁻ level and 3407.4γ from 8416.7, 1/2 ⁺ level. T _{1/2} : lifetime τ=11 fs 3 from (p,γ) (1976Me12 , DSAM).
5158.6 29	3/2 ⁺ ,5/2 ⁺		Lm	R	WX XREF: m(5175) E(level): weighted average of 5158.9 29 from (α,p), 5163 15 from (³ He,d), 5150 20 from ³⁶ Ar(d, ³ He), and 5155 11 from ³⁶ Ar(pol d, ³ He). J ^π : L=2 from 0 ⁺ in (pol d, ³ He) and (d, ³ He). XREF: m(5175)
5163.34 22	7/2 ⁻		m 0		J ^π : spin=7/2 from the feeding 3466γ(θ) from 8630, 7/2 ⁻ level and RUL in (p,γ); parity=- from 3466γ, M1+E2 from 8630, 7/2 ⁻ level.
5214.5 20	(5/2 ⁺) [#]	<4.85 fs	LM 0		XREF: M(5230) E(level): weighted average of 5209.5 28 from (α,p),

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Adopted Levels, Gammas (continued)

³⁵ Cl Levels (continued)					
E(level) [†]	J ^π [‡]	T _{1/2} or Γ ^{&}	XREF		Comments
					5230 20 from (α,py), and 5215.8 15 from (p,γ). J ^π : other: 3/2,5/2 from py(θ) in (α,py). T _{1/2} : lifetime τ<7 fs from (p,γ) (1976Me12, DSAM). E(level): weighted average of 5402.0 29 from (α,p) and 5403.5 10 from (p,γ). Other: 5409 12 from (³ He,d). J ^π : L=1 from 0 ⁺ in (³ He,d); 2400.6γ to 5/2 ⁺ rules out 1/2 ⁻ based on RUL, since it would require a large B(M2)(W.u.) exceeding RUL. T _{1/2} : lifetime τ=17 fs 4 from (p,γ) (1976Me12, DSAM). The contribution of Iγ to unknown levels adds up to 36% of Iγ(4183)=100 18 from (p,γ). E(level): weighted average of 5407.4 8 from (²⁸ Si,αpy), 5407.64 27 from (²⁴ Mg,αpy), 5407.3 3 from (¹⁶ O,αpy), 5406.68 19 from (¹⁴ N,αpny), and 5407.1 15 from (α,py). J ^π : spin=11/2 from py(θ) in (p,γ); parity from 1059γ, M1+E2 to 9/2 ⁻ , 4347 level. T _{1/2} : lifetime τ=0.4 ps 1 from (α,py) (1974Lo17, DSAM). Other: τ<1.1 ps from (²⁸ Si,αpy) (2007Ks01, DSAM). E(level): weighted average of 5531 5 from (α,p) and 5535 20 from (α,py). XREF: m(5600)w(5600)x(5583) E(level): weighted average of 5586.0 26 from (α,p) and 5586 2 from (p,γ). J ^π : L=2 from 0 ⁺ in (pol d, ³ He) and L+1/2 transfer from analyzing power. XREF: m(5600)w(5600)x(5583) E(level): weighted average of 5594.6 26 from (α,p) and 5599.7 15 from (p,γ). J ^π : L(d, ³ He)=2 from 0 ⁺ . T _{1/2} : lifetime τ=3 fs 1 from (p,γ) (1976Me12, DSAM). XREF: m(5650)r(5660)w(5600)y(5650) E(level): from (α,p). XREF: m(5650)r(5660)y(5650) E(level): weighted average of 5645 4 from (α,p) and 5646 2 from (p,γ). J ^π : from primary transitions in (p,γ): 1903γ from 7548.9, 7/2 ⁻ level, 2355γ from 8001.0, (7/2 ⁺) level, 3261γ from 8906.8, 5/2 ⁺ level. T _{1/2} : lifetime τ=4 fs 1 from (p,γ) in 1976Me12 with DSAM. T=3/2 XREF: m(5650)r(5660)w(5600)y(5650) E(level): weighted average of 5653 4 from (α,p) and 5655 2 from (p,γ). J ^π : L=2 from 0 ⁺ in (³ He,d); primary 2142γ from 7797.2, 1/2 ⁻ level in (p,γ). Note that a weak primary 1903γ from 7548.9, 7/2 ⁻ level in
5403.3 10	3/2 ⁻	11.8 fs 28	L	O R	
5407.07 ^a 21	11/2 ⁻	0.28 ps 7	FGHI	LM	
5531 5			LM	O	
5586.0 20	5/2 ⁺		Lm	O wx	
5598.4 22	3/2 ⁺ ,5/2 ⁺	2.1 fs 7	Lm	O wx	
5633.6 30			Lm	r w y	
5646 2	(5/2,7/2,9/2 ⁺)	2.8 fs 7	Lm	O r y	
5655 2	(3/2) ⁺	13.9 fs 35	Lm	O r w y	

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)				
E(level) [†]	J ^π [‡]	T _{1/2} or Γ ^{&}	XREF	Comments
5682.2 20	1/2 ⁻ , 3/2 ⁻		Lm O R	(p,γ) favors 5/2 ⁺ . T _{1/2} : lifetime τ=20 fs 5 from (p,γ) (1976Me12, DSAM). The contribution of Iγ to unknown levels adds up to 18% of Iγ(2652)=100 6 from (p,γ). XREF: m(5650) E(level): weighted average of 5679.7 34 from (α,p), 5683 2 from (p,γ), and 5684 12 from (³ He,d). J ^π : L=1 from 0 ⁺ in (³ He,d). The contribution of Iγ to unknown levels adds up to 18% of Iγ(1504)=100 from (p,γ). E(level): others: 5731.2 25 from (α,p), 5720 20 from ³⁶ Ar(d, ³ He), and 5720 11 from ³⁶ Ar(pol d, ³ He). J ^π : L=2 from 0 ⁺ in (pol d, ³ He) and (d, ³ He) and L+1/2 transfer from analyzing power. The contribution of Iγ to unknown levels adds up to 100% of Iγ(3960)=100 11 from (p,γ). E(level): weighted average of 5757 4 from (α,p), 5758 3 from (p,γ), and 5760 12 from (³ He,d). J ^π : L(³ He,d)=(0,1) from 0 ⁺ . The contribution of Iγ to unknown levels adds up to 56% of Iγ(5758)=100 22 from (p,γ). E(level): weighted average of 5808.1 31 from (α,p) and 5806 2 from (p,γ). J ^π : primary 2890.2γ from 8696.9, 3/2 ⁻ level; 5806.1γ to 3/2 ⁺ g.s. T _{1/2} : lifetime τ=5 fs 1 from (p,γ) (1976Me12, DSAM). XREF: M(5850) E(level): weighted average of 5829 4 from (α,p) and 5850 25 from (α,pγ). J ^π : other: spin=5/2,9/2 from pγ(θ) in (α,pγ). E(level): weighted average of 5927.1 7 from (²⁸ Si,αpγ), 5926.9 5 from (¹⁶ O,αpγ), 5926.47 29 from (¹⁴ N,αpγ), 5928 3 from (α,p), and 5927.1 19 from (α,pγ). J ^π : 1579γ M1+E2, ΔJ=1 to 4348, 9/2 ⁻ in ¹² C(²⁸ Si,αpγ), also assuming spins increase with excitation energies in fusion-evaporation reactions. Spin=7/2,11/2 from pγ(θ) in (α,pγ). T _{1/2} : lifetime τ<0.4 ps from (²⁸ Si,αpγ) (2007Ks01, DSAM). E(level): weighted average of 6087.6 8 from (²⁸ Si,αpγ), 6088.05 29 from (²⁴ Mg,αpγ), 6087.2 4 from (¹⁶ O,αpγ), 6086.89 23 from (¹⁴ N,αpγ), 6084.2 29 from (α,p), and 6087.0 18 from (α,pγ). J ^π : spin=13/2 from pγ(θ) in (α,pγ); parity from 680γ, M1+E2 to 11/2 ⁻ , 5406 level in (α,pγ). T _{1/2} : lifetime τ=8.9 ps 8: weighted average of 9.3 ps 8 from (α,pγ) in 1974Lo17 with RDM and 7.7 ps 14 from (¹⁴ N,αpγ) in 1976Wa11. XREF: l(6104)w(6140)x(6136) E(level): weighted average of 6104 4 from (α,p)
5723.52 18	5/2 ⁺		L O WX	
5758 3	(1/2,3/2 ⁻)		L O R	
5806.6 20	(1/2,3/2,5/2)	3.5 fs 7	L O	
5830 4	(5/2 ⁻) [#]		LM	
5926.65 29	(11/2) ⁻	<0.28 ps	F HI LM	
6087.31 24	13/2 ⁻	6.2 ps 6	FGHI LM	
6106.2 21	(3/2,5/2 ⁺)	8.3 fs 21	1 O WX	

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Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

E(level) [†]	J ^π [‡]	XREF		Comments
				and 6106.2 21 from (p,γ). J ^π : 4886.5γ to 1/2 ⁺ , 4342.8γ to 5/2 ⁺ ; primary 1675.5γ from 7781.7, (5/2 ⁻) level. T _{1/2} : lifetime τ=12 fs 3 from (p,γ) in 1976Me12 with DSAM. XREF: w(6140)x(6136) J ^π : L=2 from 0 ⁺ in (d, ³ He) and (pol d, ³ He), and L+1/2 transfer from analyzing power.
6141 4	5/2 ⁺	L	WX	XREF: n(6200) E(level): other: 6180 3 from (α,p). J ^π : 6180γ to 3/2 ⁺ g.s.; primary 1438γ from 7618.7, 5/2 ⁺ level in (p,γ). Apart from the γ branch I _γ (6180)=100 observed in (p,γ), the contribution of I _γ to unknown levels adds up to 122% of I _γ (6180).
6181 3	(1/2 ⁺ to 7/2 ⁺)	L n0		XREF: n(6200) E(level): other: 6180 3 from (α,p). J ^π : 6180γ to 3/2 ⁺ g.s.; primary 1438γ from 7618.7, 5/2 ⁺ level in (p,γ). Apart from the γ branch I _γ (6180)=100 observed in (p,γ), the contribution of I _γ to unknown levels adds up to 122% of I _γ (6180).
6224.4 30		L n		XREF: n(6200) J ^π : L(α,d)=6 for a group at 6200 10.
6283 3	(5/2 ⁺) [#]	L	R	E(level): weighted average of 6283 3 from (α,p) and 6284 4 from (³ He,d). J ^π : other: L=(2) from 0 ⁺ in (³ He,d). J ^π : L=(0,1) from 0 ⁺ in (³ He,d).
6329 4	(1/2,3/2 ⁻)		R	XREF: R(6377) E(level): weighted average of 6380.8 8 from (¹⁶ O,αpγ) and 6379.3 25 from (α,p). J ^π : other: L=(2,3) from 0 ⁺ in (³ He,d) for a group at E=6377 2 based on very few data points in a very narrow range of angles in the DWBA fit.
6380.2 9	(9/2 ⁻) [#]	H L	R	
6401.2 22	(1/2 ⁻) [#]	LM		
6428 2	(1/2 ⁺) [#]	L	R	E(level): weighted average of 6429 3 from (α,p) and 6427 2 from (³ He,d). J ^π : other: L=(3) from 0 ⁺ in (³ He,d) is inconsistent.
6469 3	(1/2 ⁻ ,3/2 ⁻)	L	R	E(level): weighted average of 6475 4 from (α,p) and 6468 2 from (³ He,d). J ^π : L=(1) from 0 ⁺ in (³ He,d).
6492 2	(3/2 ⁺) [#]	L O R		E(level): weighted average of 6492 3 from (α,p), 6492 3 from (p,γ), and 6491 2 from (³ He,d). J ^π : other: L=(2) from 0 ⁺ in (³ He,d).
6546 2	(1/2 ⁺) [#]	L	R	E(level): weighted average of 6549 3 from (α,p) and 6545 2 from (³ He,d). J ^π : other: L=(0,1) from 0 ⁺ in (³ He,d), indicating (1/2,3/2 ⁻). E(level): from (³ He,d). J ^π : from L=(1) from 0 ⁺ in (³ He,d).
6643 2	(1/2 ⁻ ,3/2 ⁻)	D	R	XREF: y(6677) J ^π : 2312.4γ to 9/2 ⁻ , 3497.1γ to 7/2 ⁻ in (²⁸ Si,αpγ), assuming spin increasing as excitation energy increases.
6659.1 30	(7/2 ⁺) [#]	L	y	XREF: y(6677)
6660.2 8	(11/2 ⁻)	F		E(level): weighted average of 6679.4 31 from (α,p) and 6674 2 from (³ He,d). XREF: X(6745)
6675.6 25		D L	R y	E(level): weighted average of 6761 2 from (³ He,d), 6750 20 from ³⁶ Ar(d, ³ He), and 6745 12 from ³⁶ Ar(pol d, ³ He). J ^π : L(d, ³ He)=2 from 0 ⁺ .
6761 2	3/2 ⁺ ,5/2 ⁺	D	R WX	E(level): weighted average of 6781.3 30 from (α,p) and 6778 2
6779.0 20		D L	R	

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Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

<u>E(level)[†]</u>	<u>J^π</u>	<u>T_{1/2} or Γ^{&}</u>	<u>XREF</u>			<u>Comments</u>
6800 4			D	L		from ($^3\text{He},d$). XREF: D(6802) E(level): from (α,p).
6823 2			D		R	E(level): from ($^3\text{He},d$).
6842 2			D	L	R	E(level): weighted average of 6842 4 from (α,p) and 6842 2 from ($^3\text{He},d$).
6863 3	(9/2 ⁺) [#]			L		
6866.7 6	(1/2,3/2 ⁻)		D	O	R	E(level): other: 6866 2 from ($^3\text{He},d$). J ^π : L($^3\text{He},d$)=(0,1) from 0 ⁺ .
6891.6 30	(9/2 ⁺) [#]			L		
6948.1 34	5/2 ⁺			L	WX	XREF: W(6930) E(level): weighted average of 6948.6 34 from (α,p) and 6930 20 from $^{36}\text{Ar}(d,^3\text{He})$. Other: 6953 43 from $^{36}\text{Ar}(\text{pol } d,^3\text{He})$. J ^π : L(pol d, ^3He)=2 and L+1/2 from analyzing power.
6986.3 34	(9/2 ⁺) [#]			L		
7066.2 7	(5/2 ⁺) [#]			L	O R	E(level): weighted average of 7066.2 7 from (p, γ), and 7066 2 from ($^3\text{He},d$).
7103.4 7	3/2 ⁽⁻⁾			L	O R	XREF: L(7105?) E(level): weighted average of 7103.4 7 from (p, γ), and 7103 2 from ($^3\text{He},d$). J ^π : spin=3/2 from $\gamma(\theta)$ in (p, γ); parity from L($^3\text{He},d$)=(1,3) from 0 ⁺ .
7121.7 24				L		
7178.6 8	1/2 ⁽⁺⁾				O	T=3/2 J ^π : spin=1/2 from $\gamma(\theta)$ in (p, γ); parity from IAS in ^{35}S .
7179 4	(7/2 ⁺) [#]			L	N	E(level): weighted average of 7180 4 from $^{32}\text{S}(\alpha,p)$ and 7170 10 from $^{33}\text{S}(\alpha,d)$. J ^π : also L=6 from 3/2 ⁺ in $^{33}\text{S}(\alpha,d)$, indicating (7/2;17/2) ⁺ .
7185	5/2 ⁺				O R X	XREF: R(7178)X(7181) J ^π : L(pol d, ^3He)=2 and L+1/2 from analyzing power for a group at E=7181 60. Other: L($^3\text{He},d$)=(2) from 0 ⁺ for a group at E=7178 2 considered the same level.
7194.6 7	1/2 ⁻	0.027 keV 8		OP	R	J ^π : from resonance analysis of measured $\sigma(\theta)$ (1971BrXT) in $^{34}\text{S}(p,p),(p,p'\gamma)$:resonances. Other: 1/2,3/2 from $\gamma(\theta)$ in (p, γ).
7213.7 25				L		y
7225.5 8	5/2				O	y
7228.2 20	(3/2 ⁺) [#]			L	R	y
						XREF: R(7227)y(7250) E(level): weighted average of 7230.9 30 from $^{32}\text{S}(\alpha,p)$, 7227 2 from $^{34}\text{S}(^3\text{He},d)$. J ^π : other: L=(0,1) from 0 ⁺ in ($^3\text{He},d$) is inconsistent.
7234.0 8	5/2 ⁽⁺⁾				O	J ^π : spin=5/2 from $\gamma(\theta)$ in (p, γ); primary 6014.1 γ to 1/2 ⁺ .
7269.2 1					O	
7272.7 9	1/2 ⁻	0.014 keV 4		OP	R y	XREF: P(7274)R(7273)y(7250)

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)				
E(level) [†]	J ^π [‡]	T _{1/2} or Γ ^{&}	XREF	Comments
				J ^π : from resonance analysis of measured $\sigma(\theta)$ (1971BrXT) in $^{34}\text{S}(\text{p},\text{p}), (\text{p},\text{p}'\gamma)$: resonances. Other: (1/2,3/2) from $\gamma(\theta)$ in (p, γ).
7348.3	(7/2) [#]		L	
7361.9 8	3/2 ⁽⁺⁾		L 0 R	E(level): weighted average of 7362.3 from (α ,p), 7362.0 8 from (p, γ), and 7361.2 from (^3He ,d). J ^π : spin=3/2 from $\gamma(\theta)$ in (p, γ); 6141.9 γ (M1+E2) to 1/2 ⁺ .
7396.4 10	7/2 ⁽⁻⁾		0 R	E(level): weighted average of 7396.0 10 from (p, γ), and 7398.2 from (^3He ,d). J ^π : spin=7/2 from $\gamma(\theta)$ in (p, γ); 4233.3 γ (M1+E2) to 7/2 ⁻ .
7413.8 30			L	
7450.9 8	3/2		L 0	E(level): weighted average of 7447.3 from (α ,p) and 7451.1 6 from (p, γ). J ^π : from $\gamma(\theta)$ in (p, γ).
7496			0	
7501.1 8	(5/2 ⁺ , 7/2, 9/2 ⁺)		1 0	XREF: l(7503) J ^π : primary 3558.0 γ to 9/2 ⁺ , 4498.2 γ to 5/2 ⁺ .
7502.9 7			1 0	XREF: l(7503)
7518.8 7	7/2		0	J ^π : from $\gamma(\theta)$ in (p, γ).
7548.9 6	7/2 ⁻	<0.69 fs	0P	T=3/2 XREF: P(7550) J ^π : from resonance analysis of measured $\sigma(\theta)$ in $^{34}\text{S}(\text{p},\text{p}), (\text{p},\text{p}'\gamma)$: resonances. Spin=7/2 also from $\gamma(\theta)$ in (p, γ). T _{1/2} : lifetime $\tau < 1$ fs from (p, γ) (1973Fa07, DSAM). Other: $\Gamma_p = 4.6$ eV 10 and $\Gamma = 5$ eV +4-2 from $^{34}\text{S}(\text{p},\text{p})$ elastic scattering excitation function analysis (1971BrXT). Other Γ_p , Γ_γ , Γ values/ratios are mostly deduced from (p, γ) resonance strengths, which have large discrepancies. See comments in the (p, γ) dataset for details.
7561.3 6	1/2, 3/2		L 0	E(level): other: 7566.5 35 from (α ,p). J ^π : from $\gamma(\theta)$ in (p, γ).
7565.20	5/2 ⁺			X J ^π : L=2 from 0 ⁺ in (pol d, ^3He) and L+1/2 transfer from analyzing power.
7590.4			L	
7600.8 8	5/2 ⁺	<13.9 fs	0	J ^π : spin=5/2 from $\gamma(\theta)$ in (p, γ); 7599.9 γ M1+E2 to 3/2 ⁺ . T _{1/2} : lifetime $\tau < 20$ fs from (p, γ) (1971Wi13, DSAM), which is equivalent to $\Gamma > 0.033$ eV.
7618.7 5	5/2 ⁺		0	J ^π : spin=5/2 from $\gamma(\theta)$ in (p, γ); 7617.8 γ M1+E2 to 3/2 ⁺ .
7648.1 30			L	
7656.6 8	(1/2, 3/2, 5/2 ⁺)		0	J ^π : primary 3688.8 γ to 1/2 ⁺ .
7670.10	(7/2 to 17/2) ⁺		N	J ^π : L(α ,d)=6 from 3/2 ⁺ .
7671.9 8	5/2 ⁺		1 0p	XREF: p(7686) J ^π : spin=5/2 from $\gamma(\theta)$ in (p, γ); 4668.8 γ , M1+E2 to 5/2 ⁺ ; 5908.2 γ , M1+E2 to 5/2 ⁺ ; other: 3324.0 γ to 9/2 ⁻ seems to favor 5/2 ⁻ .
7685.5 8	3/2 ⁻		0P	J ^π : spin=3/2 from $\gamma(\theta)$ in (p, γ) and 3/2 ⁻ from R-Matrix analysis in (p,p): resonances. $\Gamma_p = 446$ eV 15 from (p,p): resonances.

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)				
E(level) [†]	J ^π [‡]	T _{1/2} or Γ ^{&}	XREF	Comments
7693.9 8	(1/2 ⁺ , 3/2, 5/2)		0	J ^π : primary 7693.0γ to 3/2 ⁺ , 3634.6γ to (3/2) ⁻ , 4690.9γ to 5/2 ⁺ .
7706.4 8	5/2 ⁺		L OP	E(level): other: 7709.9 25 from (α,p). J ^π : spin=5/2 from γ(θ) in (p,γ) and 3/2 ⁺ , 5/2 ⁺ from R-Matrix analysis in (p,p):resonances.
7744.8 9	7/2 ⁻		L 0	Γ _p =4 eV 1 in (p,p). E(level): other: 7744.1 30 from (α,p). J ^π : spin=7/2 from γ(θ) in (p,γ); 4581.6γ M1+E2 to 7/2 ⁻ .
7748	(7/2 to 17/2) ⁺		NO	XREF: N(7750) E(level): from (p,γ). J ^π : L(α,d)=6 from 3/2 ⁺ .
7777.1 9	(5/2 ⁺)		L 0	E(level): from (p,γ). Other: 7770.3 33 from (α,p). J ^π : based on primary transitions: 4613.9γ to 7/2 ⁻ , 5131.0γ to 7/2 ⁺ , 6557.1γ to 1/2 ⁺ .
7781.7 11	(5/2 ⁻)		0	J ^π : primary 3433.8γ to 9/2 ⁻ ; relatively weak 6561.7γ to 1/2 ⁺ .
7797.2 9	1/2 ⁻	0.031 keV 10	L OP	E(level): weighted average of 7799.8 30 from (α,p), 7797.1 9 from (p,γ), and 7796 3 from $^{34}\text{S}(p,p), (p,p'\gamma)$: resonances. J ^π : spin=1/2 from γ(θ) in (p,γ); (1/2) ⁻ from R-Matrix analysis of in (p,p):resonances.
7837.2 10	3/2 ⁻	3.7 keV 4	OP	T=3/2 E(level): weighted average of 7837.2 10 from (p,γ), and 7837 3 from $^{34}\text{S}(p,p), (p,p'\gamma)$:resonances. J ^π : spin=3/2 from γ(θ) in (p,γ) and 3/2 ⁻ from R-Matrix analysis in (p,p):resonances. Γ: other: lifetime τ<5 fs from (p,γ) (1971Wi13, DSAM).
7839.0 9	(5/2 ⁺)		0	J ^π : primary 6619.0γ to 1/2 ⁺ , 3895.9γ to 9/2 ⁺ .
7868.7 10	(1/2 ⁺ , 3/2, 5/2 ⁺)		L 0	E(level): other: 7868.5 35 from (α,p). J ^π : primary 6648.6γ to 1/2 ⁺ , 2654.1γ to (5/2 ⁺).
7873.03 ^b 30	13/2 ⁺		F HI NO	XREF: N(7870) E(level): weighted average of 7873.3 8 from ($^{28}\text{Si}, \alpha p \gamma$), 7873.4 4 from ($^{16}\text{O}, \alpha p \gamma$), and 7872.78 30 from ($^{14}\text{N}, \alpha p n \gamma$). J ^π : 1946.35γ E1+M2, ΔJ=1 to 5927, (11/2) ⁻ ; 1786.2γ D(+Q) to 13/2 ⁻ ; band head.
7880.8 9	3/2 ⁺ , 5/2 ⁺	0.008 keV 4	OP	J ^π : from γ(θ) in (p,γ) and R-Matrix analysis in (p,p):resonances.
7885			0	
7889.1 15			0	
7899.2 7	(5/2)		1 0	XREF: l(7901) J ^π : primary 2735.8γ to 7/2 ⁻ , 3018.1γ to

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)					
E(level) [†]	$J^{\pi\ddagger}$	$T_{1/2}$ or Γ ^{&}	XREF		Comments
7903			L	0	$7/2^{(+)}$, 3721.1 γ to $3/2^{-}$, 5204.9 γ to $3/2^{+}$. XREF: I(7901)
7923.3 9	($3/2^{+}$, $5/2^{+}$)			0	J^{π} : primary 5277.2 γ to $7/2^{+}$, 6703.2 γ to $1/2^{+}$.
7930				0	
7949				0	
7970.3 10	($5/2^{-}$)			0	J^{π} : primary 3622.3 γ to $9/2^{-}$, weak 6750.2 γ to $1/2^{+}$.
7978.6 35			L		
7987.8 10	$3/2^{+}$			0	J^{π} : spin= $3/2$ from $\gamma(\theta)$ in (p, γ); 7986.8 γ M1+E2 to $3/2^{+}$.
7995.6 10	($5/2$)			0	J^{π} : primary 2592.2 γ to $3/2^{-}$, 7994.6 γ to $3/2^{+}$, 2832.1 γ to $7/2^{-}$, 5349.5 γ to $7/2^{+}$.
8001.0 9	($7/2^{+}$)		L	0	E(level): other: 8000.7 41 from (α ,p). J^{π} : from primary transitions in (p, γ): 3653.0 γ to $9/2^{-}$, 4057.9 to $9/2^{+}$, 3827.3 γ to $5/2^{-}$, 6237.4 γ to $5/2^{+}$, 8000.0 γ to $3/2^{+}$. XREF: P(8004)W(8010)X(8007)
8005.0 10	$5/2^{+}$	0.011 keV 5	OP	WX	J^{π} : L(d, ^3He)=2 from 0^{+} ; spin= $5/2$ from $\gamma(\theta)$ in (p, γ). J^{π} : L(α ,d)=6 from $3/2^{+}$.
8010 10	($7/2$ to $17/2$) ⁺		N		
8019.2 35			L		
8035.7 10	$1/2^{-}$, $3/2^{-}$	0.026 keV 7		OP	E(level): other: 8034 3 from $^{34}\text{S}(\text{p},\text{p}), (\text{p},\text{p}'\gamma)$:resonances.
8038.4 11	$1/2^{+}$, $3/2^{+}$	0.30 keV 2		OP	E(level): weighted average of 8038.5 11 from (p, γ), and 8038 3 from $^{34}\text{S}(\text{p},\text{p}), (\text{p},\text{p}'\gamma)$:resonances. J^{π} : spin= $3/2$ from $\gamma(\theta)$ in (p, γ), but $1/2^{+}$ from R-Matrix analysis in (p,p); 8037.5 γ M1+E2 to $3/2^{+}$.
8075.9 10	($5/2$)		L	0	E(level): weighted average of 8075.1 46 from (α ,p) and 8075.9 10 from (p, γ). J^{π} : primary 3897.8 γ to $3/2^{-}$, 8074.9 γ to $3/2^{+}$; 2912.4 γ to $7/2^{-}$, 3962.0 γ to $7/2^{+}$.
8080				0	
8096.1 11	($5/2$)			0	J^{π} : primary 2932.6 γ to $7/2^{-}$, 5450.0 γ to $7/2^{+}$; 3918.0 γ to $3/2^{-}$, 4177.4 γ to $3/2^{+}$.
8100 10	($7/2$ to $17/2$) ⁺		N		J^{π} : L(α ,d)=6 from $3/2^{+}$.
8106.4 11	$3/2^{+}$		L	0	E(level): other: 8103.8 42 from (α ,p). J^{π} : spin= $3/2$ from $\gamma(\theta)$ in (p, γ); 8105.4 γ M1+E2 to $3/2^{+}$.
8113.3 11	($1/2^{+}$, $3/2$, $5/2^{+}$)		L	0	XREF: L(8121?) J^{π} : primary 2897.4 γ ($5/2^{+}$), 4145.4 γ to $1/2^{+}$.
8147.2 11	$3/2^{-}$	0.56 keV 6		OP	XREF: P(8150) J^{π} : from R-Matrix analysis in (p,p); also proposed by 1976Me12 in (p, γ) based on strengths of primary γ rays. Γ : from $^{34}\text{S}(\text{p},\text{p}), (\text{p},\text{p}'\gamma)$:resonances in 1977Ou01.
8148 3	$1/2^{-}$	2.66 keV 27		P	
8150 3	$3/2^{-}$	0.56 keV 6		P	
8156.8 11	($5/2^{+}$)			0	J^{π} : primary 3978.7 γ to $3/2^{-}$, 8155.8 γ to $3/2^{+}$, 4213.6 γ to $9/2^{+}$.
8171.7 42			L		

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)					
E(level) [†]	J ^π [‡]	T _{1/2} or Γ ^{&}	XREF		Comments
8179.4 12	(3/2 ⁻ , 5/2, 7/2 ⁺)		0		J ^π : primary 5016.1γ to 7/2 ⁻ , 5484.9γ to 3/2 ⁺ .
8208.2 10	5/2 ⁺	0.033 keV 10	OP	WX	T=3/2 XREF: W(8210)X(8204) E(level): other: 8209 3 from $^{34}\text{S}(\text{p},\text{p}), (\text{p}, \text{p}'\gamma)$:resonances. J ^π : L(pol d, ^3He)=2 and L+1/2 from analyzing power.
8211 3	1/2 ⁺	0.094 keV 15	L OP		XREF: O(8209) E(level): from (p,p):resonances. Other: 8210.2 46 from (α,p).
8216.3 10	5/2 ⁺	0.014 keV 3	1 OP	WX	XREF: w(8210)x(8204) J ^π : from γ(θ) in (p,γ) and R-Matrix analysis in (p,p):resonances.
8242.4 12	3/2 ⁻	0.140 keV 15	1 OP		XREF: l(8246.0)P(8243) J ^π : from γ(θ) in (p,γ) and R-Matrix analysis in (p,p):resonances.
8251 5	(3/2 ⁺ , 5/2 ⁺)		1 0		XREF: l(8246.0)
8269.0 5	5/2 ⁺	0.005 keV 3	L OP		J ^π : primary 4138γ to 7/2 ⁺ , 7031γ to 1/2 ⁺ . XREF: L(8270.6) E(level): other: 8271 3 from $^{34}\text{S}(\text{p},\text{p}), (\text{p}, \text{p}'\gamma)$:resonances.
8277.2 5	(5/2 ⁺)	0.006 keV 3	OP		J ^π : 3/2 ⁺ , 5/2 ⁺ from R-Matrix analysis in (p,p):resonances; primary 5105.5γ to 7/2 ⁻ rules out 3/2 ⁺ , since it would require Mult=M2 which is unlikely based on RUL and Γ=0.005 keV.
8282.4 5	(3/2 ⁻ , 5/2)		0		XREF: P(8279) J ^π : 3/2 ⁺ , 5/2 ⁺ from R-Matrix analysis in (p,p):resonances; primary 3939.2γ to 9/2 ⁻ disfavors both 3/2 ⁺ and 5/2 ⁺ , but 3/2 ⁺ is much less likely.
8287.82 30	1/2 ⁻	0.04 keV 1	L OP		J ^π : primary 3118.9γ to 7/2 ⁻ , 4104.2γ to 3/2 ⁻ , 8281.4γ to 3/2 ⁺ . XREF: L(8292)P(8290)
8297.2 9	3/2 ⁻	0.073 keV 15	OP		J ^π : from γ(θ) in (p,γ) and R-Matrix analysis in (p,p):resonances.
8298.2 5	3/2 ⁺		1 0		XREF: P(8300) XREF: l(8292)
8318.1 3	(5/2 ⁺)		L 0		J ^π : spin=3/2 from γ(θ) in (p,γ); 8297.1γ M1+E2 to 3/2 ⁺ . XREF: L(8321.0)
8319.6 ^a 5	15/2 ⁻	12.5 fs 7	F H		J ^π : primary 4350.2γ to 1/2 ⁺ , 5154.8γ to 7/2 ⁻ , 5671.9γ to 7/2 ⁺ . E(level): weighted average of 8319.4 9 from ($^{28}\text{Si}, \alpha\text{p}\gamma$) and 8319.7 5 from ($^{16}\text{O}, \alpha\text{p}\gamma$). J ^π : 2911.9γ E2, ΔJ=2 to 11/2 ⁻ , 2232.7γ to 13/2 ⁻ ; band assignment. T _{1/2} : lifetime τ=18 fs 1 from ($^{28}\text{Si}, \alpha\text{p}\gamma$) (2013Bi10, DSAM).
8323.1 13	(3/2 ⁺ , 5/2 ⁺)		L 0		T _{1/2} : from (HI,xnγ). XREF: L(8321.0)
8346 5			0		J ^π : primary 4210.4γ to 7/2 ⁺ , 7103.0γ to 1/2 ⁺ .

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)					
E(level) [†]	J^π [‡]	$T_{1/2}$ or Γ ^{&}	XREF		Comments
8381.8 8	5/2 ⁺	0.023 keV 7	OP		XREF: P(8383) J^π : from $\gamma(\theta)$ in (p, γ) and R-Matrix analysis in (p,p):resonances.
8389.4 16 8402.9 5	5/2 ⁻	0.002 keV 1	L O L OP		XREF: L(8398.8)P(8404) E(level): others: 8398.8 38 from (α ,p), and 8404 3 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances. J^π : 5/2 ⁻ , 7/2 ⁻ from R-Matrix analysis in (p,p):resonances; its primary 5708.5 γ to 3/2 ⁺ , 2694 level rules out 7/2 ⁻ based on RUL and $\Gamma=0.002$ keV.
8405.8 14	(5/2 ⁻)	0.001 keV 1	OP		E(level): weighted average of 8405.7 14 from (p, γ), and 8406 3 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances. J^π : (5/2 ⁻ , 7/2 ⁻) from R-Matrix analysis in (p,p):resonances; primary 8404.6 γ to 3/2 ⁺ .
8409.4 18	1/2 ⁻	0.125 keV 15	OP		E(level): weighted average of 8409.6 18 from $^{34}\text{S}(\text{p},\gamma)$ and 8409 3 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances.
8416.7 5	1/2 ⁺	0.026 keV 7	OP		E(level): weighted average of 8416.7 5 from (p, γ), and 8418 3 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances. J^π : 1/2 ⁺ from R-Matrix analysis in (p,p):resonances; but primary 4473.5 γ to 9/2 ⁺ and 7196.5 γ to 1/2 ⁺ in (p, γ) gives (5/2 ⁺), which may indicate two different resonances with close energies.
8430.3 5 8434.8 5 8464.3 5	(3/2) ⁺ (1/2 ⁺ , 3/2, 5/2 ⁺)	0.090 keV 15	L O OP O		E(level): other: 8429.2 40 from (α ,p). XREF: P(8435) J^π : primary 4496.4 γ to 1/2 ⁺ , 3249.6 γ to (5/2 ⁺).
8484.4 5	(3/2) ⁺	0.012 keV 5	1 OP		E(level): weighted average of 8484.4 5 from (p, γ), and 8486 3 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances. J^π : 3/2 ⁺ , 5/2 ⁺ from R-Matrix analysis in (p,p):resonances; 3/2 ⁺ proposed by 2001Vo24 in (p, γ) based on proposed isobaric analog to 2940, 3/2 ⁺ level in ^{35}S .
8486.2 5	3/2 ⁻	0.150 keV 15	1 OP		E(level): weighted average of 8486.1 5 from (p, γ), and 8488 3 from $^{34}\text{S}(\text{p,p}),(\text{p,p}'\gamma)$:resonances.
8487.5 5	15/2 ⁻	<69 fs	F H		E(level): weighted average of 8487.8 10 from ($^{28}\text{Si},\alpha\text{p}\gamma$) and 8487.4 5 from ($^{16}\text{O},\alpha\text{p}\gamma$). J^π : 3080.0 γ E2, $\Delta J=2$ to 11/2 ⁻ , 2399.8 γ D(+Q) to 13/2 ⁻ . $T_{1/2}$: lifetime $\tau<0.1$ ps from ($^{28}\text{Si},\alpha\text{p}\gamma$) (2007Ks01, DSAM).
8506.5 5 8514.3 5 8533.9 5	1/2 ⁻	0.150 keV 15	L O OP L O		E(level): other: 8505.1 47 from (α ,p). XREF: P(8515) E(level): weighted average of 8532.4 43 from (α ,p) and 8533.9 5 from (p, γ).
8558.2? 13	(13/2 ⁻)	<0.14 ps	F		XREF: F(?) J^π : 2631 γ D(+Q), $\Delta J=1$ to 11/2 ⁻ in

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Adopted Levels, Gammas (continued)

³⁵ Cl Levels (continued)						
E(level) [†]	J ^π [‡]	T _{1/2} or Γ ^{&}	XREF			Comments
8571.9 5	(5/2) ⁺	0.08 keV 1	L	OP	X	(²⁸ Si,αpγ). T _{1/2} : lifetime τ<0.2 ps from (²⁸ Si,αpγ) (2007Ks01, DSAM). T=(3/2) XREF: P(8573)X(8580) E(level): weighted average of 8571.9 5 from (p,γ), and 8573 3 from ³⁴ S(p,p),(p,p'γ):resonances. Other: 8567.9 37 from (α,p). XREF: P(8582)
8580.5 5	1/2 ⁺	0.75 keV 8		OP		XREF: P(8582)
8585.8 5				O		
8590.1 5	5/2 ⁺	0.003 keV 2		OP		XREF: P(8592) E(level): weighted average of 8590.0 5 from (p,γ), and 8592 3 from ³⁴ S(p,p),(p,p'γ):resonances.
8612.0 5	3/2 ⁺ ,5/2 ⁺		1	O	W	J ^π : 3/2 ⁺ ,5/2 ⁺ from R-Matrix analysis in (p,p):resonances; primary 5426.6γ to 7/2 ⁻ ruled out 3/2 ⁺ based on RUL and Γ=0.003 keV. XREF: W(8610) J ^π : L(d, ³ He)=2 from 0 ⁺ .
8613.7 5	5/2 ⁺	0.175 keV 20	1	OP		XREF: P(8616)
8618.2 5	3/2 ⁺ ,5/2 ⁺	0.002 keV 1	1	OP		XREF: P(8620) E(level): weighted average of 8618.1 5 from (p,γ), and 8620 3 from ³⁴ S(p,p),(p,p'γ):resonances.
8630.1 3	7/2 ⁻	0.001 keV 1		O		J ^π : 4282.1γ M1+E2 to 9/2 ⁻ , 6866.2γ D+Q to 5/2 ⁺ , spin=3/2,7/2 from γ(θ) in (p,γ).
8633 3	(3/2,5/2 ⁺)	0.001 keV 1		P		
8640	(5/2 ⁺)			O		J ^π : spin=(5/2) from γ(θ) in (p,γ); primary 4697γ to 9/2 ⁺ .
8641.4 5	3/2 ⁺ ,5/2 ⁺	0.003 keV 2		OP		XREF: P(8643)
8654 4			L			
8686.2 5	5/2 ⁻	0.001 keV 1		OP		E(level): weighted average of 8686.2 5 from (p,γ), and 8687 3 from ³⁴ S(p,p),(p,p'γ):resonances.
8688.3 5	1/2 ⁺	0.20 keV 2		OP		J ^π : 5/2 ⁻ ,7/2 ⁻ from R-Matrix analysis in (p,p):resonances; primary 7446.0γ to 1219, 1/2 ⁺ disfavors both 5/2 ⁻ and 7/2 ⁻ , but 7/2 ⁻ is much less likely. XREF: P(8689)
8691 3	1/2 ⁻	6.4 keV 7		P		
8696.9 5	3/2 ⁻	0.8 keV 1	L	OP		XREF: L(8698)P(8698)
8700 10	(7/2 to 17/2) ⁺			N		J ^π : L(α,d)=6 from 3/2 ⁺ .
8706.3 5				O		
8717.8 5			L	O		E(level): weighted average of 8720.9 37 from (α,p) and 8717.7 5 from (p,γ).
8750.9 5	3/2 ⁻	0.30 keV 3		OP		E(level): weighted average of 8750.9 5 from (p,γ), and 8750 3 from ³⁴ S(p,p),(p,p'γ):resonances.
8767.0 5				O		
8772.9 5	1/2 ⁻	0.57 keV 6		OP		XREF: P(8775)
8779.8 5	3/2 ⁻	0.214 keV 25		OP		XREF: P(8781) E(level): weighted average of 8779.8 5 from (p,γ), and 8781 3 from ³⁴ S(p,p),(p,p'γ):resonances.

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)				
E(level) [†]	J ^π [‡]	T _{1/2} or Γ ^{&}	XREF	Comments
8787.2 5	(5/2 ⁺)	0.001 keV 1	L OP	E(level): others: 8786.3 38 from (α,p), and 8788 3 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$:resonances. J ^π : (3/2 ⁺ , 5/2 ⁺) from R-Matrix analysis in (p,p):resonances; primary 3623.7γ and 5623.8γ to 7/2 ⁻ .
8788.3 ^b 6	(15/2 ⁺)	<0.28 ps	F H	XREF: F(?) J ^π : 915.4γ D, ΔJ=1 to 13/2 ⁺ ; band assignment. T _{1/2} : lifetime τ<0.4 ps from ($^{28}\text{Si}, \alpha\text{p}\gamma$) (2007Ks01, DSAM).
8798.4 5	(3/2 ⁺ , 5/2 ⁺)	0.001 keV 1	OP	E(level): weighted average of 8798.4 5 from (p,γ), and 8799 3 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$:resonances.
8820.9 5	(1/2 ⁺ to 7/2 ⁺)		0	J ^π : primary 5817.6γ and 7057.0γ to 5/2 ⁺ ; 6126.3γ and 8819.7γ to 3/2 ⁺ , which yields (1/2 ⁺ , 3/2 [±] , 5/2 [±] , 7/2 ⁺).
8824.2 5	1/2 ⁺	1.70 keV 17	OP	XREF: P(8825) E(level): weighted average of 8824.2 5 from (p,γ), and 8825 3 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$:resonances.
8829.3 5	1/2 ⁻	12.2 keV 12	OP	E(level): weighted average of 8829.3 5 from (p,γ), and 8829 3 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$:resonances. J ^π : 1/2 ⁻ from R-Matrix analysis in (p,p):resonances; but primary 6183.0γ to 7/2 ⁺ gives J≥3/2, which may indicate a different level.
8830 3	1/2 ⁺	0.080 keV 15	P	
8833.1 5	(5/2 ⁻ , 7/2)		0	J ^π : primary 3618.4γ to (5/2 ⁺), 4063.0γ to (5/2 ⁻); 4720.4γ to 7/2 ⁺ , 5669.6γ to 7/2 ⁻ .
8838.1 5	5/2 ⁻ , 7/2 ⁻	0.001 keV 1	1 OP	XREF: P(8839)
8844.4 ^b 4	17/2 ⁺	5.9 ps 11	F HI N	XREF: N(8840) E(level): weighted average of 8845.1 9 from ($^{28}\text{Si}, \alpha\text{p}\gamma$), 8844.6 5 from ($^{16}\text{O}, \alpha\text{p}\gamma$), and 8844.2 4 from ($^{14}\text{N}, \alpha\text{p}\gamma$). J ^π : 971.38γ E2, ΔJ=2 to 13/2 ⁺ , 524.8γ to 15/2 ⁻ ; band assignment. T _{1/2} : weighted average of 6.1 ps 11 from ($^{16}\text{O}, \alpha\text{p}\gamma$) (1991Ja11, DSAM) and 5.5 ps 14 from ($^{14}\text{N}, \alpha\text{p}\gamma$) (1976Wa11, DSAM).
8856.2 5	5/2 ⁺	0.010 keV 5	OP	E(level): weighted average of 8856.1 5 from (p,γ), and 8858 3 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$:resonances. J ^π : 3/2 ⁺ , 5/2 ⁺ from R-Matrix analysis in (p,p):resonances; primary 5692.6γ to 7/2 ⁻ rules out 3/2 ⁺ based on RUL and Γ=0.010 keV.
8868.6 5	3/2 ⁺ , 5/2 ⁺	0.027 keV 10	OP	E(level): weighted average of 8868.6 5 from (p,γ), and 8870 3 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$:resonances.
8884.1 5			0	
8886.1 5	(5/2)		L 0	E(level): other: 8887.5 41 from (α,p). J ^π : primary 3722.6γ to 7/2 ⁻ , 6239.8γ to 7/2 ⁺ ; 4707.9γ to 3/2 ⁻ , 4967.2γ to 3/2 ⁺ .

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)					
E(level) [†]	J ^π [‡]	T _{1/2} or Γ ^{&}	XREF		Comments
8893.3 5	(5/2 ⁺ , 7/2 ⁺)		0		J ^π : primary 4950.0γ to 9/2 ⁺ , 6198.7γ to 3/2 ⁺ .
8904.8 5	(1/2, 3/2, 5/2 ⁺)		0		J ^π : primary 4726.6γ to 3/2 ⁻ , 6210.2γ to 3/2 ⁺ , 7684.5γ to 1/2 ⁺ .
8906.8 5	5/2 ⁺	0.002 keV 1	OP	x	XREF: P(8909)x(8921) J ^π : 3/2 ⁺ , 5/2 ⁺ from R-Matrix analysis in (p,p):resonances; primary 4963.5γ to 9/2 ⁺ rules out 3/2 ⁺ based on RUL and Γ=0.002 keV.
8919.8 5	(5/2 ⁻ , 7/2, 9/2 ⁺)		0		XREF: x(8921) J ^π : primary 4571.7γ to 9/2 ⁻ , 7155.9γ to 5/2 ⁺ .
8933.3 5			0		
8953.1 5	3/2 ⁺	0.075 keV 15	OP		XREF: P(8955) E(level): weighted average of 8953.0 5 from (p,γ), and 8955 3 from ³⁴ S(p,p),(p,p'γ):resonances. J ^π : spin=3/2 from γ(θ) in (p,γ) and (3/2) ⁺ from R-Matrix analysis in (p,p):resonances.
8957.9 5	(1/2 ⁻ , 3/2 ⁻)	0.04 keV 1	L	OP	XREF: P(8959) E(level): others: 8957.7 44 from (α,p) and 8959 3 from (p,p):resonances.
8981.9 5	5/2 ⁻ , 7/2 ⁻	0.003 keV 2	OP		XREF: P(8983) E(level): weighted average of 8981.9 5 from (p,γ), and 8983 3 from ³⁴ S(p,p),(p,p'γ):resonances.
8984.2 5	5/2 ⁺	0.025 keV 10	OP		XREF: P(8985) J ^π : 3/2 ⁺ , 5/2 ⁺ from R-Matrix analysis in (p,p):resonances; primary 5820.7γ to 7/2 ⁻ rules out 3/2 ⁺ based on RUL and Γ=0.025 keV.
8988.4 5			0		
8992.4 5			0		
8996.7 5	5/2 ⁻ , 7/2 ⁻	0.002 keV 1	L	OP	XREF: L(8996.9)P(8998)
9001.1 5			0		
9019.3 5	3/2 ⁻	3.27 keV 30	OP		XREF: P(9021) Γ: weighted average of 3.1 keV 3 from (p,γ) and 3.50 keV 35 from (p,p):resonances.
9024.5 5			1	0	
9029.9 5	(5/2) ⁺	0.04 keV 1	1	OP	XREF: P(9031)
9033.1 5			0		
9038.4 5	1/2 ⁻	0.292 keV 30	OP		XREF: P(9040) E(level): weighted average of 9038.3 5 from (p,γ), and 9040 3 from ³⁴ S(p,p),(p,p'γ):resonances.
9048.4 5	5/2 ⁻ , 7/2 ⁻	0.001 keV 1	OP		XREF: P(9048)
9050 3	(5/2) ⁺	0.095 keV 15	P		
9081.5 4	5/2 ⁺	59 eV 10	L	OP	XREF: L(9085.7)P(9084) E(level): weighted average of 9081.4 4 from (p,γ), and 9084 3 from ³⁴ S(p,p),(p,p'γ):resonances. J ^π : spin=5/2 from γ(θ) in (p,γ) and (5/2) ⁺ from R-Matrix analysis in (p,p):resonances.
9088.5 5			L	OP	Γ: weighted average of 65 eV 20 from (p,γ) and 57 eV 10 from (p,p):resonances. XREF: L(9085.7)P(9089) E(level): weighted average of 9088.5 5 from

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)				
E(level) [†]	J ^π [‡]	T _{1/2} or Γ ^{&}	XREF	Comments
				(p,γ), and 9089 4 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$:resonances.
9099.3 5			0	
9100.4 5	3/2 ⁻	0.20 keV 2	OP	XREF: P(9101)
9107.4 5			L OP	XREF: L(9109.3)P(9108) E(level): weighted average of 9107.4 5 from (p,γ), and 9108 4 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$:resonances.
9110.1 5			L OP	XREF: L(9109.3)P(9114) E(level): weighted average of 9110.0 5 from (p,γ), and 9114 4 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$:resonances.
9124.0 5			OP	XREF: P(9120)
9127 4	5/2 [@]		J P	E(level): from (p,p):resonances. Other: 9127 9 from $^{31}\text{P}(\alpha, \text{p})$:resonances.
9135.1 5		2.0 keV 4	Op	XREF: p(9136) Γ: from (p,γ).
9138.3 5			Op	XREF: p(9136)
9147 4	(7/2 to 17/2) ⁺		N P	XREF: N(9150)P(9147) J ^π : L(α,d)=6 from 3/2 ⁺ .
9155.7 5			0	
9157.1 5	5/2 ⁻		OP	XREF: P(9159) J ^π : spin=5/2 from γ(θ) in (p,γ); 5993.7γ M1+E2 to 7/2 ⁻ .
9161 4	1/2 [@]		J 1	XREF: l(9160.0)
9163.3 5	(3/2 ⁻)		1 OP	XREF: l(9160.0)P(9168) E(level): other: 9168 4 from (p,p):resonances.
9184.2 5			Op	XREF: p(9185)
9188.8 5			Op	XREF: p(9185)
9194.1 5			L OP	E(level): others: 9194.0 39 from (α,p), and 9196 4 from (p,p):resonances.
9207 4	9/2 ⁺		P	
9222 4	5/2 ⁺		P	X XREF: X(9220) J ^π : L(pol d, ^3He)=2 and L+1/2 from analyzing power for a group at E=9220 120.
9227 4	5/2 ⁻		P	
9241 4	7/2 ⁺		P	
9255 4	1/2 [@]		J P	E(level): from (p,p):resonances. Other: 9256 4 from $^{31}\text{P}(\alpha, \text{p})$:resonances.
9264.9 39			L	
9271 4			P	
9276 4			P	
9284 4	5/2 ⁻		P	
9294 4			P	
9299 4			P	
9316.2 39			L	
9325 4	5/2 ⁺		P	
9335 4	7/2 ⁺		L P	E(level): weighted average of 9336 4 from $^{34}\text{S}(\text{p,p}), (\text{p,p}'\gamma)$:resonances and 9334 5 from $^{32}\text{S}(\alpha, \text{p})$.
9345 4			P	
9349 4			P	
9355 4			P	
9376 4			L	

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)				
E(level) [†]	J ^π [‡]	T _{1/2} or Γ ^{&}	XREF	Comments
9400 9	1/2 [@]		J	
9450 10	(7/2 to 17/2) ⁺		N	J ^π : L(α,d)=6 from 3/2 ⁺ .
9456 4	3/2 [@]		J	
9476 4			L	
9481.0 18	3/2 [@]		J	
9508 4	(1/2 ⁻ , 3/2, 5/2 ⁺)		L X	XREF: X(9514) J ^π : L(pol d, ³ He)=(1,2) from 0 ⁺ .
9551 6	5/2 [@]		J	
9673 6	1/2 [@]		J	
9712.7 27	1/2 [@]		J L	E(level): weighted average of 9713.0 27 from ³¹ P(α,p),(α,n):resonances and 9711.8 44 from (α,p).
9740 4			L	
9751.1 27	7/2 [@]		J	
9799 5			L	
9814.0 27	5/2 ⁽⁺⁾		J X	XREF: X(9820) J ^π : spin=5/2 from angular distributions in ³¹ P(α,p):resonances; L=(1,2) from 0 ⁺ in (pol d, ³ He) gives (1/2 ⁻ , 3/2, 5/2 ⁺).
9836 5			L	
9869.8 27	1/2 [@]		J L	XREF: L(9870)
9900.8 27	3/2 [@]		J	
9923 4	3/2 ^{-@}		J	
9958 6	3/2 [@]		J	
9969 5	(1/2, 3/2) [@]		J L	XREF: J(9968) E(level): from (α,p) and (α,n):resonances.
10031 5	(1/2 ⁺ , 3/2 ⁻) [@]		J	
10058 5	3/2 [@]		J	
10075 5	(1/2) [@]		J L	E(level): from (α,p) and (α,n):resonances.
10090 5	5/2 ⁺ [@]		J	
10132 5	3/2 ^{-@}		J	
10163 5	[@]		L	
10172 5	3/2 ⁻ , 5/2 ^{-@}		J	
10179 5			L	
10181.1 ^a 5	19/2 ⁻	42.3 fs 28	F H	E(level): weighted average of 10181.1 10 from (²⁸ Si, αpγ) and 10181.1 5 from (¹⁶ O, αpγ). J ^π : 1693.5γ E2, ΔJ=2 to 15/2 ⁻ ; 1336.3γ D, ΔJ=1 to 17/2 ⁺ ; band assignment. T _{1/2} : lifetime τ=61 fs 4 from (²⁸ Si, αpγ) (2013Bi10, DSAM).
10215 5	3/2 [@]		J L	XREF: L(10218.0)
10221.9 11	(17/2 ⁻)	<0.28 ps	F H	XREF: F(?) E(level): from (¹⁶ O, αpγ). J ^π : 1377.8γ to 17/2 ⁺ consistent with ΔJ=0. T _{1/2} : lifetime τ<0.4 ps from (²⁸ Si, αpγ) (2007Ks01, DSAM).
10236 5	3/2 ^{-@}		J	
10278 5	1/2 [@]		J	

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Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

E(level) [†]	J ^π [‡]	T _{1/2} or Γ ^{&}	XREF	Comments
10295 5			J	
10322 5	5/2 ⁺ @		J	
10340 5	(3/2,5/2) @		J	
10359 5	3/2 @		J	
10385 5			J	
10395 5			L	
10399 5	3/2 ⁺ , 7/2 ⁺ @		J	
10431 5	5/2 ⁺ @		J	
10465 5	5/2 ⁺ @		J L	E(level): weighted average of 10467 5 from $^{31}\text{P}(\alpha, \text{p}), (\alpha, \text{n})$:resonances and 10463.2 49 from (α, p) .
10486 5	5/2 ⁺ @		J	
10517 6			L	
10534 5	1/2 ⁺ , 5/2 ⁺ @		J	
10548 6			L	
10584 5	3/2 ⁺ @		J L	E(level): weighted average of 10587 5 from $^{31}\text{P}(\alpha, \text{p}), (\alpha, \text{n})$:resonances and 10579.5 54 from (α, p) .
10642 5	5/2 ⁺ @		J L	E(level): weighted average of 10641 5 from $^{31}\text{P}(\alpha, \text{p}), (\alpha, \text{n})$:resonances and 10643.0 50 from (α, p) .
10678 5	3/2 ⁺ , 5/2 ⁺ @		J	
10696 5	7/2 ⁻ @		J	
10733 5	3/2 ⁺ @		J L	E(level): weighted average of 10734 5 from $^{31}\text{P}(\alpha, \text{p}), (\alpha, \text{n})$:resonances and 10732.1 50 from (α, p) .
10760 5	3/2 ⁺ , 7/2 ⁻ @		J L	E(level): weighted average of 10760 5 from $^{31}\text{P}(\alpha, \text{p}), (\alpha, \text{n})$:resonances and 10759.1 55 from (α, p) .
10800 5	5/2 ⁺ @		J	
10816 5	1/2 ⁺ , 3/2 ⁻ @		J	
10844 5	3/2 ⁻ @		J	
10858.5 ^b 8	19/2 ⁺	<0.28 ps	F H	XREF: F(?) J ^π : 2069.8γ E2, ΔJ=2 to 15/2 ⁺ , 2014.7γ D, ΔJ=1 to 17/2 ⁺ ; band assignment. T _{1/2} : lifetime τ<0.4 ps from ($^{28}\text{Si}, \alpha \text{py}$) (2007Ks01, DSAM).
10870 5	(5/2, 7/2) @		J	
10886 5			J	
10909 5	3/2 ⁺ , 5/2 ⁻ @		J	
10928 5	7/2 @		J	
10942 5			J	
10964 5	5/2 ⁺ @		J	
10973 5			J	
10998 5	3/2 ⁺ , 7/2 ⁻ @		J	
11034 5	3/2 ⁺ @		J	
11063 5	1/2 ⁺ , 9/2 ⁺ @		J	
11082 5	5/2 ⁺ , 9/2 ⁺ @		J	

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Adopted Levels, Gammas (continued)

^{35}Cl Levels (continued)				
E(level) [†]	J ^π [‡]	T _{1/2} or Γ ^{&}	XREF	Comments
11105 5			J	
11123 5	3/2 [@]		J	
11142 5	5/2 ⁺ [@]		J	
11154 5			J	
11169 5			J	
11179 5	3/2 ⁻ , 5/2 ⁺ [@]		J	
11185 5			J	
11198 5	3/2 [@]		J	
11230 5			J	
11246 5	3/2 [@]		J	
11265 5	3/2 ⁻ [@]		J	
11287 5			J	
11307 5			J	
11314 5	1/2 ⁻ , 5/2 ⁺ [@]		J	
11327 5	1/2 ⁻ , 5/2 ⁺ [@]		J	
11357 5			J	
11375 5			J	
11390 5			J	
11410 5			J	
11421 5	3/2 ⁻ [@]		J	
11434 5			J	
11459.1 ^b 7	21/2 ⁺	43.0 fs 21	F H J	E(level): weighted average of 11459.2 12 from (²⁸ Si,αγ) and 11459.0 7 from (¹⁶ O,αγ). Other: 11460 5 from ³¹ P(α,p),(α,n):resonances. J ^π : 2614.5γ E2, ΔJ=2 to 17/2 ⁺ ; band assignment. T _{1/2} : lifetime τ=62 fs 3 from (²⁸ Si,αγ) (2013Bi10, DSAM).
11473 5	1/2 ⁻ , 5/2 ⁻ [@]		J	
11492 5			J	
11505 5	3/2 ⁻ , 5/2 ⁻ [@]		J	
11525 5			J	
11540 5			J	
11551 5	3/2 ⁻ , 5/2 ⁺ [@]		J	
11565 5			J	
11589 5			J	
11609 5	5/2 ⁺ , 7/2 ⁻ [@]		J	
11627 5	1/2 ⁺ , 5/2 ⁺ [@]		J	
11638 5			J	
11651 5			J	
11684 5	1/2 ⁺ , 5/2 ⁺ [@]		J	
11773 5	3/2 ⁻ , 7/2 ⁻ [@]		J	
11783 5	3/2 ⁻ [@]		J	
1.25×10 ⁴ 5			J	X T=3/2 XREF: J(12900) E(level): from (pol d, ³ He).
12572.2 ^a 6	23/2 ⁻	33 fs 4	F H	E(level): weighted average of 12572.2 12 from (²⁸ Si,αγ) and 12572.2 6 from (¹⁶ O,αγ). J ^π : 2390.8γ E2, ΔJ=2 to 19/2 ⁻ , 1113.3γ D, ΔJ=1 to 21/2 ⁺ ; band assignment.

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Adopted Levels, Gammas (continued) ^{35}Cl Levels (continued)

<u>E(level)[†]</u>	<u>J^π[‡]</u>	<u>T_{1/2} or Γ^{&}</u>	<u>XREF</u>	<u>Comments</u>
13900				T _{1/2} : lifetime $\tau=48$ fs 6 from ($^{28}\text{Si},\alpha\gamma$) (2013Bi10, DSAM).
16306.4 ^a 16	(27/2 ⁻)	<9.0 fs	F J	T=3/2 J ^π : 3734 γ to 23/2 ⁻ ; band assignment. T _{1/2} : lifetime $\tau<13$ fs from ($^{28}\text{Si},\alpha\gamma$) (2013Bi10, DSAM).

[†] Most of Adopted Levels are reported in (p, γ) dataset with precise E(level) values, deduced based on measured γ -ray energies which however are not reported in references in (p, γ) dataset, while E(level) values from E γ values with uncertainties in other datasets are less precise and the numbers of γ rays are much less than that of (p, γ) datasets. Therefore, a least-squares fit to available E γ values with uncertainties would not produce E(level) values more precise and complete than those already reported in (p, γ) and the evaluators have adopted E(level) values from (p, γ) or averages of values based on E γ data from all available datasets where applicable, as noted.

[‡] For proton-resonance levels populated in (p,p):resonances, J^π assignments are from R-Matrix analysis, unless otherwise noted.

[#] From FRESKO-DWBA analysis of measured $\sigma(\theta)$ of (α ,p) (2019Se09).

[@] From angular distributions in $^{31}\text{P}(\alpha,\text{p})$:resonances.

[&] Sources of T_{1/2} values are listed under comments; Γ from $^{34}\text{S}(\text{p,p}),(\text{p,p}')\gamma$:resonances, unless otherwise noted.

^a Band(A): Band based on f_{7/2} orbital.

^b Band(B): Band based on 13/2⁺.

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	Comments
1219.38	1/2 ⁺	1219.4 1	100	0.0	3/2 ⁺	M1+E2	0.093 10	3.66×10 ⁻⁵ 5	B(M1)(W.u.)=0.101 +14-11; B(E2)(W.u.)=2.2 +6-5 E _γ : weighted average of 1219.3 2 from ³⁵ Ar ε decay and 1219.4 1 from (p,p'γ), and 1219.4 6 from (α,pγ). Mult.,δ: ΔJ=1 from γ(θ) in (p,p'γ), E1+M2 ruled out by RUL. δ from adopted B(E2)†=0.00076 7 measured in Coulomb excitation and adopted lifetime τ=0.172 ps 20.
1763.15	5/2 ⁺	543.77 1763.16 10	<0.2 100	1219.38 0.0	1/2 ⁺ 3/2 ⁺	[E2] M1+E2	-2.85 10	0.000323 5 0.0002124 30	I _γ : from (p,γ). Other: <0.5 from (α,pγ). B(M1)(W.u.)=0.00122 +14-12; B(E2)(W.u.)=12.1 +11-9 E _γ : weighted average of 1763.0 2 from ³⁵ Ar ε decay, 1763.1 2 from (²⁴ Mg,αpγ), 1763.3 2 from (¹⁶ O,αpγ), 1763.15 10 from (¹⁴ N,αpnγ), and 1763.2 1 from (p,p'γ). Other: 1763.2 6 from (α,pγ). Mult.: from γγ(ADO) in (¹⁶ O,αpγ) with ΔJ=1 and γ(lin pol) in (²⁸ Si,αpγ). δ: weighted average of -3.0 1 from (α,pγ), -2.66 12 from (p,p'γ), -2.95 45 from Coulomb excitation, -2.6 4 from (¹⁶ O,αpγ), all from γ(θ) or pγ(θ). Other: -0.42 17 from γγ(θ) in (p,γ); 1.8 +10-4; deduced by the evaluators from adopted B(E2)†=0.0106 7 and T _{1/2} =0.36 ps 3.
2645.67	7/2 ⁺	882.80 11	11.4 10	1763.15	5/2 ⁺	M1+E2	+0.21 5	5.29×10 ⁻⁵ 9	B(M1)(W.u.)=0.0196 +35-27; B(E2)(W.u.)=4.2 +23-18 E _γ : weighted average of 882.9 1 from (²⁴ Mg,αpγ), 882.7 5 from (¹⁶ O,αpγ), 882.32 35 from (¹⁴ N,αpnγ), 882.4 10 from (α,pγ), and 882.3 3 from (p,p'γ). I _γ : unweighted average of 8.7 6 from (²⁸ Si,αpγ), 15.2 13 from (²⁴ Mg,αpγ), 11.08 28 from (¹⁶ O,αpγ), 11.1 22 from (α,pγ), 9.5 11 from (p,γ), and 13 5 from (p,p'γ). Mult.: D+Q from pγ(θ) in (α,pγ); E1+M2 ruled out by RUL. δ: weighted average of +0.25 5 from (α,pγ) and +0.17 5 from (p,p'γ). I _γ : from (α,pγ). Other: <2.2 from (p,γ). B(E2)(W.u.)=3.6 +6-4 E _γ : weighted average of 2645.5 4 from (²⁴ Mg,αpγ), 2645.8 5 from (¹⁶ O,αpγ), 2645.79 30 from
		1426.26 2645.70 20	<1.2 100.0 11	1219.38 0.0	1/2 ⁺ 3/2 ⁺	[M3] E2		7.52×10 ⁻⁵ 11 0.000633 9	

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.#	$\delta^\#$	α^a	Comments
									$(^{14}\text{N},\alpha\text{p}\gamma)$, and 2645.7 2 from $(\text{p},\text{p}'\gamma)$. Other: 2646.2 15 from $(\alpha,\text{p}\gamma)$. I_γ : others: 100.0 31 from $(^{24}\text{Mg},\alpha\text{p}\gamma)$, 100.0 22 from $(^{16}\text{O},\alpha\text{p}\gamma)$, 100.0 22 from $(\alpha,\text{p}\gamma)$, and 100.0 20 from $(\text{p},\text{p}'\gamma)$. Other: 100 6 from $(^{28}\text{Si},\alpha\text{p}\gamma)$. Mult.: from $\gamma\gamma(\theta)(\text{DCO})$ and $\gamma\gamma(\text{lin pol})$ in $(^{28}\text{Si},\alpha\text{p}\gamma)$; $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in $(\text{p},\text{p}'\gamma)$. $\text{B}(\text{M1})(\text{W.u.})=0.110 +26-19$; $\text{B}(\text{E2})(\text{W.u.})=3.9 +33-22$ E_γ : weighted average of 930.7 2 from ^{35}Ar ε decay, 930.5 15 from $(\alpha,\text{p}\gamma)$, and 931.3 3 from $(\text{p},\text{p}'\gamma)$. I_γ : weighted average of 14.3 18 from ^{35}Ar ε decay, 18.2 26 from $(\alpha,\text{p}\gamma)$, 16.3 13 from (p,γ) , and 18.6 25 from $(\text{p},\text{p}'\gamma)$. Mult., δ : from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in $(\text{p},\text{p}'\gamma)$. $\text{B}(\text{M1})(\text{W.u.})=0.0116 +37-29$; $\text{B}(\text{E2})(\text{W.u.})=8.0 +45-39$ E_γ : weighted average of 1474.8 3 from ^{35}Ar ε decay and 1475 2 from $(\text{p},\text{p}'\gamma)$. I_γ : weighted average of 9.3 10 from ^{35}Ar ε decay, 11.7 26 from $(\alpha,\text{p}\gamma)$, 9.5 13 from (p,γ) , and 8.3 25 from $(\text{p},\text{p}'\gamma)$. Mult., δ : from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in $(\text{p},\text{p}'\gamma)$. $\text{B}(\text{M1})(\text{W.u.})=0.026 +6-4$; $\text{B}(\text{E2})(\text{W.u.})=0.92 +26-19$ E_γ : weighted average of 2693.7 2 from ^{35}Ar ε decay and 2694.1 6 from $(\text{p},\text{p}'\gamma)$. Other: 2693.2 20 from $(\alpha,\text{p}\gamma)$. I_γ : weighted average of 100.0 17 from ^{35}Ar ε decay, 100 4 from $(\alpha,\text{p}\gamma)$, 100.0 25 from (p,γ) , and 100.0 30 from $(\text{p},\text{p}'\gamma)$. Mult.: from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in $(\text{p},\text{p}'\gamma)$. δ : weighted average of +0.17 8 from $(\alpha,\text{p}\gamma)$ and +0.26 2 from $(\text{p},\text{p}'\gamma)$. I_γ : from $(\alpha,\text{p}\gamma)$. I_γ : from $(\alpha,\text{p}\gamma)$. I_γ : from $(\text{p},\text{p}'\gamma)$. $\text{B}(\text{M1})(\text{W.u.})=0.011 +8-5$ if M1, $\text{B}(\text{E2})(\text{W.u.})=26 +19-12$ if E2. $\text{B}(\text{E2})(\text{W.u.})=6.4 +35-22$ I_γ : from $(\text{p},\text{p}'\gamma)$. $\text{B}(\text{M1})(\text{W.u.})=0.034 +14-7$; $\text{B}(\text{E2})(\text{W.u.})=0.09 +7-4$ E_γ : weighted average of 3002.1 4 from ^{35}Ar ε decay,
2694.00	3/2 ⁺	930.9 2	16.3 13	1763.15	5/2 ⁺	M1+E2	+0.09 3	4.68×10 ⁻⁵ 7	
		1474.8 3	9.5 10	1219.38	1/2 ⁺	M1+E2	-0.63 21	8.4×10 ⁻⁵ 4	
		2693.7 2	100.0 17	0.0	3/2 ⁺	M1+E2	+0.26 2	0.000553 8	
3002.64	5/2 ⁺	308.64	<5	2694.00	3/2 ⁺	[M1,E2]		0.0014 9	
		356.97	<1	2645.67	7/2 ⁺	[M1,E2]		9	
		1239.47	2.2 11	1763.15	5/2 ⁺	[M1,E2]		4.4×10 ⁻⁵ 6	
		1783.21	3.3 11	1219.38	1/2 ⁺	[E2]		0.0002268 32	
		3002.3 4	100.0 10	0.0	3/2 ⁺	M1+E2	+0.08 2	0.000671 9	

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	Comments
3162.87	7/2 ⁻	160.67 20	1.5 4	3002.64	5/2 ⁺	[E1]		0.00330 5	3002.5 15 from ($\alpha,\text{p}\gamma$), and 3002.5 5 from ($\text{p},\text{p}'\gamma$). I_γ : weighted average of 100.0 19 from ^{35}Ar ε decay and 100.0 10 from ($\text{p},\text{p}'\gamma$). Mult.: from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in ($\text{p},\text{p}'\gamma$). δ : weighted average of +0.09 3 from ($\alpha,\text{p}\gamma$) and +0.07 2 from ($\text{p},\text{p}'\gamma$). Other: -0.22 1 from (p,γ). $B(\text{E1})(\text{W.u.})=6.6\times 10^{-5}$ 17 E_γ : weighted average of 161 1 from ($^{16}\text{O},\alpha\text{p}\gamma$) and 160.66 20 from ($^{14}\text{N},\alpha\text{p}\text{n}\gamma$). I_γ : unweighted average of 1.10 30 from ($^{28}\text{Si},\alpha\text{p}\gamma$) and 1.89 22 from (p,γ). $B(\text{E1})(\text{W.u.})=1.61\times 10^{-5}$ 20 E_γ : weighted average of 517.7 1 from ($^{24}\text{Mg},\alpha\text{p}\gamma$), 517.3 5 from ($^{16}\text{O},\alpha\text{p}\gamma$), 517.26 10 from ($^{14}\text{N},\alpha\text{p}\text{n}\gamma$), 517.4 15 from ($\alpha,\text{p}\gamma$), and 517.0 8 from ($\text{p},\text{p}'\gamma$). I_γ : unweighted average of 9.2 5 from ($^{28}\text{Si},\alpha\text{p}\gamma$), 14.3 8 from ($^{24}\text{Mg},\alpha\text{p}\gamma$), 11.1 13 from ($^{16}\text{O},\alpha\text{p}\gamma$), 11.1 33 from ($\alpha,\text{p}\gamma$), 8.9 11 from (p,γ), and 19 8 from ($\text{p},\text{p}'\gamma$). Mult.: D from $\gamma\gamma(\theta)(\text{DCO})$ in ($^{28}\text{Si},\alpha\text{p}\gamma$); $\Delta\pi=\text{yes}$ from level scheme.
		517.47 11	12.3 16	2645.67	7/2 ⁺	(E1)		0.0001097 15	$B(\text{E1})(\text{W.u.})=1.88\times 10^{-8}$ +31-34; $B(\text{M2})(\text{W.u.})=0.0085$ +43-38 I_γ : weighted average of 0.50 20 from ($^{16}\text{O},\alpha\text{p}\gamma$) and 0.33 5 from (p,γ). Mult., δ : D+Q from $\gamma(\theta)$ in (p,γ) (1971Pr11); $\Delta\pi=\text{yes}$ from level scheme.
		1399.9 7	0.34 5	1763.15	5/2 ⁺	(E1+M2)	+0.44 12	0.000181 12	$B(\text{M2})(\text{W.u.})=0.255$ 9; $B(\text{E3})(\text{W.u.})=4.2$ +11-9 E_γ : weighted average of 3162.7 4 from ($^{24}\text{Mg},\alpha\text{p}\gamma$), 3162.6 3 from ($^{16}\text{O},\alpha\text{p}\gamma$), 3162.43 10 from ($^{14}\text{N},\alpha\text{p}\text{n}\gamma$), and 3162.6 1 from ($\text{p},\text{p}'\gamma$), and 3162.6 15 from ($\alpha,\text{p}\gamma$). I_γ : weighted average of 100 5 from ($^{28}\text{Si},\alpha\text{p}\gamma$), 100.0 30 from ($^{24}\text{Mg},\alpha\text{p}\gamma$), 100.0 20 from ($^{16}\text{O},\alpha\text{p}\gamma$), 100.0 33 from ($\alpha,\text{p}\gamma$), 100.0 11 from (p,γ), and 100.0 20 from ($\text{p},\text{p}'\gamma$). Mult.: from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in (p,γ). δ : weighted average of +0.16 1 from ($^{28}\text{Si},\alpha\text{p}\gamma$), +0.18 7 from ($^{16}\text{O},\alpha\text{p}\gamma$), and +0.25 4 from (p,γ). Others: -0.14 2 from ($\alpha,\text{p}\gamma$) and -0.22 5 from ($\text{p},\text{p}'\gamma$).
		1943.43	<0.22	1219.38	1/2 ⁺	[E3]		0.0001742 24	
		3162.53 10	100.0 11	0.0	3/2 ⁺	M2+E3	+0.17 2	0.000523 7	

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	α^a	Comments
3918.47	3/2 ⁺	1272.78	<3.7	2645.67	7/2 ⁺	[E2]	5.56×10 ⁻⁵ 8	E_γ : from ³⁵ Ar ε decay (1.7755 s). I_γ : weighted average of 29 12 from ³⁵ Ar ε decay and 23.9 28 from (p, γ). B(M1)(W.u.)=0.075 +32-18 if M1, B(E2)(W.u.)=61 +26-14 if E2. I_γ : from (α ,p γ). B(M1)(W.u.)=0.030 +15-9 if M1, B(E2)(W.u.)=16 +8-5 if E2. I_γ : weighted average of 100 6 from ³⁵ Ar ε decay, 100 5 from (α ,p γ), and 100.0 19 from (p, γ). δ : -0.21 2 or +20 +10-5 from $\gamma(\theta)$ in (p, γ). B(M1)(W.u.)=0.052 +22-11 if M1, B(E2)(W.u.)=13 +6-3 if E2. E_γ : from (α ,p γ). I_γ : weighted average of 6.2 31 from (²⁸ Si, α p γ), 8.1 21 from (α ,p γ), and 8.7 22 from (p, γ). B(M1)(W.u.)=0.0042 +13-12 if M1, B(E2)(W.u.)=9.3 +29-27 if E2. B(E2)(W.u.)=8.7 +17-13 E_γ : weighted average of 2180 1 from (¹⁶ O, α p γ) and 2180.2 15 from (α ,p γ). I_γ : weighted average of 100.0 21 from (α ,p γ) and 100.0 22 from (p, γ). Other: 100 11 from (²⁸ Si, α p γ). B(M1)(W.u.)=0.019 +8-6 if M1, B(E2)(W.u.)=45 +18-13 if E2. B(M1)(W.u.)=0.071 +23-14 E_γ : from ³⁵ Ar ε decay (1.7755 s). I_γ : other: 100 7 from ³⁵ Ar ε decay. I_γ : weighted average of 22 5 from ³⁵ Ar ε decay and 24 4 from (p, γ). B(M1)(W.u.)=0.0054 +19-13 if M1, B(E2)(W.u.)=1.31 +46-30 if E2. B(E1)(W.u.)=2.1×10 ⁻⁴ +8-7 B(E1)(W.u.)=1.9×10 ⁻⁴ +7-6 I_γ : weighted average of 8 4 from (α ,p γ) and 5.1 16 from (p, γ). B(E1)(W.u.)=0.00179 +34-22 I_γ : weighted average of 100 4 from (α ,p γ) and 100.0 21 from (p, γ).
		2155.1 15	24.2 28	1763.15	5/2 ⁺	[M1,E2]	0.00036 4	
		2698.98	19 5	1219.38	1/2 ⁺	[M1,E2]	0.00060 6	
		3918.24	100.0 19	0.0	3/2 ⁺	M1+E2	0.00108 7	
3942.9	9/2 ⁺	1297.4 15	8.0 21	2645.67	7/2 ⁺	[M1,E2]	5.2×10 ⁻⁵ 8	E_γ : from ³⁵ Ar ε decay (1.7755 s). I_γ : weighted average of 29 12 from ³⁵ Ar ε decay and 23.9 28 from (p, γ). B(M1)(W.u.)=0.075 +32-18 if M1, B(E2)(W.u.)=61 +26-14 if E2. I_γ : from (α ,p γ). B(M1)(W.u.)=0.030 +15-9 if M1, B(E2)(W.u.)=16 +8-5 if E2. I_γ : weighted average of 100 6 from ³⁵ Ar ε decay, 100 5 from (α ,p γ), and 100.0 19 from (p, γ). δ : -0.21 2 or +20 +10-5 from $\gamma(\theta)$ in (p, γ). B(M1)(W.u.)=0.052 +22-11 if M1, B(E2)(W.u.)=13 +6-3 if E2. E_γ : from (α ,p γ). I_γ : weighted average of 6.2 31 from (²⁸ Si, α p γ), 8.1 21 from (α ,p γ), and 8.7 22 from (p, γ). B(M1)(W.u.)=0.0042 +13-12 if M1, B(E2)(W.u.)=9.3 +29-27 if E2. B(E2)(W.u.)=8.7 +17-13 E_γ : weighted average of 2180 1 from (¹⁶ O, α p γ) and 2180.2 15 from (α ,p γ). I_γ : weighted average of 100.0 21 from (α ,p γ) and 100.0 22 from (p, γ). Other: 100 11 from (²⁸ Si, α p γ). B(M1)(W.u.)=0.019 +8-6 if M1, B(E2)(W.u.)=45 +18-13 if E2. B(M1)(W.u.)=0.071 +23-14 E_γ : from ³⁵ Ar ε decay (1.7755 s). I_γ : other: 100 7 from ³⁵ Ar ε decay. I_γ : weighted average of 22 5 from ³⁵ Ar ε decay and 24 4 from (p, γ). B(M1)(W.u.)=0.0054 +19-13 if M1, B(E2)(W.u.)=1.31 +46-30 if E2. B(E1)(W.u.)=2.1×10 ⁻⁴ +8-7 B(E1)(W.u.)=1.9×10 ⁻⁴ +7-6 I_γ : weighted average of 8 4 from (α ,p γ) and 5.1 16 from (p, γ). B(E1)(W.u.)=0.00179 +34-22 I_γ : weighted average of 100 4 from (α ,p γ) and 100.0 21 from (p, γ).
		2180.1 10	100.0 21	1763.15	5/2 ⁺	E2	0.000414 6	
3967.6	1/2 ⁺	1273.6	2.7 6	2694.00	3/2 ⁺	[M1,E2]	4.9×10 ⁻⁵ 7	E_γ : from ³⁵ Ar ε decay (1.7755 s). I_γ : other: 100 7 from ³⁵ Ar ε decay. I_γ : weighted average of 22 5 from ³⁵ Ar ε decay and 24 4 from (p, γ). B(M1)(W.u.)=0.0054 +19-13 if M1, B(E2)(W.u.)=1.31 +46-30 if E2. B(E1)(W.u.)=2.1×10 ⁻⁴ +8-7 B(E1)(W.u.)=1.9×10 ⁻⁴ +7-6 I_γ : weighted average of 8 4 from (α ,p γ) and 5.1 16 from (p, γ). B(E1)(W.u.)=0.00179 +34-22 I_γ : weighted average of 100 4 from (α ,p γ) and 100.0 21 from (p, γ).
		2204.4	≤1.3	1763.15	5/2 ⁺	[E2]	0.000426 6	
		2748.5 6	100 6	1219.38	1/2 ⁺	[M1]	0.000569 8	
		3967.4	23 4	0.0	3/2 ⁺	[M1,E2]	0.00109 7	
4059.1	(3/2) ⁻	1365.1	1.3 4	2694.00	3/2 ⁺	[E1]	0.0001824 26	E_γ : from ³⁵ Ar ε decay (1.7755 s). I_γ : other: 100 7 from ³⁵ Ar ε decay. I_γ : weighted average of 22 5 from ³⁵ Ar ε decay and 24 4 from (p, γ). B(M1)(W.u.)=0.0054 +19-13 if M1, B(E2)(W.u.)=1.31 +46-30 if E2. B(E1)(W.u.)=2.1×10 ⁻⁴ +8-7 B(E1)(W.u.)=1.9×10 ⁻⁴ +7-6 I_γ : weighted average of 8 4 from (α ,p γ) and 5.1 16 from (p, γ). B(E1)(W.u.)=0.00179 +34-22 I_γ : weighted average of 100 4 from (α ,p γ) and 100.0 21 from (p, γ).
		2295.9	5.5 16	1763.15	5/2 ⁺	[E1]	0.000848 12	
		2839.6	100.0 21	1219.38	1/2 ⁺	[E1]	1.17×10 ⁻³ 2	
		4058.9	<2.1	0.0	3/2 ⁺	[E1]	1.70×10 ⁻³ 2	

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.#	$\delta^\#$	α^a	Comments
4112.4	7/2 ⁺	2349.2	89 6	1763.15	5/2 ⁺	M1+E2	-0.16 2	0.000405 6	B(M1)(W.u.)=0.0159 +48-31; B(E2)(W.u.)=0.28 +12-8 I _γ : other: 16 5 from (α,pγ).
		4112.1	100 6	0.0	3/2 ⁺	E2(+M3)	0.00 10	1.21×10 ⁻³ 2	B(E2)(W.u.)=0.76 +22-15 I _γ : other: 100 5 from (α,pγ).
4173.43	(5/2 ⁺)	1479.40	45 14	2694.00	3/2 ⁺	[M1,E2]		9.1×10 ⁻⁵ 13	B(M1)(W.u.)=0.048 +20-12 if M1, B(E2)(W.u.)=82 +34-21 if E2. B(E2)(W.u.)=82 +34-21 upper bound exceeds RUL=100 if E2.
		1527.72	<17	2645.67	7/2 ⁺	[M1,E2]		0.000106 15	
		2410.19	28 9	1763.15	5/2 ⁺	[M1,E2]		0.00048 5	B(M1)(W.u.)=0.0068 +33-19 if M1, B(E2)(W.u.)=4.5 +21-13 if E2.
		2953.92	≤5.2	1219.38	1/2 ⁺	[E2]		0.000771 11	
		4173.16	100 17	0.0	3/2 ⁺	[M1,E2]		0.00116 7	B(M1)(W.u.)=0.0047 +15-7 if M1, B(E2)(W.u.)=1.02 +33-14 if E2.
4177.9	3/2 ⁻	1175.2	<0.82	3002.64	5/2 ⁺	[E1]		6.06×10 ⁻⁵ 8	
		1483.9	13.1 33	2694.00	3/2 ⁺	[E1]		0.000265 4	B(E1)(W.u.)=5.7×10 ⁻⁴ +16-14
		1532.2	<0.82	2645.67	7/2 ⁺	[M2]		6.18×10 ⁻⁵ 9	
		2414.7	<1.6	1763.15	5/2 ⁺	[E1]		0.000921 13	
		2958.4	51 8	1219.38	1/2 ⁺	(E1+M2)	+0.11 4	1.22×10 ⁻³ 2	B(E1)(W.u.)=2.8×10 ⁻⁴ +5-4; B(M2)(W.u.)=1.7 +16-11 Mult.,δ: D+Q from γ(θ) in (p,γ); Δπ=yes from level scheme. B(M2)(W.u.)=1.7 +16-11 upper bound exceeds RUL=3.
		4177.6	100 8	0.0	3/2 ⁺	(E1+M2)	+0.06 3	1.74×10 ⁻³ 3	B(E1)(W.u.)=1.93×10 ⁻⁴ +28-20; B(M2)(W.u.)=0.18 +24-14 Mult.,δ: D+Q from γ(θ) in (p,γ); Δπ=yes from level scheme.
4347.76	9/2 ⁻	1184.90 20	100.0 29	3162.87	7/2 ⁻	M1+E2	-0.36 3	3.57×10 ⁻⁵ 5	B(M1)(W.u.)=0.007 +7-3; B(E2)(W.u.)=2.5 +27-10 E _γ : weighted average of 1185.0 2 from (²⁴ Mg,αpγ), 1184.9 3 from (¹⁶ O,αpγ), and 1184.80 20 from (¹⁴ N,αpnγ). Other: 1184.6 10 from (α,pγ). I _γ : weighted average of 100 4 from (²⁸ Si,αpγ), 100.0 31 from (²⁴ Mg,αpγ), 100 4 from (¹⁶ O,αpγ), 100.0 31 from (α,pγ), and 100.0 29 from (p,γ). Mult.: from γγ(DCO) and γγ(lin pol) in (²⁸ Si,αpγ), ΔJ=1. δ: weighted average of -0.36 3 from (α,pγ), -0.3 1 from (²⁸ Si,αpγ), and -0.65 30 from (¹⁶ O,αpγ).
		1701.88 25	46.8 24	2645.67	7/2 ⁺	E1+M2	-0.018 12	0.000434 6	B(E1)(W.u.)=3.7×10 ⁻⁵ +39-14; B(M2)(W.u.)=0.019

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	α^a	Comments
								+47-17 E_γ : weighted average of 1702.0 3 from ($^{24}\text{Mg}, \alpha\text{py}$), 1701.7 5 from ($^{16}\text{O}, \alpha\text{py}$), and 1701.85 25 from ($^{14}\text{N}, \alpha\text{pny}$). Other: 1702.1 15 from (α, py). I_γ : unweighted average of 45.6 24 from ($^{28}\text{Si}, \alpha\text{py}$), 39.5 24 from ($^{24}\text{Mg}, \alpha\text{py}$), 52.0 14 from ($^{16}\text{O}, \alpha\text{py}$), 52 7 from (α, py), and 44.9 29 from (p, γ). Mult.: from $\gamma\gamma(\text{DCO})$ and $\gamma\gamma(\text{lin pol})$ in ($^{28}\text{Si}, \alpha\text{py}$) with $\Delta J=1$ and $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in (α, py). δ : from (p, γ). Others: +0.5 +5-3 from ($^{28}\text{Si}, \alpha\text{py}$), +0.5 4 from ($^{16}\text{O}, \alpha\text{py}$).
4347.76	9/2 ⁻	2584.51 4347.8 8	≤ 4.4 12.4 11	1763.15 0.0	5/2 ⁺ 3/2 ⁺	[M2] [E3]	0.000329 5 0.001000 14	B(E3)(W.u.)=44 +46-17 E_γ, I_γ : from ($^{16}\text{O}, \alpha\text{py}$). E_γ : other: 4618 2 from (α, py). Mult., δ : from $\text{py}(\theta)$ with -0.08 14 for J=3/2; +0.36 15 for J=5/2 in (α, py). If J(4624=5/2), Mult would be M1+E2 based on RUL.
4624.3	3/2, 5/2	4624.0	100	0.0	3/2 ⁺	D+Q		
4769.86	(5/2 ⁻)	1606.95 1767.17 2075.79 3550.29 4769.51	100 16 54 16 <15 <15 <15	3162.87 3002.64 2694.00 1219.38 0.0	7/2 ⁻ 5/2 ⁺ 3/2 ⁺ 1/2 ⁺ 3/2 ⁺	[M1, E2] [E1] [E1] [M2] [E1]	0.000133 19 0.000483 7 0.000702 10 0.000645 9 1.95×10 ⁻³ 3	B(M1)(W.u.)=0.040 +23-5 if M1, B(E2)(W.u.)=58 +34-7 if E2. B(E1)(W.u.)=4.6×10 ⁻⁴ +30-9
4839.10	(1/2 ⁺ , 3/2)	1836.41 2145.03 2193.36 ^b 3075.81 3619.52 4838.74	<12 <6.8 <5.1 100 9 <12 70 9	3002.64 2694.00 2645.67 1763.15 1219.38 0.0	5/2 ⁺ 3/2 ⁺ 7/2 ⁺ 5/2 ⁺ 1/2 ⁺ 3/2 ⁺			
4854.4	(1/2, 3/2)	3634.8 4854.0	100 7 33 7	1219.38 0.0	1/2 ⁺ 3/2 ⁺			
4880.96	7/2 ⁽⁺⁾	1878.27 2186.89 2235.21	15 5 <16 47 8	3002.64 2694.00 2645.67	5/2 ⁺ 3/2 ⁺ 7/2 ⁺	[M1, E2] [E2] [M1, E2]	0.000240 30 0.000417 6 0.00040 4	B(M1)(W.u.)=0.051 +30-16 if M1, B(E2)(W.u.)=55 +32-17 if E2. I_γ : other: 52 9 from (α, py) for a doublet. B(M1)(W.u.)=0.095 +43-18 if M1, B(E2)(W.u.)=72 +33-14 if E2. B(E2)(W.u.)=72 +33-14 upper bound exceeds RUL=100 if E2. E_γ : from (α, py). I_γ : weighted average of 100 9 from (α, py) and 100 8 from (p, γ). B(M1)(W.u.)=0.075 +33-12 if M1, B(E2)(W.u.)=29 +13-5 if E2.
		3117.4 20	100 8	1763.15	5/2 ⁺	[M1, E2]	0.00078 6	
		4880.60	<8.1	0.0	3/2 ⁺	[E2]	1.46×10 ⁻³ 2	

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	Comments
5009.1	(1/2,3/2)	835.7	<25	4173.43	(5/2 ⁺)				
		950.0	<10	4059.1	(3/2) ⁻				
		1846.2 ^b	<50	3162.87	7/2 ⁻				
		2006.4	<20	3002.64	5/2 ⁺				
		2315.0	<25	2694.00	3/2 ⁺				
		2363.3 ^b	<5	2645.67	7/2 ⁺				
		3245.8	<10	1763.15	5/2 ⁺				
		3789.5	<7	1219.38	1/2 ⁺				
		5008.7	100	0.0	3/2 ⁺				
5163.34	7/2 ⁻	2000.41	93 12	3162.87	7/2 ⁻	D+Q	+0.44 20		
		2160.63	23 9	3002.64	5/2 ⁺				
		3400.01	100 12	1763.15	5/2 ⁺	D(+Q)	0.00 3		
		3943.72 ^b	<23	1219.38	1/2 ⁺				
		5162.93	<23	0.0	3/2 ⁺				
5214.5	(5/2 ⁺)	205.4	<23	5009.1	(1/2,3/2)				
		1041.1	<60	4173.43	(5/2 ⁺)				
		1155.4	<10	4059.1	(3/2) ⁻				
		2051.6	<15	3162.87	7/2 ⁻				
		2211.8	<25	3002.64	5/2 ⁺				
		2520.4	<15	2694.00	3/2 ⁺				
		2568.7	<15	2645.67	7/2 ⁺				
		3451.2	<6	1763.15	5/2 ⁺				
		3994.9	<4	1219.38	1/2 ⁺				
		5214.1	100	0.0	3/2 ⁺	[M1,E2]		0.00147 8	
5403.3	3/2 ⁻	2400.6	46 9	3002.64	5/2 ⁺	[E1]		0.000913 13	B(E1)(W.u.)=0.0011 +6-2
		3640.0	<18	1763.15	5/2 ⁺	[E1]		1.54×10 ⁻³ 2	
		4183.7	100 18	1219.38	1/2 ⁺	[E1]		1.75×10 ⁻³ 2	B(E1)(W.u.)=4.5×10 ⁻⁴ +20-6
		5402.9	<13	0.0	3/2 ⁺	[E1]		2.14×10 ⁻³ 3	
5407.07	11/2 ⁻	1059.32 20	14.5 22	4347.76	9/2 ⁻	M1+E2	+0.14 3	3.67×10 ⁻⁵ 5	B(M1)(W.u.)=0.0082 +30-20; B(E2)(W.u.)=0.54 +35-23
									E_γ : weighted average of 1059.3 2 from (²⁴ Mg, $\alpha\gamma$), 1059.5 2 from (¹⁶ O, $\alpha\gamma$), and 1059.17 20 from (¹⁴ N, $\alpha p n \gamma$). Other: 1059.3 10 from ($\alpha, p \gamma$).
									I_γ : unweighted average of 12.0 12 from (²⁸ Si, $\alpha\gamma$), 16.9 17 from (²⁴ Mg, $\alpha\gamma$), 19.2 5 from (¹⁶ O, $\alpha\gamma$), and 9.9 22 from ($\alpha, p \gamma$).
									Mult.: from $\gamma\gamma$ (DCO) and $\gamma\gamma$ (lin pol) in (²⁸ Si, $\alpha\gamma$), $\Delta J=1$.
									δ : weighted average of +0.12 3 from (¹⁶ O, $\alpha\gamma$),

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)									Comments
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	
5407.07	11/2 ⁻	2243.71 23	100.0 22	3162.87	7/2 ⁻	E2		0.000444 6	+0.2 1 from ($^{28}\text{Si}, \alpha p \gamma$), and +0.25 8 from ($\alpha, p \gamma$). B(E2)(W.u.)=4.6 +15-9 E _γ : weighted average of 2244.2 3 from ($^{24}\text{Mg}, \alpha p \gamma$), 2243.3 2 from ($^{16}\text{O}, \alpha p \gamma$), 2244.00 25 from ($^{14}\text{N}, \alpha p n \gamma$), and 2244 3 from ($\alpha, p \gamma$). I _γ : weighted average of 100.0 33 from ($^{28}\text{Si}, \alpha p \gamma$), 100.0 32 from ($^{24}\text{Mg}, \alpha p \gamma$), 100.0 23 from ($^{16}\text{O}, \alpha p \gamma$), and 100.0 22 from ($\alpha, p \gamma$). Mult.: from $\gamma\gamma$ (DCO) and $\gamma\gamma$ (lin pol) in ($^{28}\text{Si}, \alpha p \gamma$), $\Delta J=2$.
5586.0	5/2 ⁺	2940.2 3822.6 4366.3 5585.5	67 17 100 17 <17 <17	2645.67 7/2 ⁺ 1763.15 5/2 ⁺ 1219.38 1/2 ⁺ 0.0 3/2 ⁺					
5598.4	3/2 ⁺ , 5/2 ⁺	3835.0	37 14	1763.15 5/2 ⁺		[M1, E2]		0.00105 7	B(M1)(W.u.)=0.050 +29-19 if M1, B(E2)(W.u.)=13 +8-5 if E2.
		5597.9	100 14	0.0 3/2 ⁺		[M1, E2]		0.00157 8	B(M1)(W.u.)=0.044 +23-12 if M1, B(E2)(W.u.)=5.3 +28-14 if E2.
5646	(5/2, 7/2, 9/2 ⁺)	2483 2643 3883	100	3162.87 7/2 ⁻ 3002.64 5/2 ⁺ 1763.15 5/2 ⁺					
5655	(3/2) ⁺	5646 2652	<8 100 6	0.0 3/2 ⁺ 3002.64 5/2 ⁺		[M1, E2]		0.00058 5	B(M1)(W.u.)=0.076 +30-14 if M1, B(E2)(W.u.)=41 +16-7 if E2.
		3892	7.5 25	1763.15 5/2 ⁺		[M1, E2]		0.00107 7	B(M1)(W.u.)=0.0018 +10-6 if M1, B(E2)(W.u.)=0.45 +25-15 if E2.
5682.2	1/2 ⁻ , 3/2 ⁻	4435 1504.3	<7.5 100	1219.38 1/2 ⁺ 4177.9 3/2 ⁻		[M1, E2]		0.00125 8	
5723.52	5/2 ⁺	3029.38 3960.13	63 8 100 11	2694.00 3/2 ⁺ 1763.15 5/2 ⁺					
5758	(1/2, 3/2 ⁻)	1580 4538 5758	67 22 <13 100 22	4177.9 3/2 ⁻ 1219.38 1/2 ⁺ 0.0 3/2 ⁺					
5806.6	(1/2, 3/2, 5/2)	5806.1	100	0.0 3/2 ⁺					
5830	(5/2 ⁻)	2667	100	3162.87 7/2 ⁻		D+Q			I _γ : from ($\alpha, p \gamma$). δ : -2.2 5 or -0.6 1 for J=5/2 from $\gamma(\theta)$ in ($\alpha, p \gamma$).
5926.65	(11/2) ⁻	1579.09 30	100 11	4347.76 9/2 ⁻		M1+E2	-0.2 1	0.0001074 22	E _γ : weighted average of 1578.9 5 from ($^{16}\text{O}, \alpha p \gamma$), 1579.15 30 from ($^{14}\text{N}, \alpha p n \gamma$), and 1579.3 15 from ($\alpha, p \gamma$).

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

<u>E_i(level)</u>	<u>J^{π}_i</u>	<u>E_{γ}[†]</u>	<u>I_{γ}[‡]</u>	<u>E_f</u>	<u>J^{π}_f</u>	<u>Mult.#</u>	<u>δ[#]</u>	<u>α^a</u>	<u>Comments</u>
									Mult., δ : D+Q and δ from $\gamma\gamma(\theta)$ (DCO) in (²⁸ Si, $\alpha\gamma$) with $\Delta J=1$; E1+M2 ruled out by RUL. Other: -0.8 4 from $\gamma(\theta)$ in (α,γ).
5926.65	(11/2) ⁻	1983.7	11.1 22	3942.9	9/2 ⁺	[E1]		0.000639 9	I _{γ} : from (²⁸ Si, $\alpha\gamma$).
		2763.66		3162.87	7/2 ⁻				
6087.31	13/2 ⁻	680.35 10	100.0 21	5407.07	11/2 ⁻	M1+E2	+0.020 15	8.70×10 ⁻⁵ 12	B(M1)(W.u.)=0.0107 10 E _{γ} : weighted average of 680.4 1 from (²⁴ Mg, $\alpha\gamma$), 680.5 5 from (¹⁶ O, $\alpha\gamma$), and 680.22 15 from (¹⁴ N, $\alpha\text{pn}\gamma$). Other: 679.9 10 from (α,γ). I _{γ} : weighted average of 100.0 31 from (²⁸ Si, $\alpha\gamma$), 100.0 21 from (¹⁶ O, $\alpha\gamma$), and 100 6 from ($\alpha,\text{p}\gamma$). Mult., δ : from $\gamma(\theta)$ and $\gamma(\text{lin pol})$ in ($\alpha,\text{p}\gamma$). Other: $\delta=+0.1$ 1 from $\gamma\gamma(\text{DCO})$ in (²⁸ Si, $\alpha\gamma$), $\Delta J=1$.
		1739.4 4	6.9 9	4347.76	9/2 ⁻	E2		0.0002076 29	B(E2)(W.u.)=0.054 +9-8 E _{γ} : weighted average of 1739.6 5 from (¹⁶ O, $\alpha\gamma$) and 1739.3 4 from (¹⁴ N, $\alpha\text{pn}\gamma$). I _{γ} : weighted average of 6.3 6 from (²⁸ Si, $\alpha\gamma$), 7.9 9 from (¹⁶ O, $\alpha\gamma$), and 18 6 from ($\alpha,\text{p}\gamma$). Mult.: from $\gamma\gamma(\theta)$ (DCO) and $\gamma\gamma(\text{lin pol})$ in (²⁸ Si, $\alpha\gamma$), $\Delta J=2$.
6106.2	(3/2,5/2 ⁺)	4342.8	10.3 35	1763.15	5/2 ⁺				
		4886.5	62 17	1219.38	1/2 ⁺				
		6105.6	100 17	0.0	3/2 ⁺				
6181	(1/2 ⁺ to 7/2 ⁺)	6180	100	0.0	3/2 ⁺				
6380.2	(9/2 ⁻)	296 1		6087.31	13/2 ⁻				I _{γ} : from (¹⁶ O, $\alpha\gamma$).
		971 1		5407.07	11/2 ⁻				I _{γ} : from (¹⁶ O, $\alpha\gamma$).
6492	(3/2 ⁺)	6491	100	0.0	3/2 ⁺				
6660.2	(11/2 ⁻)	2312.4		4347.76	9/2 ⁻				
		3497.1		3162.87	7/2 ⁻				
7066.2	(5/2 ⁺)	1902.8	1.8	5163.34	7/2 ⁻				
		2892.6	1.0	4173.43	(5/2 ⁺)				
		2953.7	1.2	4112.4	7/2 ⁺				
		3007.0	2.9	4059.1	(3/2) ⁻				
		3903.1	12	3162.87	7/2 ⁻				
		4063.3	6.1	3002.64	5/2 ⁺				

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

<u>E_i(level)</u>	<u>J^{π}_i</u>	<u>E_{γ}^{$\dagger$}</u>	<u>I_{γ}^{$\ddagger$}</u>	<u>E_f</u>	<u>J^{π}_f</u>	<u>Mult.[#]</u>	<u>δ[#]</u>
7066.2	(5/2 ⁺)	4371.9	10	2694.00	3/2 ⁺		
		4420.2	3.3	2645.67	7/2 ⁺		
		5302.6	29	1763.15	5/2 ⁺		
		5846.3	37	1219.38	1/2 ⁺		
		7065.4	100	0.0	3/2 ⁺		
7103.4	3/2 ⁽⁻⁾	2925.4	2.3	4177.9	3/2 ⁻		
		3184.8	2.3	3918.47	3/2 ⁺		
		4100.5	4.7	3002.64	5/2 ⁺		
		4409.1	17	2694.00	3/2 ⁺	D(+Q)	0.00 3
		5339.8	6.3	1763.15	5/2 ⁺		
		5883.5	100	1219.38	1/2 ⁺	D(+Q)	0.00 3
		7102.6	23	0.0	3/2 ⁺		
7178.6	1/2 ⁽⁺⁾	1072.4	7.9	6106.2	(3/2,5/2 ⁺)		
		1775.3	4.5	5403.3	3/2 ⁻		
		2339.4	4	4839.10	(1/2 ⁺ ,3/2)		
		3005.0	2.1	4173.43	(5/2 ⁺)		
		3119.4	58	4059.1	(3/2) ⁻		
		3210.8	24	3967.6	1/2 ⁺		
		3260.0	7.9	3918.47	3/2 ⁺		
		4484.3	11	2694.00	3/2 ⁺		
		5958.7	45	1219.38	1/2 ⁺		
		7177.8	100	0.0	3/2 ⁺		
		1970	<1.1	5214.5	(5/2 ⁺)		
		2176	<1.1	5009.1	(1/2,3/2)		
7185	5/2 ⁺	3011	<11	4173.43	(5/2 ⁺)		
		3126	57 13	4059.1	(3/2) ⁻		
		4022	<6.5	3162.87	7/2 ⁻		
		4182	<6.5	3002.64	5/2 ⁺		
		4491	4.4 22	2694.00	3/2 ⁺		
		4539	13 5	2645.67	7/2 ⁺		
		5421	9 7	1763.15	5/2 ⁺		
		5965	35 15	1219.38	1/2 ⁺		
		7184	100 22	0.0	3/2 ⁺		
		2185.4	1.2	5009.1	(1/2,3/2)		
		2340.1	3.0	4854.4	(1/2,3/2)		
		2355.4	2.7	4839.10	(1/2 ⁺ ,3/2)		
7194.6	1/2 ⁻	3016.6	24	4177.9	3/2 ⁻		
		3135.4	4.5	4059.1	(3/2) ⁻		
		3226.8	2.7	3967.6	1/2 ⁺		
		3276.0	6	3918.47	3/2 ⁺		
		4500.3	1.6	2694.00	3/2 ⁺		
		5431.0	0.75	1763.15	5/2 ⁺		

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. #	$\delta^\#$	Comments
7194.6	1/2 ⁻	5974.7	100	1219.38	1/2 ⁺			
		7193.8	3.0	0.0	3/2 ⁺			
7225.5	5/2	1119.3	3.1	6106.2	(3/2,5/2 ⁺)			
		1418.9		5806.6	(1/2,3/2,5/2)			
		1543.3	3.4	5682.2	1/2 ⁻ ,3/2 ⁻			
		1571	1.4	5655	(3/2 ⁺)			
		1822.2	1.5	5403.3	3/2 ⁻			
		2010.9	12	5214.5	(5/2 ⁺)			
		3047.5	14	4177.9	3/2 ⁻			
		3166.3	3.4	4059.1	(3/2 ⁻)			
		3306.9	2.5	3918.47	3/2 ⁺			
		4222.6		3002.64	5/2 ⁺			
		4531.2	17	2694.00	3/2 ⁺	D+Q	+0.36 4	
		5461.9	12	1763.15	5/2 ⁺	D+Q	-0.36 7	
		7224.7	100	0.0	3/2 ⁺	D+Q	+0.36 4	
7234.0	5/2 ⁽⁺⁾	1579	0.22	5655	(3/2 ⁺)			
		3056.0		4177.9	3/2 ⁻			
		4539.7	1.9	2694.00	3/2 ⁺			
		4588.0	3.2	2645.67	7/2 ⁺			
		5470.4	0.22	1763.15	5/2 ⁺			
		6014.1	1.9	1219.38	1/2 ⁺			
		7233.2	100	0.0	3/2 ⁺	D+Q	-0.10	
7272.7	1/2 ⁻	1515	1.7	5758	(1/2,3/2 ⁻)			
		1869.4	0.29	5403.3	3/2 ⁻			
		2263.5	2.0	5009.1	(1/2,3/2)			
		2433.5	1.5	4839.10	(1/2 ⁺ ,3/2)			
		3099.1	0.58	4173.43	(5/2 ⁺)			
		3213.4	1.3	4059.1	(3/2 ⁻)			
		3304.9	1.2	3967.6	1/2 ⁺			
		4578.4	1.6	2694.00	3/2 ⁺			
		6052.8	35	1219.38	1/2 ⁺			
		7271.9	100	0.0	3/2 ⁺			
7361.9	3/2 ⁽⁺⁾	1707	0.41	5655	(3/2 ⁺)			
		2352.7	0.82	5009.1	(1/2,3/2)			
		2522.7	1.4	4839.10	(1/2 ⁺ ,3/2)			
		3183.8	0.69	4177.9	3/2 ⁻			
		3302.6	2.7	4059.1	(3/2 ⁻)			
		3394.1	4.1	3967.6	1/2 ⁺			
		3443.3	1.1	3918.47	3/2 ⁺			
		4359.0	0.41	3002.64	5/2 ⁺			
		4667.6	0.69	2694.00	3/2 ⁺			
		5598.3	11	1763.15	5/2 ⁺	D+Q	δ : >+2.5 or -0.01 10 from $\gamma(\theta)$ in (p, γ).	

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	Comments
7361.9	3/2 ⁽⁺⁾	6141.9	100	1219.38	1/2 ⁺	(M1+E2)			Mult.: D+Q from $\gamma(\theta)$ in (p, γ); E1+M2 is less likely for this strong primary transition. δ : -0.05 3 or $+2.5$ 3 in (p, γ). δ : $-0.16 < \delta(Q/D) < -0.12$ or $+7.5 < \delta(Q/D) < +10$ in (p, γ).
		7361.1	14	0.0	3/2 ⁺	D+Q			
7396.4	7/2 ⁽⁻⁾	3048.5	23	4347.76	9/2 ⁻				Mult.: D+Q from $\gamma(\theta)$ in (p, γ); E1+M2 is less likely for this strong primary transition.
		3283.8	17	4112.4	7/2 ⁺				
		3453.3	11	3942.9	9/2 ⁺				
		4233.3	100	3162.87	7/2 ⁻	(M1+E2)	+0.28 25	1.12×10^{-3} 3	
		4393.5	30	3002.64	5/2 ⁺	D(+Q)	+0.011 18		
		4750.4	28	2645.67	7/2 ⁺				
		5632.8	4.3	1763.15	5/2 ⁺				
7450.9	3/2	3532.2	6.9 34	3918.47	3/2 ⁺				
		4448.0	13.7 14	3002.64	5/2 ⁺				
		4756.6	11.0 11	2694.00	3/2 ⁺				
		5687.3	5.5 28	1763.15	5/2 ⁺				
		6230.9	100 10	1219.38	1/2 ⁺				
7501.1	(5/2 ⁺ , 7/2, 9/2 ⁺)	3558.0	46 6	3942.9	9/2 ⁺				
		4498.2	23.6 24	3002.64	5/2 ⁺				
		4855.1	100 11	2645.67	7/2 ⁺				
		5737.5	12.7 13	1763.15	5/2 ⁺				
7502.9		1396.7 @	14 @	6106.2	(3/2, 5/2 ⁺)				
		1904.4 @	4 @	5598.4	3/2 ⁺ , 5/2 ⁺				
		2493.7 @	4 @	5009.1	(1/2, 3/2)				
		3324.8 @	6 @	4177.9	3/2 ⁻				
		3443.6 @	100 @	4059.1	(3/2) ⁻				
		3559.8 @	4 @	3942.9	9/2 ⁺				
		4339.7 @	8 @	3162.87	7/2 ⁻				
		4500.0 @	8 @	3002.64	5/2 ⁺				
		4856.9 @	14 @	2645.67	7/2 ⁺				
		5739.2 @	1.4 @	1763.15	5/2 ⁺				
		6282.9 @	36 @	1219.38	1/2 ⁺				
		7502.0 @	0.6 @	0.0	3/2 ⁺				
7518.8	7/2	1920.3	2.9 15	5598.4	3/2 ⁺ , 5/2 ⁺				
		1932.7	4.4 22	5586.0	5/2 ⁺				
		3170.9	2.9 15	4347.76	9/2 ⁻				
		3575.7	2.9 15	3942.9	9/2 ⁺				

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.#	$\delta^\#$	α^a	Comments
7518.8	7/2	4355.6	100 10	3162.87	7/2 ⁻	D(+Q)	+0.1 2		
		4872.8	4.4 22	2645.67	7/2 ⁺				
		5755.1	29.4 30	1763.15	5/2 ⁺	D+Q	+0.098 22		
		1903	0.59 7	5646	(5/2,7/2,9/2 ⁺)				
		1962.8	0.11	5586.0	5/2 ⁺	[E1]		0.000624 9	
		2778.9	1.49 11	4769.86	(5/2 ⁻)	[M1,E2]		0.00064 6	
		4385.7	100.0 11	3162.87	7/2 ⁻	M1+E2	-0.07 2	1.16×10 ⁻³ 2	
		4545.9	2.13 11	3002.64	5/2 ⁺	(E1+M2)	+0.6 4	0.00163 22	Mult., δ : D+Q and δ from $\gamma(\theta)$ in (p, γ); $\Delta\pi$ =yes from level scheme.
		4902.9	0.72 8	2645.67	7/2 ⁺	[E1]		1.99×10 ⁻³ 3	
		5785.2	0.35 4	1763.15	5/2 ⁺	[E1]		2.24×10 ⁻³ 3	
7561.3	1/2,3/2	7548.0	0.287 32	0.0	3/2 ⁺	(M2)			Mult.: Q(+O) with $\delta(\text{O/Q})=0.1$ 2 from $\gamma(\theta)$ in (p, γ); $\Delta\pi$ =yes from level scheme; E3 component is less likely based on RUL.
		3383.2	2.6	4177.9	3/2 ⁻				
		3502.0	17	4059.1	(3/2 ⁻)				
		3593.5	1.4	3967.6	1/2 ⁺				
		3642.6	4.6	3918.47	3/2 ⁺				
		4866.9	63	2694.00	3/2 ⁺				
		6341.3	100	1219.38	1/2 ⁺				
		7560.4	97	0.0	3/2 ⁺				
		2437.4	5.8	5163.34	7/2 ⁻	[E1]		0.000934 13	
		2719.7	6.1	4880.96	7/2 ⁽⁺⁾				
7600.8	5/2 ⁺	2761.6	6.1	4839.10	(1/2 ⁺ ,3/2)				
		2830.8	3.9	4769.86	(5/2 ⁻)				
		2976.4	5.5	4624.3	3/2,5/2				
		3422.7	9.1	4177.9	3/2 ⁻	[E1]		1.44×10 ⁻³ 2	
		3427.2		4173.43	(5/2 ⁺)				
		3541.5	6.1	4059.1	(3/2 ⁻)	[E1]		1.50×10 ⁻³ 2	
		4437.6	15	3162.87	7/2 ⁻	[E1]		1.85×10 ⁻³ 3	
		4597.8	24	3002.64	5/2 ⁺	M1+E2	+0.50 22	1.25×10 ⁻³ 3	Mult.: D+Q from $\gamma(\theta)$ in (p, γ); E1+M2 is less likely for this strong primary transition.
		4906.4	55	2694.00	3/2 ⁺	(M1+E2)		0.00139 8	Mult.: D+Q from $\gamma(\theta)$ in (p, γ); $\Delta\pi$ =no from level scheme.
		4954.8	5.8	2645.67	7/2 ⁺	[M1,E2]		0.00140 8	
7618.7	5/2 ⁺	5837.1	58	1763.15	5/2 ⁺	M1+E2	+0.31 7	1.57×10 ⁻³ 2	
		6380.8	3.3	1219.38	1/2 ⁺	[E2]			
		7599.9	100	0.0	3/2 ⁺	M1+E2	-0.19 4		
		1438	0.77	6181	(1/2 ⁺ to 7/2 ⁺)				
		2404.1	2.6	5214.5	(5/2 ⁺)				
		2455.3	0.9	5163.34	7/2 ⁻				

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)										
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.#	$\delta^\#$	α^a	Comments	
7618.7	$5/2^+$	2779.5	0.9	4839.10	$(1/2^+, 3/2)$					
		3440.6	3.9	4177.9	$3/2^-$					
		3559.4	2.6	4059.1	$(3/2)^-$					
		4455.5	3.9	3162.87	$7/2^-$					
		5855.0	13	1763.15	$5/2^+$	M1+E2	+0.48 11	1.59×10^{-3} 3		
		7617.8	100	0.0	$3/2^+$	M1+E2	+2.2 3			
7656.6	$(1/2, 3/2, 5/2^+)$	1550.4	5.3 26	6106.2	$(3/2, 5/2^+)$					
		1974.3	1.8 9	5682.2	$1/2^-, 3/2^-$					
		2002	1.8 9	5655	$(3/2)^+$					
		2817.4	5.3 26	4839.10	$(1/2^+, 3/2)$					
		3478.5	1.8 9	4177.9	$3/2^-$					
		3597.3	56 6	4059.1	$(3/2)^-$					
		3688.8	3.5 18	3967.6	$1/2^+$					
		7655.7	100 11	0.0	$3/2^+$					
		7671.9	$5/2^+$	2073.4	2.1	5598.4	$3/2^+, 5/2^+$			
2508.5	7			5163.34	$7/2^-$					
2790.8	2.1			4880.96	$7/2^{(+)}$					
2901.9	1.8			4769.86	$(5/2^-)$					
3324.0	11			4347.76	$9/2^-$					
3498.3	8.8			4173.43	$(5/2^+)$					
4508.7	3.2			3162.87	$7/2^-$					
4668.9	16			3002.64	$5/2^+$	M1+E2	-1.4 5	0.00134 4		
5025.8	23			2645.67	$7/2^+$	D(+Q)	+0.3 22			
5908.2	100			1763.15	$5/2^+$	M1+E2	+0.86 14	1.64×10^{-3} 3		
7671.0	1.4			0.0	$3/2^+$					
7685.5	$3/2^-$	1927		5758	$(1/2, 3/2^-)$					
		2087.0	2.9	5598.4	$3/2^+, 5/2^+$					
		2282.1	2.9	5403.3	$3/2^-$					
		2470.9	1	5214.5	$(5/2^+)$					
		2676.3	0.86	5009.1	$(1/2, 3/2)$					
		3507.4	4.3	4177.9	$3/2^-$					
		3511.9	5.7	4173.43	$(5/2^+)$					
		3626.2	8.6	4059.1	$(3/2)^-$					
		4682.5	14	3002.64	$5/2^+$	D(+Q)	-0.01 5			
		5921.8	1	1763.15	$5/2^+$					
		6465.5	1.4	1219.38	$1/2^+$					
		7684.6	100	0.0	$3/2^+$	D+Q	+6.0 7			
		Mult.: D+Q with $\delta=+6.0$ 7 from $\gamma(\theta)$ in (p, γ);								

Mult.: D+Q with $\delta=+6.0$ 7 from $\gamma(\theta)$ in (p, γ);
 $\Delta\pi$ =yes from level scheme; but E1+M2 with such a large $\delta(M2/E1)$ is unlikely for this strong primary transition based on RUL, which might suggest a different level in (p, γ).

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.#	$\delta^\#$	α^a	Comments
7693.9	(1/2 ⁺ ,3/2,5/2)	2095.4	1.0 5	5598.4	3/2 ⁺ ,5/2 ⁺				
		3634.6	1.0 5	4059.1	(3/2) ⁻				
		4690.9	2.1 11	3002.64	5/2 ⁺				
		7693.0	100 11	0.0	3/2 ⁺				
7706.4	5/2 ⁺	2051	4.8	5655	(3/2) ⁺	M1+E2		0.00032 4	δ : <-2 or ≥ 13 from $\gamma(\theta)$ in (p, γ).
		2825.3	0.48	4880.96	7/2 ⁽⁺⁾				
		2867.2	1.6	4839.10	(1/2 ⁺ ,3/2)				
		3082.0	0.96	4624.3	3/2,5/2				
		3528.3	2.4	4177.9	3/2 ⁻				
		3593.8	2.4	4112.4	7/2 ⁺				
		3787.7	0.72	3918.47	3/2 ⁺				
		4703.4	1.2	3002.64	5/2 ⁺				
		5060.3	4.8	2645.67	7/2 ⁺	M1+E2		0.00143 8	δ : -1.01 10 or ≥ 13 from (p, γ).
		5942.7	0.48	1763.15	5/2 ⁺				
		6486.4	0.6	1219.38	1/2 ⁺				
		7705.5	100	0.0	3/2 ⁺	D(+Q)	+0.1 1		
7744.8	7/2 ⁻	2863.7	47	4880.96	7/2 ⁽⁺⁾				
		2974.8	4.7	4769.86	(5/2 ⁻)				
		3632.2	14	4112.4	7/2 ⁺				
		3801.7	23	3942.9	9/2 ⁺	(E1)		1.60 $\times 10^{-3}$ 2	Mult.: D(+Q) with $\delta=-0.16$ 17 from $\gamma(\theta)$ in (p, γ); $\Delta\pi$ =yes from level scheme; M2 component is less likely for the primary transition.
		4581.6	37	3162.87	7/2 ⁻	M1+E2	+0.18 8	1.23 $\times 10^{-3}$ 2	
		5098.7	100	2645.67	7/2 ⁺	(E1)		2.05 $\times 10^{-3}$ 3	Mult.: D(+Q) with $\delta=+0.1$ 30 from $\gamma(\theta)$ in (p, γ); $\Delta\pi$ =yes from level scheme; M2 component is less likely for the primary transition.
		5981.1	5.6	1763.15	5/2 ⁺				
		6524.8 ^b	1.4	1219.38	1/2 ⁺	[E3]			This transition is considered questionable by the evaluators since it would require Mult=E3 which is unlikely for this primary transition based on RUL.
7777.1	(5/2 ⁺)	2122	1.3	5655	(3/2) ⁺				
		2613.7	2.8	5163.34	7/2 ⁻				
		2937.9	5	4839.10	(1/2 ⁺ ,3/2)				
		3007.1	7.5	4769.86	(5/2 ⁻)				
		3599.0	18	4177.9	3/2 ⁻				
		3717.8	2.8	4059.1	(3/2) ⁻				
		4613.9	5	3162.87	7/2 ⁻				
		4774.1	10	3002.64	5/2 ⁺				
		5082.7	23	2694.00	3/2 ⁺				
		5131.0	3.3	2645.67	7/2 ⁺				

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	Comments
7777.1	(5/2 ⁺)	6013.4	100	1763.15	5/2 ⁺				
		6557.1	28	1219.38	1/2 ⁺				
		7776.2	45	0.0	3/2 ⁺				
7781.7	(5/2 ⁻)	1675.5	3.9	6106.2	(3/2,5/2 ⁺)				
		2183.2		5598.4	3/2 ⁺ ,5/2 ⁺				
		2378.3		5403.3	3/2 ⁻				
		2567.1	4.2	5214.5	(5/2 ⁺)				
		3011.7	19	4769.86	(5/2 ⁻)				
		3433.8	5.6	4347.76	9/2 ⁻				
		3608.1	100	4173.43	(5/2 ⁺)				
		3669.1	17	4112.4	7/2 ⁺				
		3722.4	3.3	4059.1	(3/2) ⁻				
		3863.0	31	3918.47	3/2 ⁺				
		4618.5	14	3162.87	7/2 ⁻				
		5135.6	22	2645.67	7/2 ⁺				
		6018.0	50	1763.15	5/2 ⁺				
		6561.7	3.3	1219.38	1/2 ⁺				
		7780.8	4.7	0.0	3/2 ⁺				
7797.2	1/2 ⁻	2039		5758	(1/2,3/2 ⁻)				
		2142	1.6	5655	(3/2) ⁺				
		2393.8	2.4	5403.3	3/2 ⁻				
		2788.0	2.4	5009.1	(1/2,3/2)				
		2942.7	1.6	4854.4	(1/2,3/2)				
		2958.0	1.3	4839.10	(1/2 ⁺ ,3/2)				
		3737.9	0.36	4059.1	(3/2) ⁻				
		6577.2	11	1219.38	1/2 ⁺				
		7796.3	100	0.0	3/2 ⁺				
7837.2	3/2 ⁻	1656 ^{&}	1.4 ^{&}	6181	(1/2 ⁺ to 7/2 ⁺)				
		2238.7 ^{&}	1.4 ^{&}	5598.4	3/2 ⁺ ,5/2 ⁺				
		2622.6 ^{&}	3.8 ^{&}	5214.5	(5/2 ⁺)				
		3659.1 ^{&}	76 ^{&}	4177.9	3/2 ⁻	(M1+E2)	-0.05 3	0.000917 13	Mult.: D+Q from (p, γ); $\Delta\pi$ =no from level scheme. B(M1)(W.u.)=1.03 $\times 10^3$ 19 exceeds RUL=10. B(E2)(W.u.)=7 $\times 10^2$ +11-6 exceeds RUL=100.
		3663.6 ^{&}	8.1 ^{&}	4173.43	(5/2 ⁺)				
		3777.9 ^{&}	5.4 ^{&}	4059.1	(3/2) ⁻				
		3894.1 ^{&b}	1.4 ^{&}	3942.9	9/2 ⁺				This γ is a primary γ from the unresolved 7837+7839 doublet goes to 9/2 ⁺ , which is unlikely from a 3/2 ⁻ , 7837 level. It is likely from the (5/2 ⁺), 7839.0 level.
		3918.5 ^{&}	0.54 ^{&}	3918.47	3/2 ⁺				

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	Comments
7837.2	3/2 ⁻	4834.2 &	11 &	3002.64	5/2 ⁺				
		6073.5 &	5.1 &	1763.15	5/2 ⁺				
		6617.2 &	100 &	1219.38	1/2 ⁺	(E1)			Mult.: D(+Q) with $\delta=+0.06$ 3 from (p, γ); $\Delta\pi$ =yes from level scheme; M2 component is less likely for this strong primary transition. B(E1)(W.u.)=6.5 +10-8 exceeds RUL=0.1.
		7836.3 &	57 &	0.0	3/2 ⁺	(E1)			Mult.: D(+Q) with $\delta=+0.02$ 3 from (p, γ); $\Delta\pi$ =yes from level scheme; M2 component is less likely for this strong primary transition. B(E1)(W.u.)=2.24 +47-44 exceeds RUL=0.1.
7839.0	(5/2 ⁺)	1658 &	1.4 &	6181	(1/2 ⁺ to 7/2 ⁺)				
		2240.5 &	1.4 &	5598.4	3/2 ⁺ , 5/2 ⁺				
		2624.4 &	3.8 &	5214.5	(5/2 ⁺)				
		3660.9 &	76 &	4177.9	3/2 ⁻				
		3665.4 &	8.1 &	4173.43	(5/2 ⁺)				
		3779.7 &	5.4 &	4059.1	(3/2 ⁻)				
		3895.9 &	1.4 &	3942.9	9/2 ⁺				
		3920.3 &	0.54 &	3918.47	3/2 ⁺				
		4836.0 &	11 &	3002.64	5/2 ⁺				
		6075.3 &	5.1 &	1763.15	5/2 ⁺				
		6619.0 &	100 &	1219.38	1/2 ⁺				
		7838.1 &	57 &	0.0	3/2 ⁺				
		2654.1	1.6	5214.5	(5/2 ⁺)				
		2859.5	4.9	5009.1	(1/2, 3/2)				
7868.7	(1/2 ⁺ , 3/2, 5/2 ⁺)	3695.1	8.2	4173.43	(5/2 ⁺)				
		6105.0	28	1763.15	5/2 ⁺				
		6648.6	21	1219.38	1/2 ⁺				
		7867.8	100	0.0	3/2 ⁺				
		1212.8		6660.2	(11/2 ⁻)				
		1786.2 5	9.6 4	6087.31	13/2 ⁻	D(+Q)	-0.6 6		E γ : from (²⁸ Si, $\alpha\gamma\gamma$) only. I γ : weighted average of 8.2 18 from (²⁸ Si, $\alpha\gamma\gamma$) and 9.7 4 from (¹⁶ O, $\alpha\gamma\gamma$). Mult., δ : from $\gamma\gamma$ (DCO) in (²⁸ Si, $\alpha\gamma\gamma$). E γ : weighted average of 1946.2 5 from (¹⁶ O, $\alpha\gamma\gamma$) and 1946.40 30 from (¹⁴ N, $\alpha\text{pn}\gamma$). I γ : weighted average of 46 5 from (²⁸ Si, $\alpha\gamma\gamma$)
7873.03	13/2 ⁺	1946.35 30	48.0 26	5926.65	(11/2 ⁻)	E1+M2	+0.2 1	0.000594 22	

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

<u>E_i(level)</u>	<u>J^{π}_i</u>	<u>E_{γ}[†]</u>	<u>I_{γ}[‡]</u>	<u>E_f</u>	<u>J^{π}_f</u>	<u>Mult.[#]</u>	<u>δ[#]</u>	<u>α^a</u>	<u>Comments</u>
7873.03	13/2 ⁺	2465.94 30	100.0 20	5407.07	11/2 ⁻	E1+M2	-0.25 10	0.00091 4	and 48.5 26 from (¹⁶ O, $\alpha\gamma$). Mult., δ : from $\gamma\gamma$ (DCO) and $\gamma\gamma$ (lin pol) in (²⁸ Si, $\alpha\gamma\gamma$), $\Delta J=1$. E _{γ} : weighted average of 2466.2 5 from (¹⁶ O, $\alpha\gamma$) and 2465.85 30 from (¹⁴ N, $\alpha p\gamma$). I _{γ} : weighted average of 100 9 from (²⁸ Si, $\alpha\gamma$) and 100.0 20 from (¹⁶ O, $\alpha\gamma$). Mult., δ : from $\gamma\gamma$ (DCO) and $\gamma\gamma$ (lin pol) in (²⁸ Si, $\alpha\gamma\gamma$), $\Delta J=1$.
7880.8	3/2 ⁺ ,5/2 ⁺	2226	40	5655	(3/2) ⁺				
		3041.6	5.7	4839.10	(1/2 ⁺ ,3/2)				
		3256.3	6.7	4624.3	3/2,5/2				
		3702.7	10	4177.9	3/2 ⁻				
		3768.2	4.3	4112.4	7/2 ⁺				
		3821.5	6.7	4059.1	(3/2) ⁻				
		3962.1	6.7	3918.47	3/2 ⁺				
		4877.8	37	3002.64	5/2 ⁺				
		5186.4	83	2694.00	3/2 ⁺				
		6117.1	100	1763.15	5/2 ⁺				
		7879.9	33	0.0	3/2 ⁺				
7899.2	(5/2)	2253	2.2	5646	(5/2,7/2,9/2 ⁺)				
		2735.8	1.3	5163.34	7/2 ⁻				
		3018.1	1.2	4880.96	7/2 ⁽⁺⁾				
		3274.7	0.74	4624.3	3/2,5/2				
		3721.1	12	4177.9	3/2 ⁻				
		3725.6	12	4173.43	(5/2 ⁺)				
		3839.9	4.4	4059.1	(3/2) ⁻				
		4736.0	5.9	3162.87	7/2 ⁻				
		5204.8	1.9	2694.00	3/2 ⁺				
		6135.5	100	1763.15	5/2 ⁺				
		7898.2	5.9	0.0	3/2 ⁺				
7923.3	(3/2 ⁺ ,5/2 ⁺)	3084.1	3.4	4839.10	(1/2 ⁺ ,3/2)				
		3153.3	4.9	4769.86	(5/2 ⁻)				
		3298.8	4.6	4624.3	3/2,5/2				
		3745.2	7.3	4177.9	3/2 ⁻				
		3749.7	12	4173.43	(5/2 ⁺)				
		3864.0	4.2	4059.1	(3/2) ⁻				
		3955.5	3.7	3967.6	1/2 ⁺				
		4004.6	2.9	3918.47	3/2 ⁺				
		4920.3	2.2	3002.64	5/2 ⁺				
		5228.9	27	2694.00	3/2 ⁺				

Adopted Levels, Gammas (continued) $\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	Comments
7923.3	$(3/2^+, 5/2^+)$	5277.2	2.2	2645.67	$7/2^+$		
		6159.6	3.7	1763.15	$5/2^+$		
		6703.2	66	1219.38	$1/2^+$		
		7922.3	100	0.0	$3/2^+$		
7970.3	$(5/2^-)$	2324	15	5646	$(5/2, 7/2, 9/2^+)$		
		2384.2	15	5586.0	$5/2^+$		
		2755.7	15	5214.5	$(5/2^+)$		
		3089.2	50	4880.96	$7/2^{(+)}$		
		3200.3	25	4769.86	$(5/2^-)$		
		3622.3	40	4347.76	$9/2^-$		
		3792.2	30	4177.9	$3/2^-$		
		3857.7	30	4112.4	$7/2^+$		
		4807.1	50	3162.87	$7/2^-$		
		4967.3	25	3002.64	$5/2^+$		
		5324.2	100	2645.67	$7/2^+$		
		6206.6	100	1763.15	$5/2^+$		
		6750.2	5	1219.38	$1/2^+$		
7987.8	$3/2^+$	1881.6	0.83	6106.2	$(3/2, 5/2^+)$		
		2333	2.8	5655	$(3/2)^+$		
		3809.7	<1.3	4177.9	$3/2^-$		I_γ : total $I_\gamma=1.3$ for γ decays to 4173 and 4178 doublet final levels.
		3814.2	<1.3	4173.43	$(5/2^+)$		I_γ : total $I_\gamma=1.3$ for γ decays to 4173 and 4178 doublet final levels.
		3928.5	1.7	4059.1	$(3/2)^-$		
		6767.7	100	1219.38	$1/2^+$	M1+E2	δ : +0.21 4 or -3.1 4 from (p, γ).
		7986.8	60	0.0	$3/2^+$	M1+E2	δ : -0.36 2 or -11 3 from (p, γ).
7995.6	$(5/2)$	2188.9	0.77	5806.6	$(1/2, 3/2, 5/2)$		
		2397.1	1.4	5598.4	$3/2^+, 5/2^+$		
		2592.2	3.9	5403.3	$3/2^-$		
		2832.1	1.9	5163.34	$7/2^-$		
		3817.5	5.1	4177.9	$3/2^-$		
		3936.3	1.4	4059.1	$(3/2)^-$		
		4832.4	9	3162.87	$7/2^-$		
		4992.6	2.6	3002.64	$5/2^+$		
		5301.2	0.26	2694.00	$3/2^+$		
		5349.5	0.77	2645.67	$7/2^+$		
		6231.9	1.2	1763.15	$5/2^+$		
		7994.6	100	0.0	$3/2^+$		
8001.0	$(7/2^+)$	2355	2.6	5646	$(5/2, 7/2, 9/2^+)$		
		2786.4	1.1	5214.5	$(5/2^+)$		
		3119.9	1	4880.96	$7/2^{(+)}$		
		3231.0	11	4769.86	$(5/2^-)$		
		3653.0	2.7	4347.76	$9/2^-$		
		3827.4	14	4173.43	$(5/2^+)$		

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	Comments
8001.0	(7/2 ⁺)	4057.9	4.3	3942.9	9/2 ⁺				
		4837.8	1.1	3162.87	7/2 ⁻				
		5354.9	2.3	2645.67	7/2 ⁺				
		6237.3	100	1763.15	5/2 ⁺				
		8000.0	2	0.0	3/2 ⁺				
		2350	3.3	5655	(3/2) ⁺				
		3123.9	1.1	4880.96	7/2 ⁽⁺⁾				
		3165.8	5.5	4839.10	(1/2 ⁺ , 3/2)				
		3826.9	<2.4	4177.9	3/2 ⁻				I_γ : total $I_\gamma=2.4$ for γ decays to 4173 and 4178 doublet final levels.
		3831.3	<2.4	4173.43	(5/2 ⁺)				I_γ : total $I_\gamma=2.4$ for γ decays to 4173 and 4178 doublet final levels.
		4841.8	27	3162.87	7/2 ⁻				
		5002.0	3.6	3002.64	5/2 ⁺				
		5310.6	3.1	2694.00	3/2 ⁺				
		5358.9	7.3	2645.67	7/2 ⁺				
		6241.3	27	1763.15	5/2 ⁺				
8035.7	1/2 ⁻ , 3/2 ⁻	6784.9	1.1	1219.38	1/2 ⁺				
		8004.0	100	0.0	3/2 ⁺				
		2229.0	10.7 11	5806.6	(1/2, 3/2, 5/2)				
		3857.6	26.8 27	4177.9	3/2 ⁻				
		3976.4	19.6 20	4059.1	(3/2) ⁻				
		5341.3	9 5	2694.00	3/2 ⁺				
		6272.0	9 5	1763.15	5/2 ⁺				
		6815.6	100 11	1219.38	1/2 ⁺				
		8034.7	3.6 18	0.0	3/2 ⁺				
		3183.8	1.1	4854.4	(1/2, 3/2)				
		3199.1	2.8	4839.10	(1/2 ⁺ , 3/2)				
		3413.9	5.3	4624.3	3/2, 5/2				
		3860.3	2.8	4177.9	3/2 ⁻				
		3979.1	5.3	4059.1	(3/2) ⁻				
		8038.4	1/2 ⁺ , 3/2 ⁺	4070.6	1.4	3967.6	1/2 ⁺		
4119.7	2.5			3918.47	3/2 ⁺				
5035.4	7			3002.64	5/2 ⁺				
5344.0	100			2694.00	3/2 ⁺	M1+E2	-0.20 2	1.44×10^{-3} 2	Mult., δ : D+Q with $\delta=-0.20$ 2 or +19 8 from (p, γ); E1+M2 and the larger $\delta(E2/M1)$ is ruled out by RUL. B(M1)(W.u.)=52.1 +43-37 exceeds RUL=10. B(E2)(W.u.)= 2.8×10^2 +6-5 exceeds RUL=100.
6274.7	26			1763.15	5/2 ⁺				
6818.3	7			1219.38	1/2 ⁺				
8037.4	14			0.0	3/2 ⁺	M1+E2	-0.13 4		B(M1)(W.u.)=2.20 44; B(E2)(W.u.)=2.2 +16-12

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)							
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. #	α^a
Comments							
Mult., δ : D+Q with $\delta=-0.13$ 4 or $+8.1$ 26 from (p, γ); E1+M2 and the larger $\delta(\text{E2/M1})$ is ruled out by RUL.							
8075.9	(5/2)	2430	20	5646	(5/2,7/2,9/2 ⁺)		
		2489.8	20	5586.0	5/2 ⁺		
		2861.3	14	5214.5	(5/2 ⁺)		
		2912.4		5163.34	7/2 ⁻		
		3194.8		4880.96	7/2 ⁽⁺⁾		
		3305.9	7.1	4769.86	(5/2 ⁻)		
		3897.8	14	4177.9	3/2 ⁻		
		3963.3	3.6	4112.4	7/2 ⁺		
		5072.9	3.6	3002.64	5/2 ⁺		
		6312.1	100	1763.15	5/2 ⁺		
		8074.9		0.0	3/2 ⁺		
8096.1	(5/2)	2289.4	5.1	5806.6	(1/2,3/2,5/2)		
		2450	13	5646	(5/2,7/2,9/2 ⁺)		
		2497.6	13	5598.4	3/2 ⁺ , 5/2 ⁺		
		2881.5	3.3	5214.5	(5/2 ⁺)		
		2932.6	5.1	5163.34	7/2 ⁻		
		3256.8	7.7	4839.10	(1/2 ⁺ , 3/2)		
		3326.1	5.1	4769.86	(5/2 ⁻)		
		3918.0	4.9	4177.9	3/2 ⁻		
		4036.8	5.1	4059.1	(3/2) ⁻		
		4177.4	4.6	3918.47	3/2 ⁺		
		4932.9	13	3162.87	7/2 ⁻		
		5401.7	13	2694.00	3/2 ⁺		
		5450.0	44	2645.67	7/2 ⁺		
		6332.3	21	1763.15	5/2 ⁺		
8106.4	3/2 ⁺	8095.1	100	0.0	3/2 ⁺		
		2507.9	1.4	5598.4	3/2 ⁺ , 5/2 ⁺		
		3251.8	1.6	4854.4	(1/2,3/2)		
		3267.1	1.4	4839.10	(1/2 ⁺ , 3/2)		
		3928.3	4.3	4177.9	3/2 ⁻		
		4047.1	3.5	4059.1	(3/2) ⁻		
		4138.5	8.1	3967.6	1/2 ⁺		
		5103.4	24	3002.64	5/2 ⁺		
		5412.0	24	2694.00	3/2 ⁺	M1+E2	0.00153 8 δ : +4.9 10 or -0.05 4.
		6342.6	8.1	1763.15	5/2 ⁺		
		6886.3	95	1219.38	1/2 ⁺		
		8105.4	100	0.0	3/2 ⁺	M1+E2	δ : +1.2 1 or -0.46 4.
8113.3	(1/2 ⁺ , 3/2, 5/2 ⁺)	1621	19	6492	(3/2 ⁺)		
		2007.0	19	6106.2	(3/2, 5/2 ⁺)		
		2306.6	3.4	5806.6	(1/2, 3/2, 5/2)		

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π
8113.3	(1/2 ⁺ , 3/2, 5/2 ⁺)	2458	2.8	5655	(3/2) ⁺
		2514.8		5598.4	3/2 ⁺ , 5/2 ⁺
		2898.7	5.6	5214.5	(5/2 ⁺)
		3939.6	22	4173.43	(5/2 ⁺)
		4054.0	4.1	4059.1	(3/2) ⁻
		4145.4	16	3967.6	1/2 ⁺
		5110.3	5.9	3002.64	5/2 ⁺
		5418.9	41	2694.00	3/2 ⁺
		6349.5	16	1763.15	5/2 ⁺
		6893.2	100	1219.38	1/2 ⁺
		8112.3	78	0.0	3/2 ⁺
8147.2	3/2 ⁻	2464.9	1.6	5682.2	1/2 ⁻ , 3/2 ⁻
		3138.0	4.8	5009.1	(1/2, 3/2)
		3969.1	13	4177.9	3/2 ⁻
		4087.8	4.8	4059.1	(3/2) ⁻
		4179.3	3.2	3967.6	1/2 ⁺
		4228.5	4.8	3918.47	3/2 ⁺
		5452.7	4.8	2694.00	3/2 ⁺
		6927.1	24	1219.38	1/2 ⁺
		8146.2	100	0.0	3/2 ⁺
8156.8	(5/2 ⁺)	2511	7.8	5646	(5/2, 7/2, 9/2 ⁺)
		2570.7	16	5586.0	5/2 ⁺
		2942.2	12	5214.5	(5/2 ⁺)
		3386.8	2.0	4769.86	(5/2 ⁻)
		3978.7	9.8	4177.9	3/2 ⁻
		4044.2	14	4112.4	7/2 ⁺
		4097.4	2.0	4059.1	(3/2) ⁻
		4213.6	3.9	3942.9	9/2 ⁺
		4993.6	9.8	3162.87	7/2 ⁻
		5153.8	12	3002.64	5/2 ⁺
		5510.7	100	2645.67	7/2 ⁺
		6393.0	2.0	1763.15	5/2 ⁺
		8155.8	5.9	0.0	3/2 ⁺
8179.4	(3/2 ⁻ , 5/2, 7/2 ⁺)	3554.9	13.2 13	4624.3	3/2, 5/2
		5016.1	11.8 12	3162.87	7/2 ⁻
		5176.4	5.9 30	3002.64	5/2 ⁺
		5484.9	4.4 22	2694.00	3/2 ⁺
		6415.6	100 10	1763.15	5/2 ⁺
		8178.4	11.8 12	0.0	3/2 ⁺
8208.2	5/2 ⁺	2553	0.9	5655	(3/2) ⁺
		3327.1	1.0	4880.96	7/2 ⁽⁺⁾
		3368.9	0.51	4839.10	(1/2 ⁺ , 3/2)

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult.#	$\delta^\#$	α^a	Comments
8208.2	$5/2^+$	4148.8	1.9	4059.1	$(3/2)^-$				
		4289.5		3918.47	$3/2^+$				
		5205.1	1.4	3002.64	$5/2^+$				
		5513.7	0.64	2694.00	$3/2^+$				
		6444.4	18	1763.15	$5/2^+$	M1+E2	+0.33 4		B(M1)(W.u.)=0.75 +35-22; B(E2)(W.u.)=7.5 +41-25
		6988.1	3.9	1219.38	$1/2^+$				
		8207.2	100	0.0	$3/2^+$	M1+E2	-0.36 4		B(M1)(W.u.)=2.0 +9-5; B(E2)(W.u.)=15 +7-4
		8216.3	$5/2^+$	2561	$(3/2)^+$				
		3446.3		4769.86	$(5/2)^-$				
		4038.2		4177.9	$3/2^-$				
		4042.6		4173.43	$(5/2^+)$				
		5053.0		3162.87	$7/2^-$	(E1)		2.04×10^{-3} 3	B(E1)(W.u.)=0.062 +21-12 Mult.: D+Q with $\delta=+0.11$ 2 from $\gamma(\theta)$ in (p, γ); $\Delta\pi=\text{yes}$ from level scheme; but M2 component is unlikely for this strong primary transition.
		5213.2		3002.64	$5/2^+$				
		5521.8		2694.00	$3/2^+$				
		6452.5		1763.15	$5/2^+$				
		8215.3		0.0	$3/2^+$	M1+E2	+0.06 3		B(M1)(W.u.)=0.55 +18-9; B(E2)(W.u.)=0.11 +16-8
		8242.4		4068.7	$(5/2^+)$				
		5239.3		3002.64	$5/2^+$				
		7022.3		1219.38	$1/2^+$				
		8251		4138	$7/2^+$				
		5248		3002.64	$5/2^+$				
		7031		1219.38	$1/2^+$				
		8250		0.0	$3/2^+$				
		8269.0		3499.0	$(5/2)^-$				
		4090.8		4177.9	$3/2^-$				
		4350.2		3918.47	$3/2^+$				
		5105.7		3162.87	$7/2^-$				
		5265.9		3002.64	$5/2^+$				
		5574.5		2694.00	$3/2^+$				
		5622.9		2645.67	$7/2^+$				
		6505.2		1763.15	$5/2^+$				
		8268.0		0.0	$3/2^+$				
		8277.2		2622	$(3/2)^+$				
		3396.1		4880.96	$7/2^{(+)}$				
		3437.9		4839.10	$(1/2^+, 3/2)$				
		3507.2		4769.86	$(5/2)^-$				
		3652.7		4624.3	$3/2, 5/2$				
		3929.2		4347.76	$9/2^-$				
		4103.5		4173.43	$(5/2^+)$				

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)						
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]
8277.2	$(5/2)^+$	4217.8	5.2	4059.1	$(3/2)^-$	
		4358.4	17	3918.47	$3/2^+$	
		5274.1	35	3002.64	$5/2^+$	
		5582.7	10	2694.00	$3/2^+$	
		5631.0	10	2645.67	$7/2^+$	
		6513.4	79	1763.15	$5/2^+$	
		8276.2	100	0.0	$3/2^+$	
		2696.3	29	5586.0	$5/2^+$	
		3067.8	29	5214.5	$(5/2^+)$	
		3118.9	11	5163.34	$7/2^-$	
8282.4	$(3/2^-, 5/2)$	4104.2	50	4177.9	$3/2^-$	
		4223.0	21	4059.1	$(3/2)^-$	
		5119.1	93	3162.87	$7/2^-$	
		6518.6	25	1763.15	$5/2^+$	
		8281.4	100	0.0	$3/2^+$	
		2633	4.6	5655	$(3/2)^+$	
		3278.6	6.8	5009.1	$(1/2, 3/2)$	
		4109.7	21	4177.9	$3/2^-$	
		4228.5	4.6	4059.1	$(3/2)^-$	
		4319.9	6.8	3967.6	$1/2^+$	
8287.82	$1/2^-$	4369.06	2.3	3918.47	$3/2^+$	
		5593.3	52	2694.00	$3/2^+$	
		7067.67	100	1219.38	$1/2^+$	(E1)
						B(E1)(W.u.)=0.069 +24-14
						Mult.: D+Q with $\delta(Q/D)=+0.21$ 2 or -3.2 2 in (p, γ); but M2 component is unlikely for this primary transition based on resonance $\Gamma=0.04$ keV.
		8286.77	30	0.0	$3/2^+$	
		3288.9	13	5009.1	$(1/2, 3/2)$	
		3417.1	17	4880.96	$7/2^{(+)}$	
		3443.6	21	4854.4	$(1/2, 3/2)$	
		4124.5	25	4173.43	$(5/2^+)$	
8298.2	$3/2^+$	4238.8	42	4059.1	$(3/2)^-$	
		5134.9	4.2	3162.87	$7/2^-$	
						This γ is questionable if $J^\pi(8298)=3/2^+$, since it would require Mult=M2 which is probably unlikely; however $J^\pi(8298)=3/2^-$ would make the strong 7078.0 γ and 8297.1 γ less likely.
		5295.1	4.2	3002.64	$5/2^+$	
		5603.7	42	2694.00	$3/2^+$	
		5652.0	25	2645.67	$7/2^+$	
		6534.4	33	1763.15	$5/2^+$	
		7078.1	100	1219.38	$1/2^+$	M1+E2
		8297.1	54	0.0	$3/2^+$	M1+E2
						δ : -1.7 1 or -0.02 2 from $\gamma(\theta)$ in (p, γ).
8318.1	$(5/2^+)$					δ : -0.15 4 or $+9.5$ 36 from $\gamma(\theta)$ in (p, γ).
		2511.4	0.9	5806.6	$(1/2, 3/2, 5/2)$	

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	Comments
8318.1	(5/2 ⁺)	2672	0.9	5646	(5/2,7/2,9/2 ⁺)				
		2719.6	1.9	5598.4	3/2 ⁺ ,5/2 ⁺				
		3548.1	0.39	4769.86	(5/2 ⁻)				
		3693.6	2.3	4624.3	3/2,5/2				
		4139.9	<3.9	4177.9	3/2 ⁻				
		4144.4	<3.9	4173.43	(5/2 ⁺)				
		4258.7	1.5	4059.1	(3/2) ⁻				
		4350.2	3.9	3967.6	1/2 ⁺				
		5154.8	0.64	3162.87	7/2 ⁻				
		5315.0	3.9	3002.64	5/2 ⁺				
		5623.6	2.1	2694.00	3/2 ⁺				
		5671.9	1.2	2645.67	7/2 ⁺				
		6554.3	2.3	1763.15	5/2 ⁺				
		7098.0	2.6	1219.38	1/2 ⁺				
		8317.0	100	0.0	3/2 ⁺				
8319.6	15/2 ⁻	2232.7 6	57.2 28	6087.31	13/2 ⁻	[M1,E2]		0.00040 4	E_γ, I_γ : from (¹⁶ O, $\alpha\gamma$). B(M1)(W.u.)=0.0577 4I if M1, B(E2)(W.u.)=43.8 3I if E2.
		2911.9 8	100 5	5407.07	11/2 ⁻	E2		0.000753 1I	B(E2)(W.u.)=20.3 12 $E_\gamma, I_\gamma, \text{Mult.}$: from (¹⁶ O, $\alpha\gamma$). Mult=Q, $\Delta J=2$ from $\gamma\gamma(\text{DCO})$; M2 ruled out by RUL.
8323.1	(3/2 ⁺ ,5/2 ⁺)	4210.4	1.2	4112.4	7/2 ⁺				
		5320.0	3.7	3002.64	5/2 ⁺				
		5676.9	4.9	2645.67	7/2 ⁺				
		6559.3	<2.4	1763.15	5/2 ⁺				
		7103.0	9.8	1219.38	1/2 ⁺				
		8322.0	100	0.0	3/2 ⁺				
8381.8	5/2 ⁺	3611.7		4769.86	(5/2 ⁻)				
		3757.3		4624.3	3/2,5/2				
		4269.1	23.5 24	4112.4	7/2 ⁺				
		4463.0	71 7	3918.47	3/2 ⁺	M1+E2	+0.15 3	1.19×10 ⁻³ 2	B(M1)(W.u.)=2.9 +13-7; B(E2)(W.u.)=13 +8-5
		5378.7	74 8	3002.64	5/2 ⁺	M1+E2	+0.02 3	1.44×10 ⁻³ 2	B(M1)(W.u.)=1.8 +14-7
		5687.3	2.9 15	2694.00	3/2 ⁺	M1+E2	-0.72 3I	0.00157 4	B(M1)(W.u.)=0.039 +32-2I; B(E2)(W.u.)=2.4 +23-18
		5735.6	15 8	2645.67	7/2 ⁺				
		6618.0	100 10	1763.15	5/2 ⁺	M1+E2	+0.06 3		B(M1)(W.u.)=1.3 +6-3; B(E2)(W.u.)=0.4 +6-3
		8380.7	9 5	0.0	3/2 ⁺	D+Q	1.65		
		3632.8	16 8	4769.86	(5/2 ⁻)				
8402.9	5/2 ⁻	4224.7	29 16	4177.9	3/2 ⁻				

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π
8402.9	$5/2^-$	4343.5	74 20	4059.1	$(3/2)^-$
		5708.4	32 16	2694.00	$3/2^+$
		5756.7	23 13	2645.67	$7/2^+$
		6639.1	48 13	1763.15	$5/2^+$
		8401.8	100 26	0.0	$3/2^+$
8405.8	$(5/2^-)$	3524.7	35 9	4880.96	$7/2^{(+)}$
		4232.1	3.7 19	4173.43	$(5/2^+)$
		4293.1	3.3 16	4112.4	$7/2^+$
		5402.7	35 9	3002.64	$5/2^+$
		6642.0	100 26	1763.15	$5/2^+$
		8404.7	56 14	0.0	$3/2^+$
		3407.4	14	5009.1	$(1/2,3/2)$
8416.7	$1/2^+$	4243.0	31	4173.43	$(5/2^+)$
		4473.5	8.6	3942.9	$9/2^+$
		5413.6	20	3002.64	$5/2^+$
		5770.5	23	2645.67	$7/2^+$
		6652.9	63	1763.15	$5/2^+$
		7196.5	26	1219.38	$1/2^+$
		8415.6	100	0.0	$3/2^+$
		2809	47 24	5655	$(3/2)^+$
		3249.6	34 17	5214.5	$(5/2^+)$
		3455.0	47 24	5009.1	$(1/2,3/2)$
8464.3	$(1/2^+,3/2,5/2^+)$	3609.7	19 10	4854.4	$(1/2,3/2)$
		3839.8	25 13	4624.3	$3/2,5/2$
		4286.1	31 15	4177.9	$3/2^-$
		4290.6	32 17	4173.43	$(5/2^+)$
		4496.4	31 15	3967.6	$1/2^+$
		4545.5	15 8	3918.47	$3/2^+$
		5461.2	100 24	3002.64	$5/2^+$
		5769.8	94 24	2694.00	$3/2^+$
		7244.1	53 24	1219.38	$1/2^+$
		8463.2	59 18	0.0	$3/2^+$
		2829	12 6	5655	$(3/2)^+$
		3603.2	15 8	4880.96	$7/2^{(+)}$
		3859.9	2.2 11	4624.3	$3/2,5/2$
		4516.5	2.6 13	3967.6	$1/2^+$
8484.4	$(3/2)^+$	4565.6	11 6	3918.47	$3/2^+$
		5481.3	15 9	3002.64	$5/2^+$
		5789.9	44 11	2694.00	$3/2^+$
		5838.2	6.1 31	2645.67	$7/2^+$
		6720.6	100 26	1763.15	$5/2^+$
		8483.3	9 5	0.0	$3/2^+$

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)											
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	Comments		
8486.2	$3/2^-$	2728	2.4 13	5758	(1/2,3/2 ⁻)						
		3861.7	3.9 20	4624.3	3/2,5/2						
		4308.0	<15	4177.9	3/2 ⁻				I_γ : total $I_\gamma=10$ 5 for γ decays to 4173 and 4178 doublet levels (1976Sp08).		
		4312.5	<15	4173.43	(5/2 ⁺)				I_γ : total $I_\gamma=10$ 5 for γ decays to 4173 and 4178 doublet levels (1976Sp08).		
		4426.8	4.1 22	4059.1	(3/2) ⁻						
		4518.3	15 9	3967.6	1/2 ⁺						
		5483.1	17 9	3002.64	5/2 ⁺						
		5791.7	41 11	2694.00	3/2 ⁺						
		6722.4	8 4	1763.15	5/2 ⁺						
		7266.0	14 7	1219.38	1/2 ⁺						
		8485.1	100 26	0.0	3/2 ⁺						
		8487.5	$15/2^-$	2399.8 8	100.0 34	6087.31	13/2 ⁻	(M1(+E2))	+0.2 +3-2	0.000427 17	Mult., δ : D+Q and δ from $\gamma\gamma(\text{DCO})$ in ($^{28}\text{Si},\alpha p\gamma$); $\Delta\pi=\text{no}$ from level scheme.
				3080.0 7	31.8 20	5407.07	11/2 ⁻	E2		0.000825 12	$E_\gamma, I_\gamma, \text{Mult.}$: from ($^{16}\text{O},\alpha p\gamma$), with Mult=Q, $\Delta J=2$ from $\gamma\gamma(\text{ADO})$ and M2 ruled out by RUL.
		8514.3	$1/2^-$	3659.7	16 8	4854.4	(1/2,3/2)				
4336.1	7 4			4177.9	3/2 ⁻						
4454.9	100 26			4059.1	(3/2) ⁻						
4546.4	85 22			3967.6	1/2 ⁺						
4595.5	24 12			3918.47	3/2 ⁺						
7294.1	37 11			1219.38	1/2 ⁺						
8513.2	100 26			0.0	3/2 ⁺						
8558.2?	(13/2 ⁻)	2631.4 ^b	100	5926.65	(11/2) ⁻	D(+Q)	-0.2 +2-3		$E_\gamma, \text{Mult.}, \delta$: from ($^{28}\text{Si},\alpha p\gamma$); $\Delta\pi=\text{no}$ from level scheme; Mult=Q, $\Delta J=1$.		
8571.9	$(5/2)^+$	2848.3	3.1 16	5723.52	5/2 ⁺						
		2917	1.9 10	5655	(3/2) ⁺						
		2973.4	1.0 6	5598.4	3/2 ⁺ , 5/2 ⁺						
		3408.4	2.6 13	5163.34	7/2 ⁻						
		3690.7	1.3 7	4880.96	7/2 ⁽⁺⁾						
		4398.2	1.0 6	4173.43	(5/2 ⁺)						
		4459.2	1.6 9	4112.4	7/2 ⁺						
		4653.1	1.6 9	3918.47	3/2 ⁺						
		5408.6	2.3 12	3162.87	7/2 ⁻						
		5568.8	6.4 33	3002.64	5/2 ⁺						
		5877.4	0.9 4	2694.00	3/2 ⁺						
		5925.7	11 6	2645.67	7/2 ⁺						
		7351.7	8 4	1219.38	1/2 ⁺						
		8570.8	100 10	0.0	3/2 ⁺						
		8580.5	1/2 ⁺	2088	8 4	6492	(3/2 ⁺)				

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π
8580.5	$1/2^+$	3741.2	2.4 13	4839.10	$(1/2^+, 3/2)$
		4402.3	8 4	4177.9	$3/2^-$
		4521.1	3.7 19	4059.1	$(3/2)^-$
		4661.7	4.3 22	3918.47	$3/2^+$
		5886.0	10 5	2694.00	$3/2^+$
		7360.3	100 11	1219.38	$1/2^+$
		8579.4	48 13	0.0	$3/2^+$
8590.1	$5/2^+$	2832	22 11	5758	$(1/2, 3/2^-)$
		2935	17 9	5655	$(3/2)^+$
		3820.0	73 18	4769.86	$(5/2^-)$
		4416.4	59 14	4173.43	$(5/2^+)$
		4477.4	16 8	4112.4	$7/2^+$
		5426.8	36 18	3162.87	$7/2^-$
		5587.0	100 27	3002.64	$5/2^+$
		5943.9	25 13	2645.67	$7/2^+$
		6826.2	64 18	1763.15	$5/2^+$
		8589.0	46 14	0.0	$3/2^+$
8612.0	$3/2^+, 5/2^+$	6848.1	100 10	1763.15	$5/2^+$
		7391.8	34 8	1219.38	$1/2^+$
		8610.9	66 16	0.0	$3/2^+$
8613.7	$5/2^+$	3732.5	2.9 14	4880.96	$7/2^{(+)}$
		4554.3	10 5	4059.1	$(3/2)^-$
		4670.5	7.1	3942.9	$9/2^+$
		4694.9	13 7	3918.47	$3/2^+$
		5610.6	4.5 23	3002.64	$5/2^+$
		5919.2	8 4	2694.00	$3/2^+$
		5967.5	41 11	2645.67	$7/2^+$
		6849.8	16	1763.15	$5/2^+$
		7393.5	3.6	1219.38	$1/2^+$
		8612.6	100 11	0.0	$3/2^+$
8618.2	$3/2^+, 5/2^+$	2894.6	6.4 33	5723.52	$5/2^+$
		2963	5.2 27	5655	$(3/2)^+$
		3763.6	4.2 21	4854.4	$(1/2, 3/2)$
		3993.7	4.6 24	4624.3	$3/2, 5/2$
		4650.3	9 5	3967.6	$1/2^+$
		4699.4	7 4	3918.47	$3/2^+$
		5615.1	49 12	3002.64	$5/2^+$
		5923.7	42 12	2694.00	$3/2^+$
		5972.0	5.2 27	2645.67	$7/2^+$
		6854.3	70 18	1763.15	$5/2^+$
		8617.1	100 24	0.0	$3/2^+$
		2906.5	0.6 4	5723.52	$5/2^+$
8630.1	$7/2^-$				

Adopted Levels, Gammas (continued)

<u>$\gamma(^{35}\text{Cl})$ (continued)</u>									
<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>$E_\gamma^\dagger$</u>	<u>$I_\gamma^\ddagger$</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Mult.[#]</u>	<u>$\delta^\#$</u>	<u>α^a</u>	<u>Comments</u>
8630.1	$7/2^-$	3466.6	17 9	5163.34	$7/2^-$	M1+E2	+0.60 10	0.000884 15	Mult., δ : D+Q and δ from $\gamma(\theta)$ in (p, γ); E1+M2 ruled out by RUL.
		3860.0	3.0 15	4769.86	$(5/2^-)$				
		4282.1	26 7	4347.76	$9/2^-$	M1+E2	-0.184 14	1.13×10^{-3} 2	Mult., δ : D+Q and δ from $\gamma(\theta)$ in (p, γ); E1+M2 ruled out by RUL.
		4456.4	8 4	4173.43	$(5/2^+)$				
		4517.4	21 11	4112.4	$7/2^+$	(E1+M2)	-0.06 2	1.86×10^{-3} 3	Mult., δ : D+Q and δ from $\gamma(\theta)$ in (p, γ); $\Delta\pi$ =yes from level scheme.
		4686.9	21 11	3942.9	$9/2^+$				
		5466.8	7 4	3162.87	$7/2^-$				
		5627.0	3.0 15	3002.64	$5/2^+$				
		5983.9	5.5 28	2645.67	$7/2^+$				
		6866.2	100 26	1763.15	$5/2^+$	(E1+M2)	-0.011 3		Mult., δ : D+Q and δ from $\gamma(\theta)$ in (p, γ); $\Delta\pi$ =yes from level scheme.
8640	$(5/2^+)$	8629.0	1.1 7	0.0	$3/2^+$				
		4697	61	3942.9	$9/2^+$				
		5477	53	3162.87	$7/2^-$				
		5637	16	3002.64	$5/2^+$				
		5946	34	2694.00	$3/2^+$				
		6876	100	1763.15	$5/2^+$				
8641.4	$3/2^+, 5/2^+$	2986	10 6	5655	$(3/2^+)$				
		3055.3	9 5	5586.0	$5/2^+$				
		3237.9	17 9	5403.3	$3/2^-$				
		3802.1	9 5	4839.10	$(1/2^+, 3/2)$				
		3871.3	86 22	4769.86	$(5/2^-)$				
		4016.9	7 4	4624.3	$3/2, 5/2$				
		4463.2	<64	4177.9	$3/2^-$				I $_\gamma$: total I $_\gamma$ =50 14 for γ decays to 4173 and 4178 doublet levels (1976Sp08).
		4467.7	<64	4173.43	$(5/2^+)$				
		4582.0	32 18	4059.1	$(3/2)^-$				I $_\gamma$: total I $_\gamma$ =50 14 for γ decays to 4173 and 4178 doublet levels (1976Sp08).
		4722.6	21 11	3918.47	$3/2^+$				
8686.2	$5/2^-$	5946.9	19 9	2694.00	$3/2^+$				
		8640.3	100 25	0.0	$3/2^+$				
		3100.1	2.8 15	5586.0	$5/2^+$				
		3282.7	4.1 21	5403.3	$3/2^-$				
		4508.0	2.6 13	4177.9	$3/2^-$				
		4626.8	4.6 24	4059.1	$(3/2)^-$				
		5522.9	4.9 25	3162.87	$7/2^-$				
		5991.7	3.7 19	2694.00	$3/2^+$				
		6922.3	6.3 32	1763.15	$5/2^+$				
		7466.0	2.5 13	1219.38	$1/2^+$				

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. #	α^a	Comments
8686.2	5/2 ⁻	8685.0	100 11	0.0	3/2 ⁺			
8696.9	3/2 ⁻	2890.2	3.6 19	5806.6	(1/2,3/2,5/2)			
		3098.4	6.0 30	5598.4	3/2 ⁺ ,5/2 ⁺			
		7476.7	15 4	1219.38	1/2 ⁺			
		8695.7	100 10	0.0	3/2 ⁺			
8750.9	3/2 ⁻	2944.2	4.2 21	5806.6	(1/2,3/2,5/2)			
		3347.4	9 5	5403.3	3/2 ⁻			
		3536.2	4.5 23	5214.5	(5/2 ⁺)			
		4572.7	4.0 21	4177.9	3/2 ⁻			
		4577.2	4.3 23	4173.43	(5/2 ⁺)			
		6056.3	5.7 28	2694.00	3/2 ⁺			
		6987.0	57 15	1763.15	5/2 ⁺			
		7530.7	3.8	1219.38	1/2 ⁺			
		8749.7	100 10	0.0	3/2 ⁺			
8779.8	3/2 ⁻	3181.2	16 8	5598.4	3/2 ⁺ ,5/2 ⁺			
		3565.1	24 12	5214.5	(5/2 ⁺)			
		4601.6	<71	4177.9	3/2 ⁻			I_γ : total $I_\gamma=56$ 15 for γ decays to 4173 and 4178 doublet levels (1976Sp08).
		4606.0	<71	4173.43	(5/2 ⁺)			I_γ : total $I_\gamma=56$ 15 for γ decays to 4173 and 4178 doublet levels (1976Sp08).
		4861.0	16 8	3918.47	3/2 ⁺			
		5776.7	100 26	3002.64	5/2 ⁺			
		6085.2	44 11	2694.00	3/2 ⁺			
		6133.6	48 11	2645.67	7/2 ⁺			
		7015.9	25 13	1763.15	5/2 ⁺			
		8778.6	41 11	0.0	3/2 ⁺			
8787.2	(5/2 ⁺)	3063.5	27 14	5723.52	5/2 ⁺			
		3623.7	76 20	5163.34	7/2 ⁻			
		4017.1	27 14	4769.86	(5/2 ⁻)			
		4609.0	36 20	4177.9	3/2 ⁻			
		5623.8	23 12	3162.87	7/2 ⁻			
		5784.1	80 20	3002.64	5/2 ⁺			
		7023.3	100 24	1763.15	5/2 ⁺			
		8786.0	32 16	0.0	3/2 ⁺			
8788.3	(15/2 ⁺)	915.4 4	100	7873.03	13/2 ⁺	(M1)	4.82×10 ⁻⁵ 7	E_γ , Mult.: from (¹⁶ O, $\alpha\gamma$), with Mult=D, $\Delta J=1$ from $\gamma\gamma$ (ADO); $\Delta\pi$ =(no) from level scheme.
8798.4	(3/2 ⁺ ,5/2 ⁺)	4739.0	13 7	4059.1	(3/2 ⁻)			
		5795.2	26 7	3002.64	5/2 ⁺			
		7034.5	11 6	1763.15	5/2 ⁺			
		7578.1	11 6	1219.38	1/2 ⁺			
		8797.2	100 10	0.0	3/2 ⁺			
8820.9	(1/2 ⁺ to 7/2 ⁺)	5817.7	17 10	3002.64	5/2 ⁺			

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)								
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. #	α^a	Comments
8820.9	(1/2 ⁺ to 7/2 ⁺)	6126.3	35 10	2694.00	3/2 ⁺			
		7057.0	40 10	1763.15	5/2 ⁺			
		8819.7	100 10	0.0	3/2 ⁺			
8824.2	1/2 ⁺	5821.0	100 24	3002.64	5/2 ⁺			
		6129.6	16.2 8	2694.00	3/2 ⁺			
		7060.3	43 11	1763.15	5/2 ⁺			
		7603.9	60 16	1219.38	1/2 ⁺			
		8823.0	51 14	0.0	3/2 ⁺			
8829.3	1/2 ⁻	3174	30 15	5655	(3/2 ⁺)			
		3990.0	3.9 21	4839.10	(1/2 ⁺ , 3/2 ⁺)			
		4059.2	4.6 24	4769.86	(5/2 ⁻)			
		4910.5	14 7	3918.47	3/2 ⁺			
		5826.1	67 18	3002.64	5/2 ⁺			
		6134.7	39 9	2694.00	3/2 ⁺			
		6183.0	100 24	2645.67	7/2 ⁺			
		7065.4	27 15	1763.15	5/2 ⁺			
		7609.0	8 4	1219.38	1/2 ⁺			
		8828.1	12 6	0.0	3/2 ⁺			
		3618.4	30 15	5214.5	(5/2 ⁺)			
		4063.0	11 6	4769.86	(5/2 ⁻)			
		4659.3	61 18	4173.43	(5/2 ⁺)			
		4720.4	20 11	4112.4	7/2 ⁺			
8833.1	(5/2, 7/2)	4773.7	48 13	4059.1	(3/2 ⁻)			
		4914.3	48 13	3918.47	3/2 ⁺			
		5669.7	16 8	3162.87	7/2 ⁻			
		5829.9	29 15	3002.64	5/2 ⁺			
		6138.5	20 10	2694.00	3/2 ⁺			
		6186.8	100 26	2645.67	7/2 ⁺			
		7069.2	39 22	1763.15	5/2 ⁺			
		8831.9	13 7	0.0	3/2 ⁺			
		3674.6		5163.34	7/2 ⁻			
		3956.9		4880.96	7/2 ⁽⁺⁾			
		4068.0	13 7	4769.86	(5/2 ⁻)			
		4490.0		4347.76	9/2 ⁻			
		4778.7		4059.1	(3/2 ⁻)			
		5674.7	46 11	3162.87	7/2 ⁻			
8844.4	17/2 ⁺	7074.2	100 10	1763.15	5/2 ⁺			
		524.8		8319.6	15/2 ⁻			
		971.38 20	100.0 30	7873.03	13/2 ⁺	E2	6.16×10 ⁻⁵ 9	B(E2)(W.u.)=16.3 +37-26 E _γ : weighted average of 971.4 5 from (¹⁶ O,αpγ) and 971.38 20 from (¹⁴ N,αpγ).

Adopted Levels, Gammas (continued)

<u>$\gamma(^{35}\text{Cl})$ (continued)</u>						
<u>$E_i(\text{level})$</u>	<u>J_i^π</u>	<u>$E_\gamma^\dagger$</u>	<u>$I_\gamma^\ddagger$</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Comments</u>
8844.4	17/2 ⁺	2757.0		6087.31	13/2 ⁻	
8856.2	5/2 ⁺	3975.0	6.7 33	4880.96	7/2 ⁽⁺⁾	
		4231.6	10 5	4624.3	3/2,5/2	
		4678.0	7 4	4177.9	3/2 ⁻	
		4937.4	22 6	3918.47	3/2 ⁺	
		5692.8	100 10	3162.87	7/2 ⁻	
		6161.6	5.5 28	2694.00	3/2 ⁺	
		6209.9		2645.67	7/2 ⁺	
		7092.3	14 8	1763.15	5/2 ⁺	
		8855.0	31 8	0.0	3/2 ⁺	
8868.6	3/2 ⁺ ,5/2 ⁺	4690.4	5.3 27	4177.9	3/2 ⁻	
		5865.4	33 8	3002.64	5/2 ⁺	
		6174.0	25 6	2694.00	3/2 ⁺	
		7104.7	9 5	1763.15	5/2 ⁺	
		7648.3	33 8	1219.38	1/2 ⁺	
		8867.4	100 25	0.0	3/2 ⁺	
8886.1	(5/2)	3231	23 12	5655	(3/2) ⁺	
		3287.5	25 13	5598.4	3/2 ⁺ ,5/2 ⁺	
		3671.4	9 4	5214.5	(5/2 ⁺)	
		3722.6	35 16	5163.34	7/2 ⁻	
		4116.0	17 9	4769.86	(5/2 ⁻)	
		4707.9	31 16	4177.9	3/2 ⁻	
		4967.3	16 8	3918.47	3/2 ⁺	
		5882.9	31 16	3002.64	5/2 ⁺	
		6239.8	100 27	2645.67	7/2 ⁺	
		7122.2	46 12	1763.15	5/2 ⁺	
		8884.9	54 16	0.0	3/2 ⁺	
8893.3	(5/2 ⁺ ,7/2 ⁺)	3307.1	4.3 22	5586.0	5/2 ⁺	
		4780.6	24 11	4112.4	7/2 ⁺	
		4950.0	11 6	3942.9	9/2 ⁺	
		5729.9	78 19	3162.87	7/2 ⁻	
		6198.7	100 24	2694.00	3/2 ⁺	
		7129.4	51 14	1763.15	5/2 ⁺	
		8892.1	2.7 14	0.0	3/2 ⁺	
8904.8	(1/2,3/2,5/2 ⁺)	2724	15 8	6181	(1/2 ⁺ to 7/2 ⁺)	
		3250	12 6	5655	(3/2) ⁺	
		3895.5	13 7	5009.1	(1/2,3/2)	
		4050.2	21 11	4854.4	(1/2,3/2)	
		4065.5	16 8	4839.10	(1/2 ⁺ ,3/2)	
		4726.6	62 17	4177.9	3/2 ⁻	
		4845.3	20 10	4059.1	(3/2) ⁻	
		6210.2	52 14	2694.00	3/2 ⁺	

Adopted Levels, Gammas (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	$\gamma(^{35}\text{Cl})$ (continued)		Comments
						Mult. #	$\delta^\#$	
8904.8	(1/2,3/2,5/2 ⁺)	7684.5	35 10	1219.38	1/2 ⁺			
		8903.6	100 24	0.0	3/2 ⁺			
8906.8	5/2 ⁺	3252		5655	(3/2) ⁺			
		3261	5.4 28	5646	(5/2,7/2,9/2 ⁺)			
		3308.2	3.2 16	5598.4	3/2 ⁺ ,5/2 ⁺			
		3320.6	5.8	5586.0	5/2 ⁺			
		3376	3.6 19	5531				
		3692.1	6.4 32	5214.5	(5/2 ⁺)			
		4136.7	8 4	4769.86	(5/2 ⁻)			
		4963.5	19 5	3942.9	9/2 ⁺			
		7142.9	100 10	1763.15	5/2 ⁺			
8919.8	(5/2 ⁻ ,7/2,9/2 ⁺)	3274	10 5	5646	(5/2,7/2,9/2 ⁺)			
		4571.7	14 7	4347.76	9/2 ⁻			
		5756.4	100 11	3162.87	7/2 ⁻			
		6273.5	46 12	2645.67	7/2 ⁺			
		7155.9	7 4	1763.15	5/2 ⁺			
8953.1	3/2 ⁺	7732.8		1219.38	1/2 ⁺	M1+E2	-0.34 2	Mult., δ : or $\delta=-5.0$ 4 from $\gamma(\theta)$ in (p, γ). E1+M2 ruled out by RUL.
		8951.9		0.0	3/2 ⁺	M1+E2	-0.39 5	Mult., δ : or $\delta=-8.2$ 7 from $\gamma(\theta)$ in (p, γ). E1+M2 ruled out by RUL.
8984.2	5/2 ⁺	3260.5	1.3 7	5723.52	5/2 ⁺			
		4144.8	2.0 10	4839.10	(1/2 ⁺ ,3/2)			
		4214.1	0.8 5	4769.86	(5/2 ⁻)			
		4359.6	0.8 5	4624.3	3/2,5/2			
		4806.0	2.5 13	4177.9	3/2 ⁻			
		4810.4	2.5 13	4173.43	(5/2 ⁺)			
		5065.3	15 8	3918.47	3/2 ⁺			
		5820.8	10 5	3162.87	7/2 ⁻			
		6289.6	3.0 15	2694.00	3/2 ⁺			
		6337.9	2.0 10	2645.67	7/2 ⁺			
		7220.3	27 7	1763.15	5/2 ⁺			
		8983.0	100 10	0.0	3/2 ⁺			
9081.5	5/2 ⁺	3357.8	1.7	5723.52	5/2 ⁺			
		4200.3	1.7	4880.96	7/2 ⁽⁺⁾			
		4242.1		4839.10	(1/2 ⁺ ,3/2)			
		4311.4		4769.86	(5/2 ⁻)			
		4456.9		4624.3	3/2,5/2			
		4903.2	3.3	4177.9	3/2 ⁻			
		4907.7	3.3	4173.43	(5/2 ⁺)			
		5162.6	15	3918.47	3/2 ⁺	D(+Q)	-0.002 9	
		5918.1	10	3162.87	7/2 ⁻			
		6386.9	3.3	2694.00	3/2 ⁺			
		6435.2	1.7	2645.67	7/2 ⁺			

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)									
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	$\delta^\#$	α^a	Comments
9081.5	5/2 ⁺	7317.5	27	1763.15	5/2 ⁺	M1+E2	+0.11 2		B(M1)(W.u.)=1.16 +31-25; B(E2)(W.u.)=0.99 +50-38
		9080.2	100	0.0	3/2 ⁺	M1+E2	+0.089 9		B(M1)(W.u.)=2.26 +49-33; B(E2)(W.u.)=0.82 +27-19
9157.1	5/2 ⁻	3942.4	17 5	5214.5	(5/2 ⁺)				
		4386.9	16 4	4769.86	(5/2 ⁻)				
		4978.8	<37	4177.9	3/2 ⁻				I $_\gamma$: total I $_\gamma$ =30 7 for γ decays to 4173 and 4178 doublet final levels (1976Sp08).
		4983.3	<37	4173.43	(5/2 ⁺)				I $_\gamma$: total I $_\gamma$ =30 7 for γ decays to 4173 and 4178 doublet final levels (1976Sp08).
		5993.7	22 6	3162.87	7/2 ⁻	M1+E2	+0.10 2	1.59×10 ⁻³ 2	
		6153.9	20 5	3002.64	5/2 ⁺				
		6462.5	3 2	2694.00	3/2 ⁺				
		6510.8	7 4	2645.67	7/2 ⁺				
		7393.1	17 5	1763.15	5/2 ⁺				
		9155.8	100 9	0.0	3/2 ⁺	(E1)			Mult.: D+Q with δ =+0.09 1 from $\gamma(\theta)$ in (p, γ); $\Delta\pi$ =yes from level scheme; M2 component is less likely.
10181.1	19/2 ⁻	1336.3 5	100.0 34	8844.4	17/2 ⁺	(E1)		0.0001637 23	B(E1)(W.u.)=0.00235 +21-18
									I $_\gamma$: weighted average of 100 17 from ($^{28}\text{Si},\alpha p\gamma$) and 100.0 34 from ($^{16}\text{O},\alpha p\gamma$).
									E $_\gamma$,Mult.: from ($^{16}\text{O},\alpha p\gamma$), with Mult=D from $\gamma\gamma$ (ADO) with ΔJ =1; $\Delta\pi$ =(yes) from level scheme.
		1693.5 5	84 9	8487.5	15/2 ⁻	E2		0.0001880 26	B(E2)(W.u.)=44.4 46
									E $_\gamma$,I $_\gamma$,Mult.: from ($^{16}\text{O},\alpha p\gamma$), with Mult=Q, ΔJ =2 from $\gamma\gamma$ (ADO) and M2 ruled out by RUL.
		1861.3 5	83 6	8319.6	15/2 ⁻	E2		0.000262 4	B(E2)(W.u.)=27.4 +26-24
									E $_\gamma$,I $_\gamma$,Mult.: from ($^{16}\text{O},\alpha p\gamma$), with Mult=Q, ΔJ =2 from $\gamma\gamma$ (ADO) and M2 ruled out by RUL.
10221.9	(17/2 ⁻)	1377.8 10	100	8844.4	17/2 ⁺				
10858.5	19/2 ⁺	2014.7 9	100 5	8844.4	17/2 ⁺	(M1)		0.000265 4	E $_\gamma$,I $_\gamma$,Mult.: from ($^{16}\text{O},\alpha p\gamma$), with Mult=D from $\gamma\gamma$ (ADO) with ΔJ =1; $\Delta\pi$ =no from level scheme.
		2069.8 10	32.5 22	8788.3	(15/2 ⁺)	E2		0.000361 5	E $_\gamma$,I $_\gamma$,Mult.: from ($^{16}\text{O},\alpha p\gamma$) with Mult=Q, ΔJ =2 from $\gamma\gamma$ (ADO); M2 ruled out by RUL.
11459.1	21/2 ⁺	2614.5 5	100	8844.4	17/2 ⁺	E2		0.000619 9	B(E2)(W.u.)=15.9 8
									E $_\gamma$,Mult.: from ($^{16}\text{O},\alpha p\gamma$), with Mult=Q, ΔJ =2 from $\gamma\gamma$ (ADO); M2 ruled out by RUL.
12572.2	23/2 ⁻	1113.3 5	24.4 18	11459.1	21/2 ⁺	(E1)		3.60×10 ⁻⁵ 5	B(E1)(W.u.)=0.00272 +41-35
									E $_\gamma$,I $_\gamma$,Mult.: from ($^{16}\text{O},\alpha p\gamma$), with Mult=D from $\gamma\gamma$ (ADO) with ΔJ =1; $\Delta\pi$ =yes from level scheme.

Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Cl})$ (continued)

$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\ddagger$	E_f	J_f^π	Mult. [#]	α^a	Comments
12572.2	23/2 ⁻	2390.8 5	100 4	10181.1	19/2 ⁻	E2	0.000515 7	B(E2)(W.u.)=26.0 +36-29 E _γ , I _γ , Mult.: from (¹⁶ O, αpγ), with Mult=Q, ΔJ=2 from γγ(ADO); M2 ruled out by RUL.
16306.4	(27/2 ⁻)	3734.0		12572.2	23/2 ⁻			

[†] Sources of values with uncertainties are listed under comments and those without uncertainties are deduced from level-energy differences, unless otherwise noted.

[‡] From (p,γ), unless otherwise noted.

[#] From γ(θ) and/or γγ(θ) in (p,γ) for γ rays originating from low-spin levels (J<9/2) and from γγ(θ)(DCO) and γ(lin pol) in (²⁸Si, αpγ) for γ rays originating from high-spin (J≥9/2) levels, unless otherwise noted. Magnetic/electric characters are determined based on γ(lin pol) or RUL with measured T_{1/2}, unless otherwise noted. For primary γ transitions from proton resonances where widths or resonance strengths are provided in (p,γ), a D+Q with definite δ(Q/D)>0 from γ(θ) is firmly assigned as M1+E2. This assignment is due to the fact that for these resonances, the proton width Γ_p is dominant over Γ_γ and thus the total width Γ is much greater than Γ_γ. The total width Γ typically falls within the same scale as the resonance strength, usually a few eV in the ³⁴S(p,γ) dataset.

@ Primary transitions probably from the unresolved 7501+7503 doublet initial levels in 1976Me12 in (p,γ).

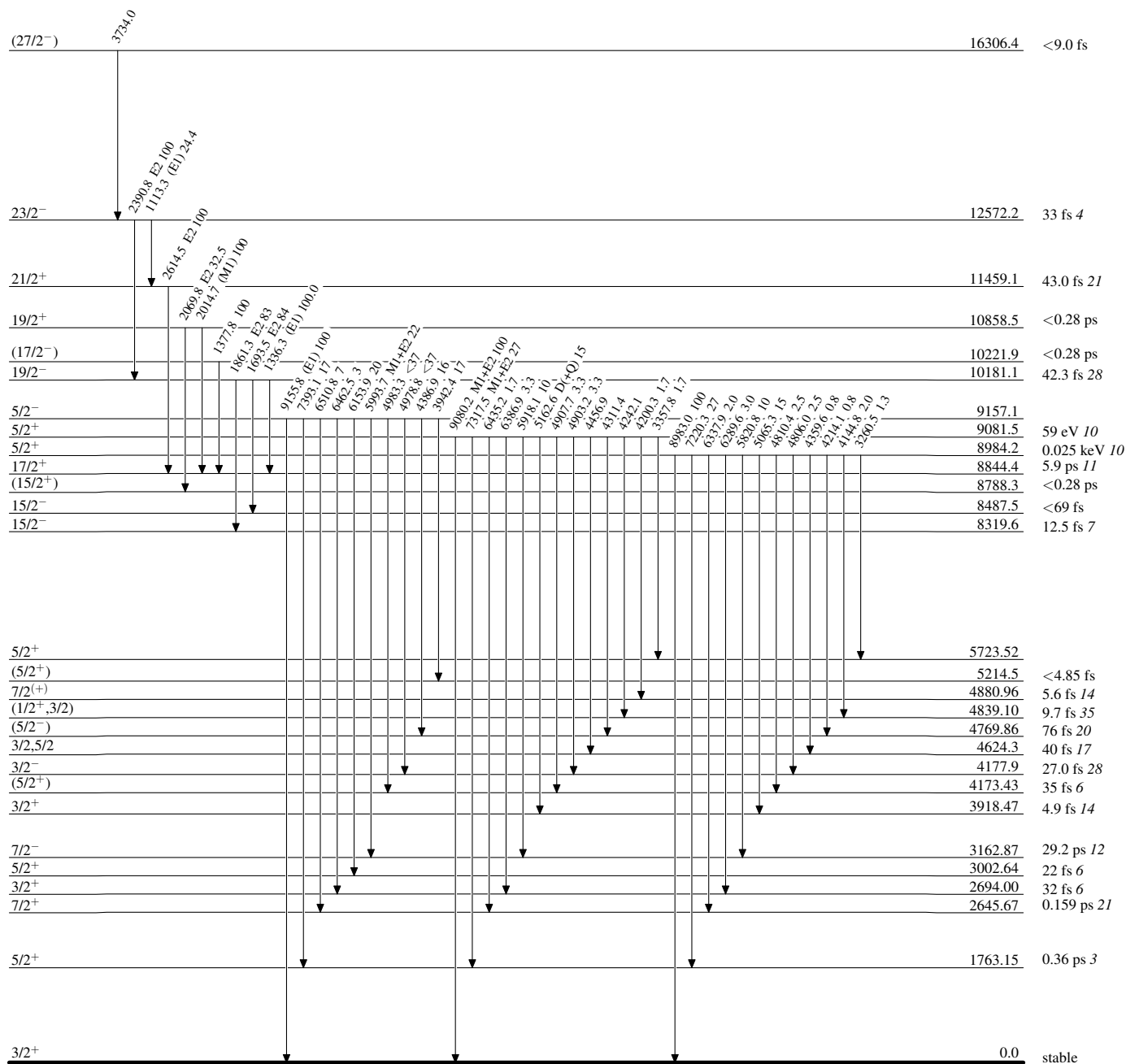
& Primary transitions probably from the unresolved 7837+7839 doublet initial levels in 1976Me12 in (p,γ).

^a Total theoretical internal conversion coefficients, calculated using the BrIcc code (2008Ki07) with "Frozen Orbitals" approximation based on γ-ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

^b Placement of transition in the level scheme is uncertain.

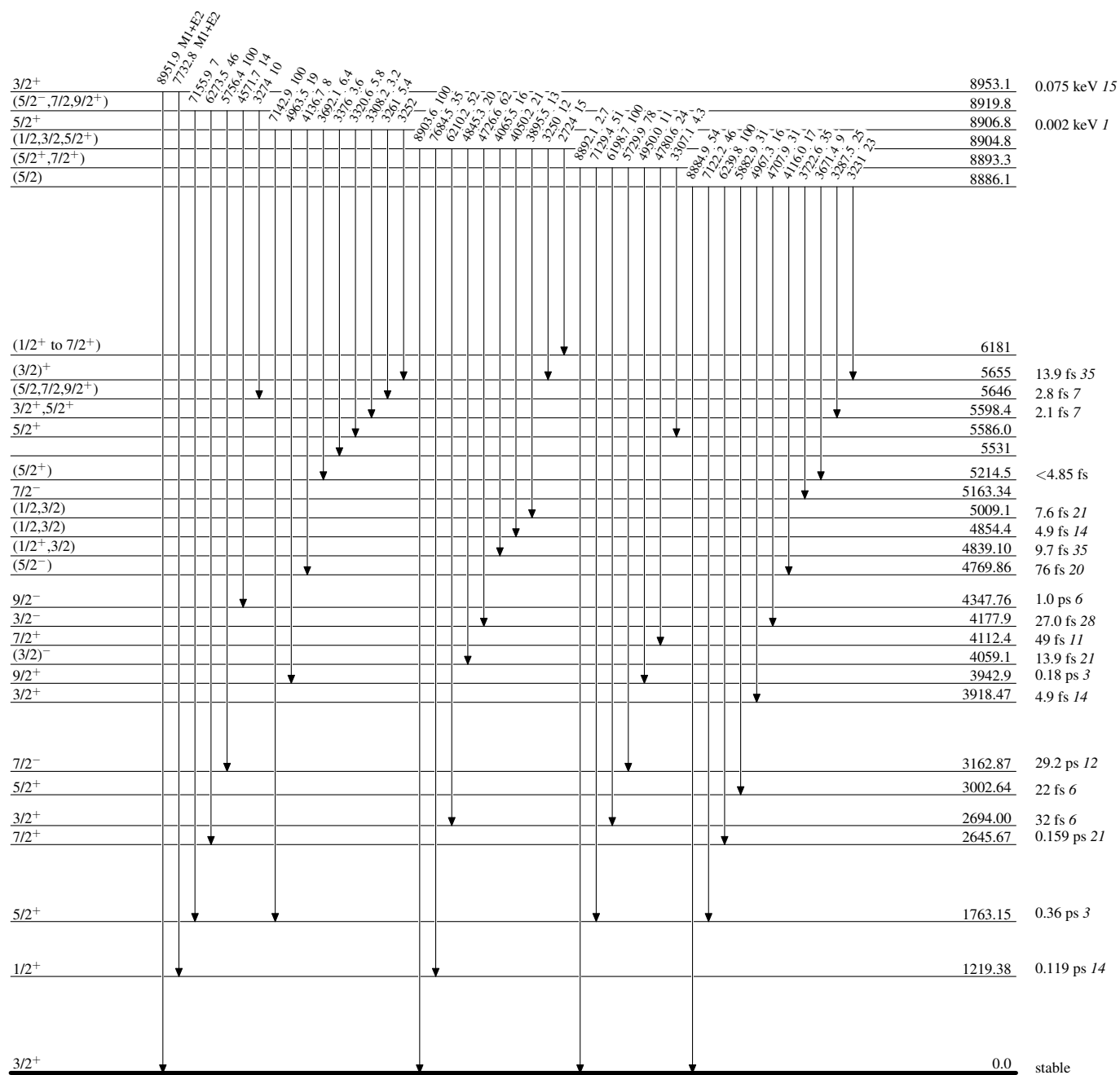
Adopted Levels, Gammas**Level Scheme**

Intensities: Relative photon branching from each level



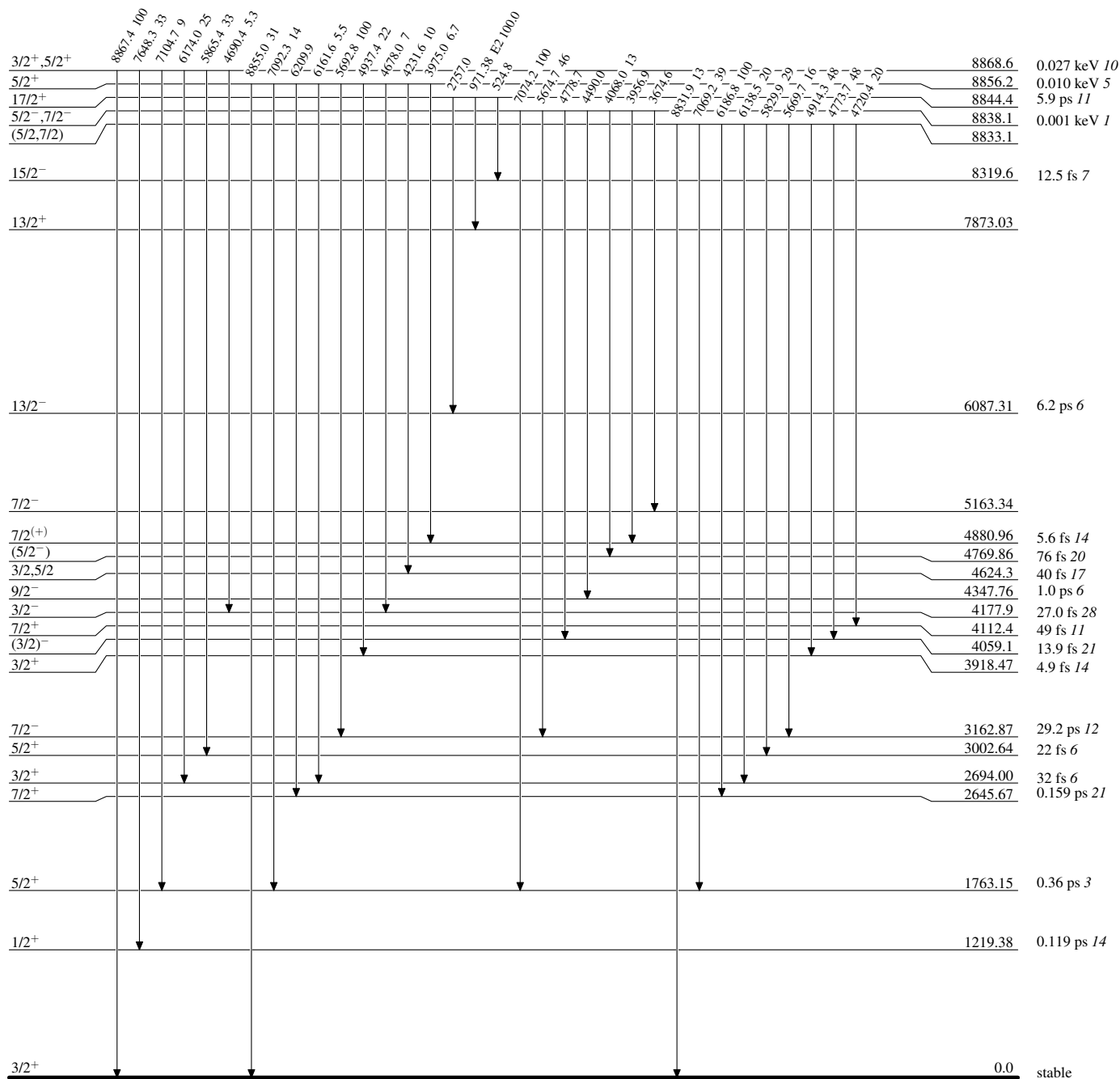
Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

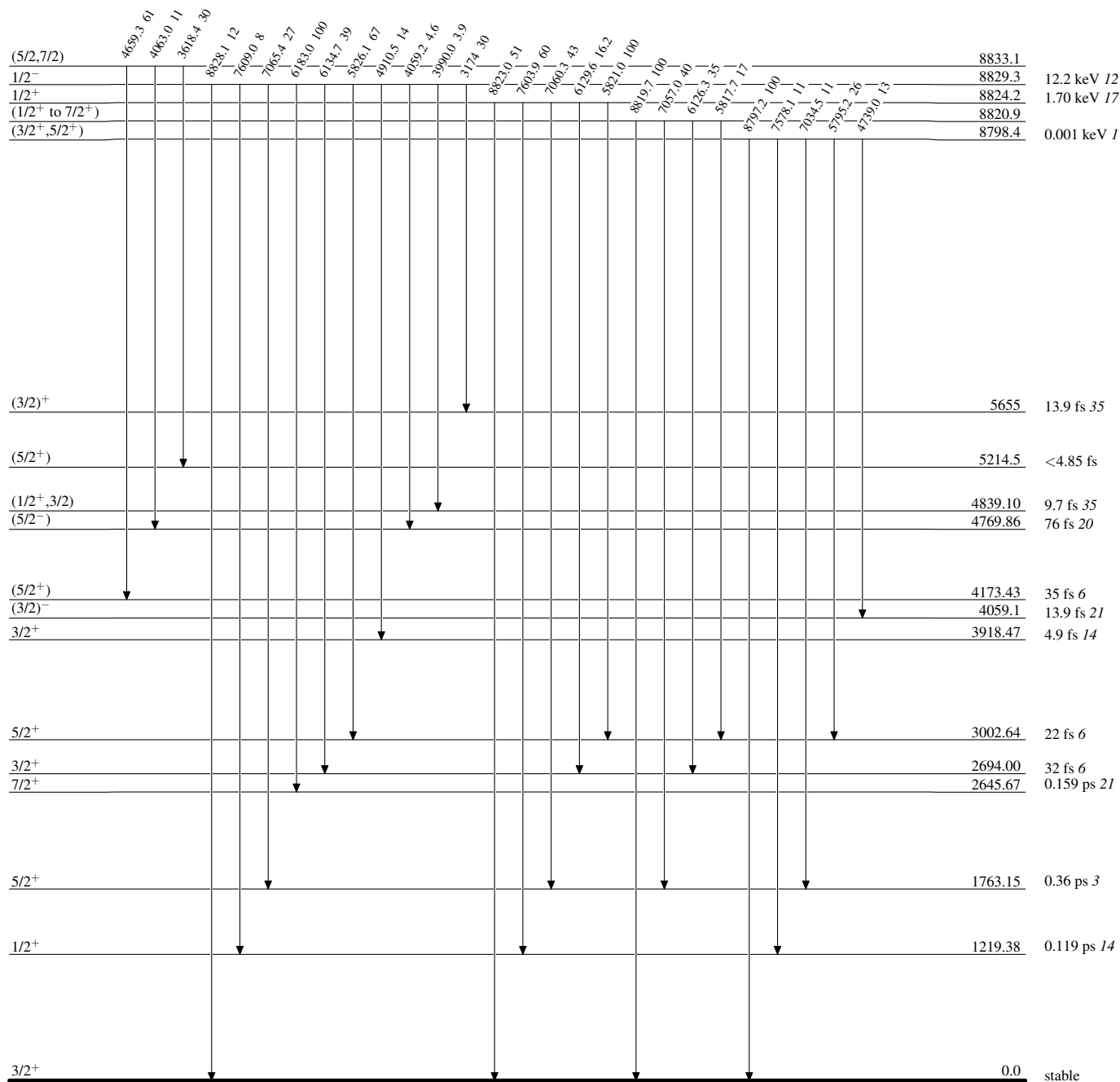
Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level



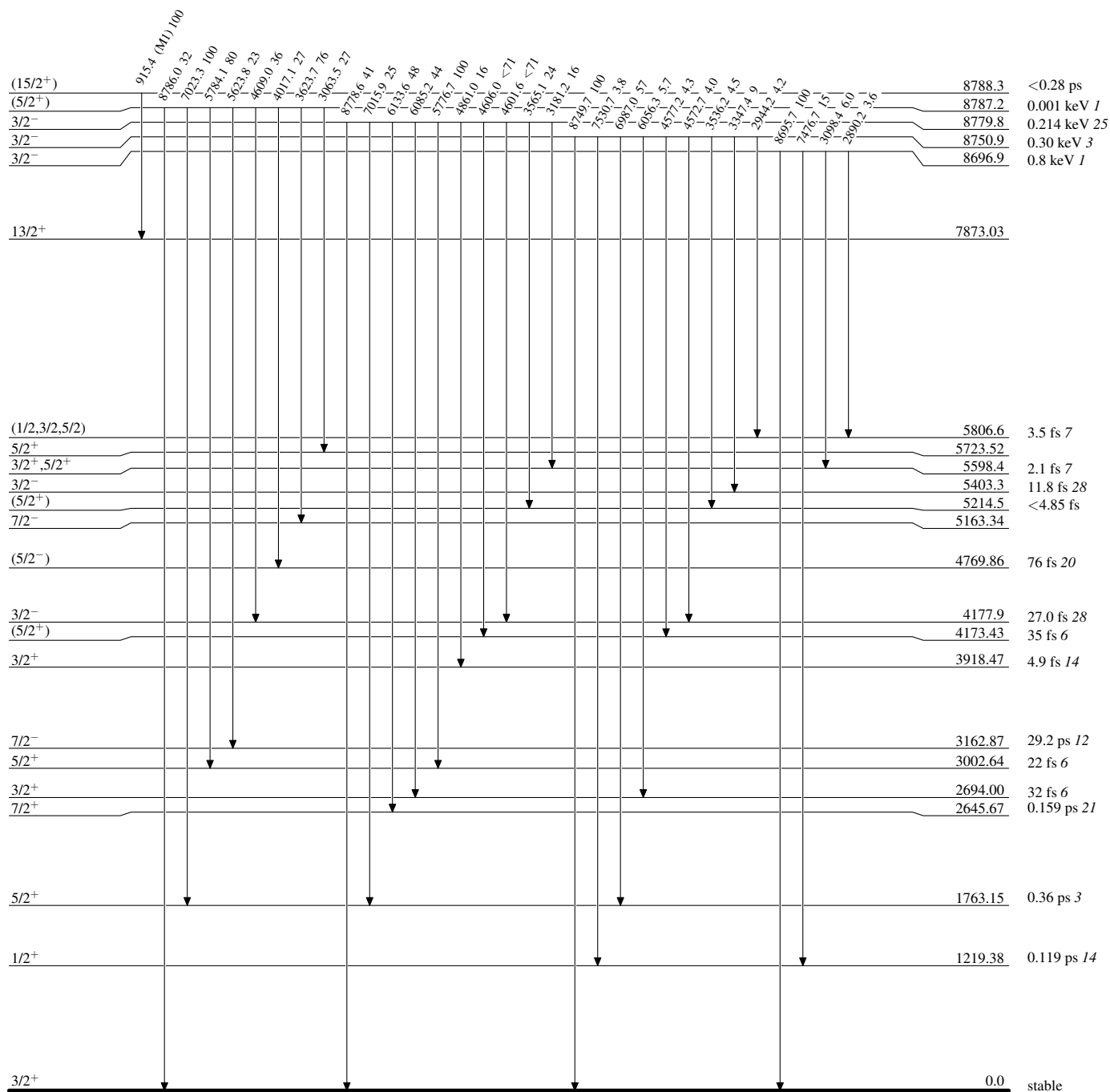
Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level



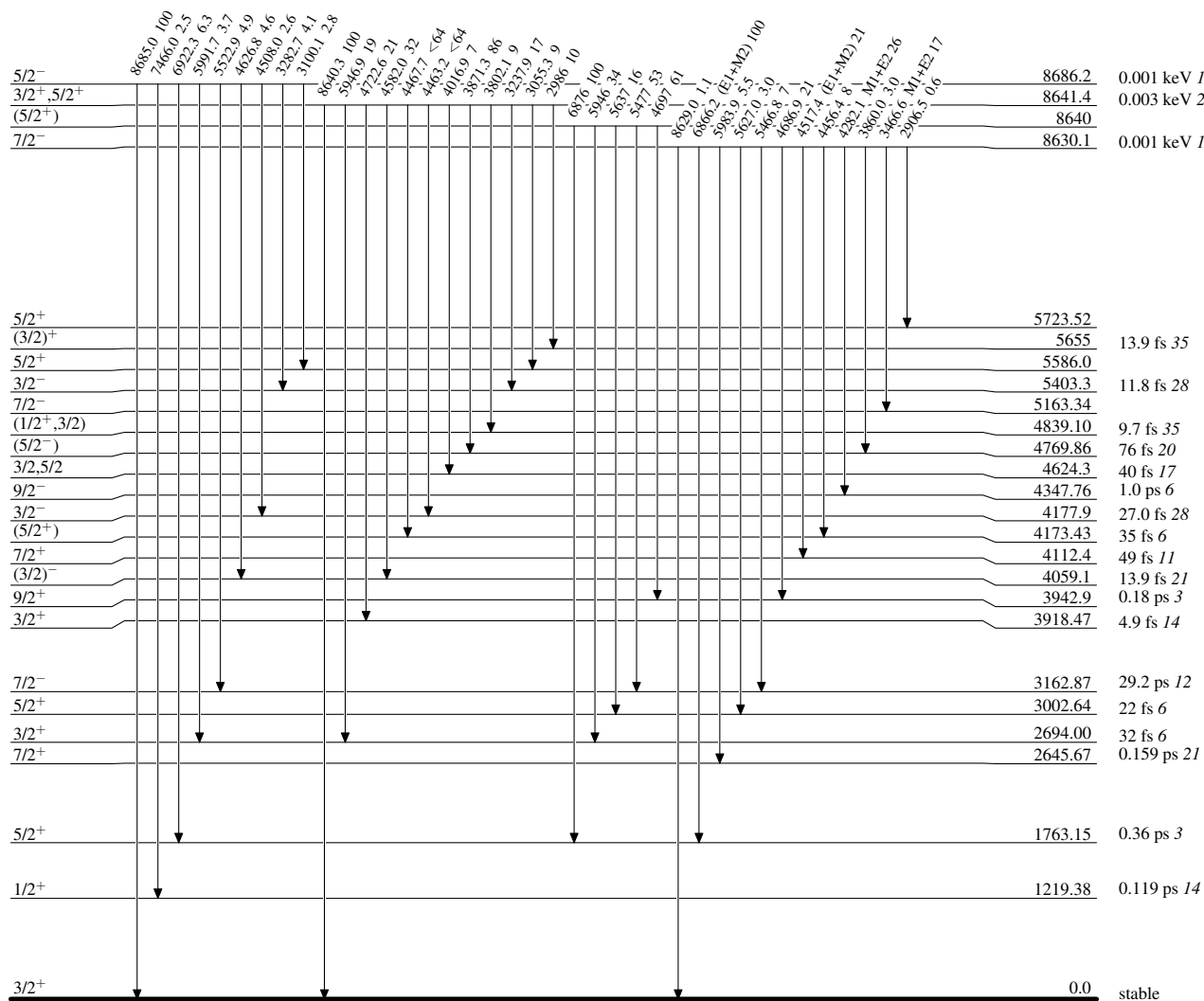
Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

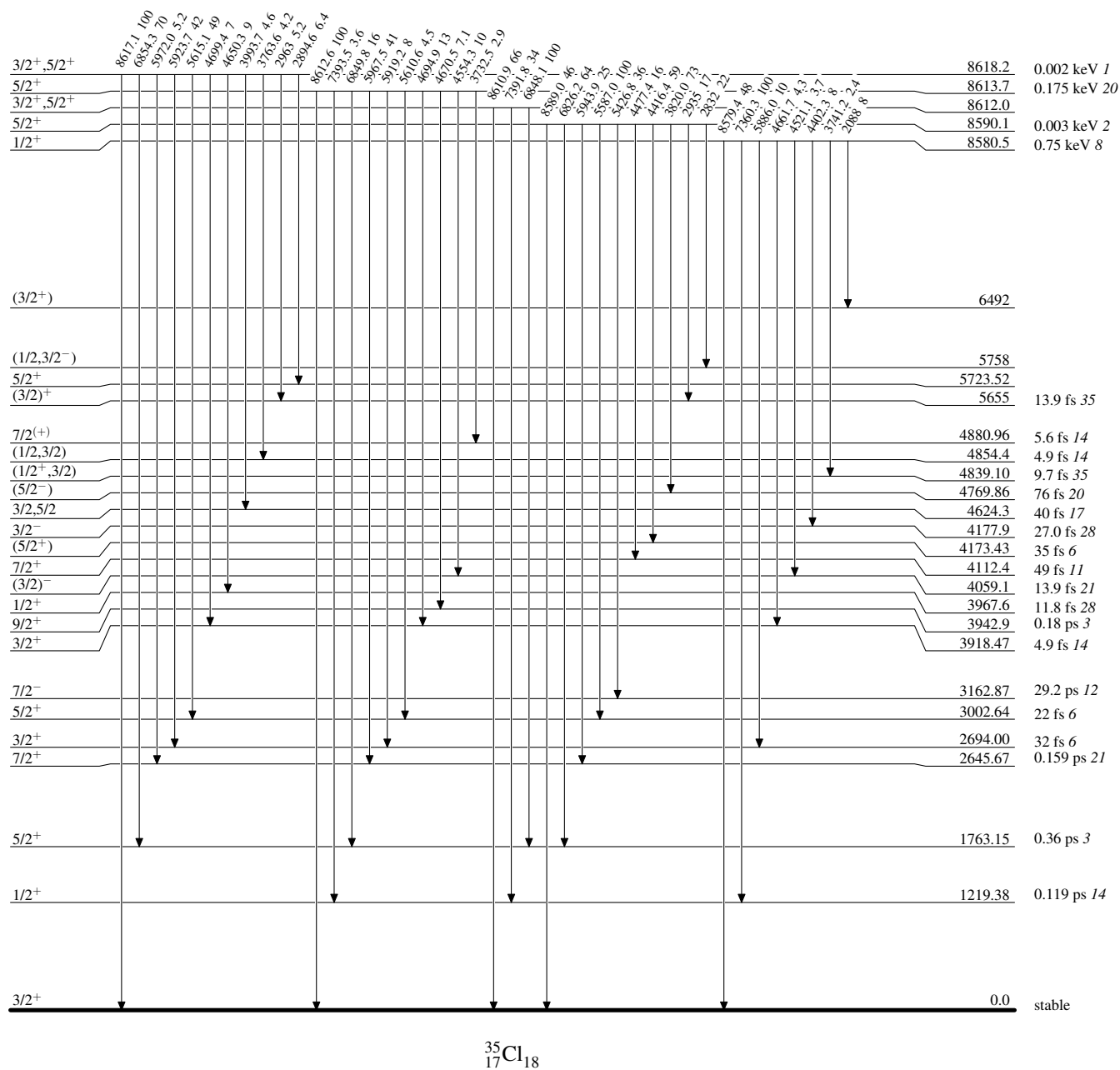
Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level

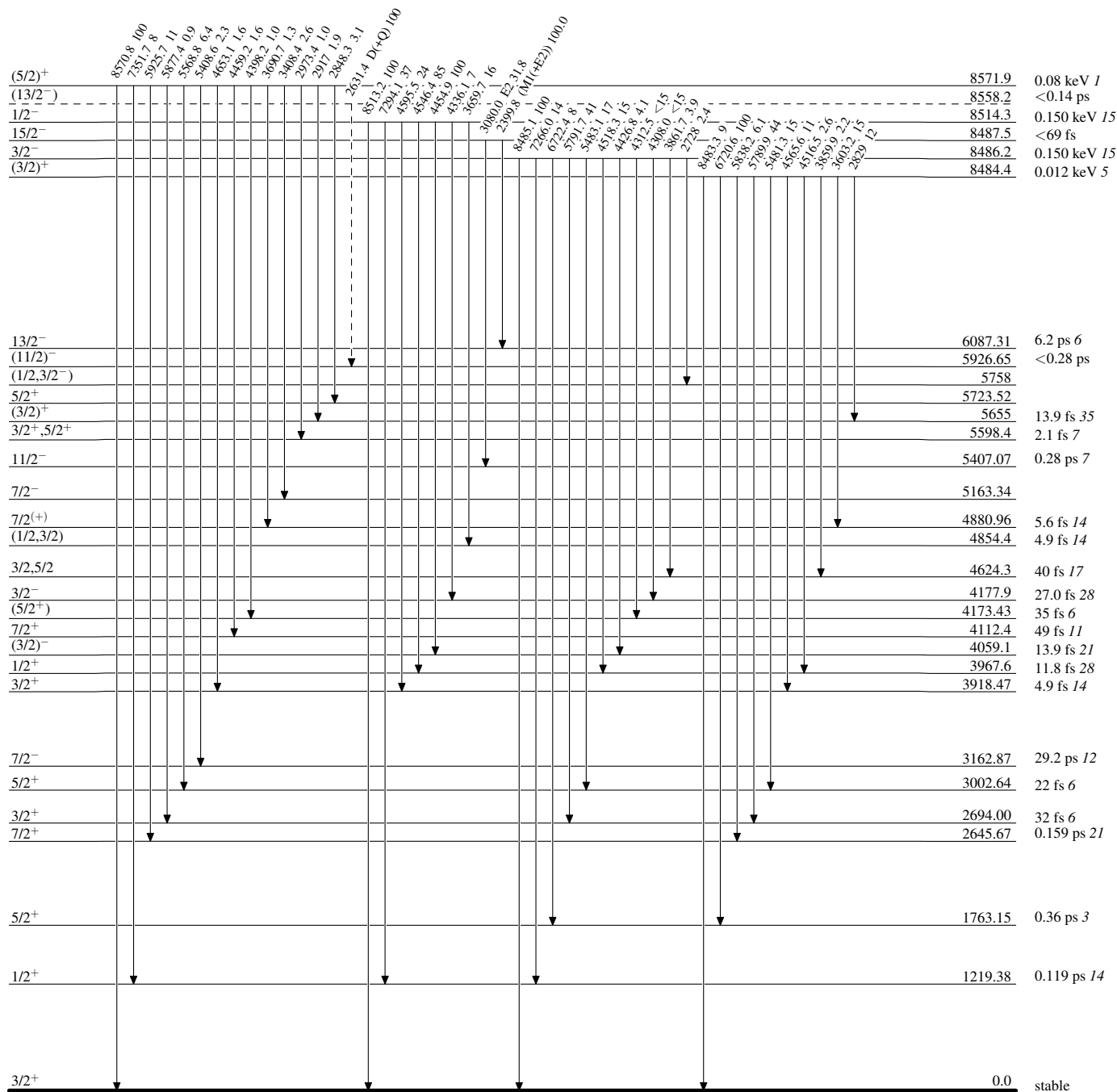


Adopted Levels, Gammas

Legend

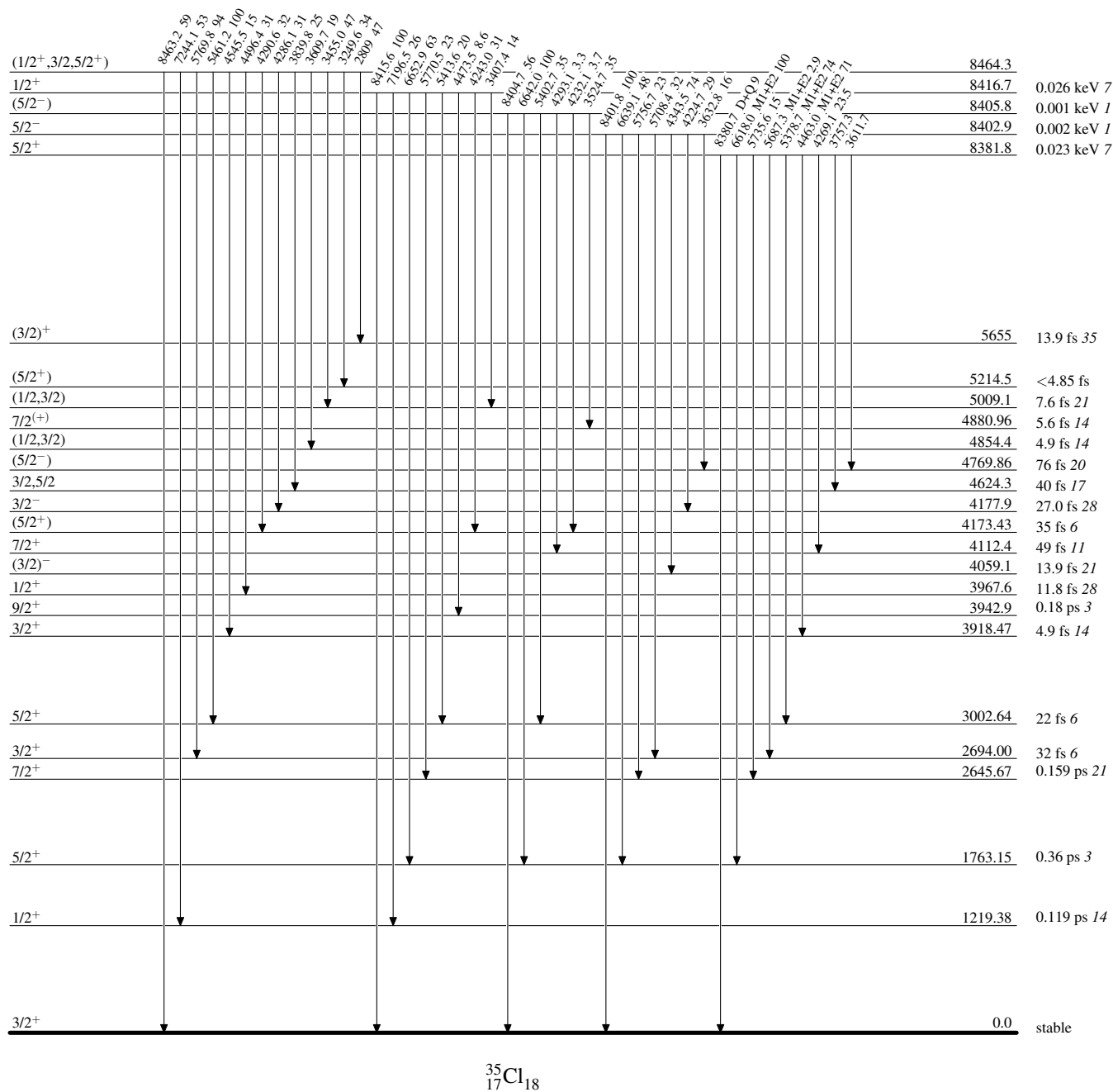
Level Scheme (continued)

Intensities: Relative photon branching from each level

-----► γ Decay (Uncertain) $^{35}_{17}\text{Cl}_{18}$

Adopted Levels, GammasLevel Scheme (continued)

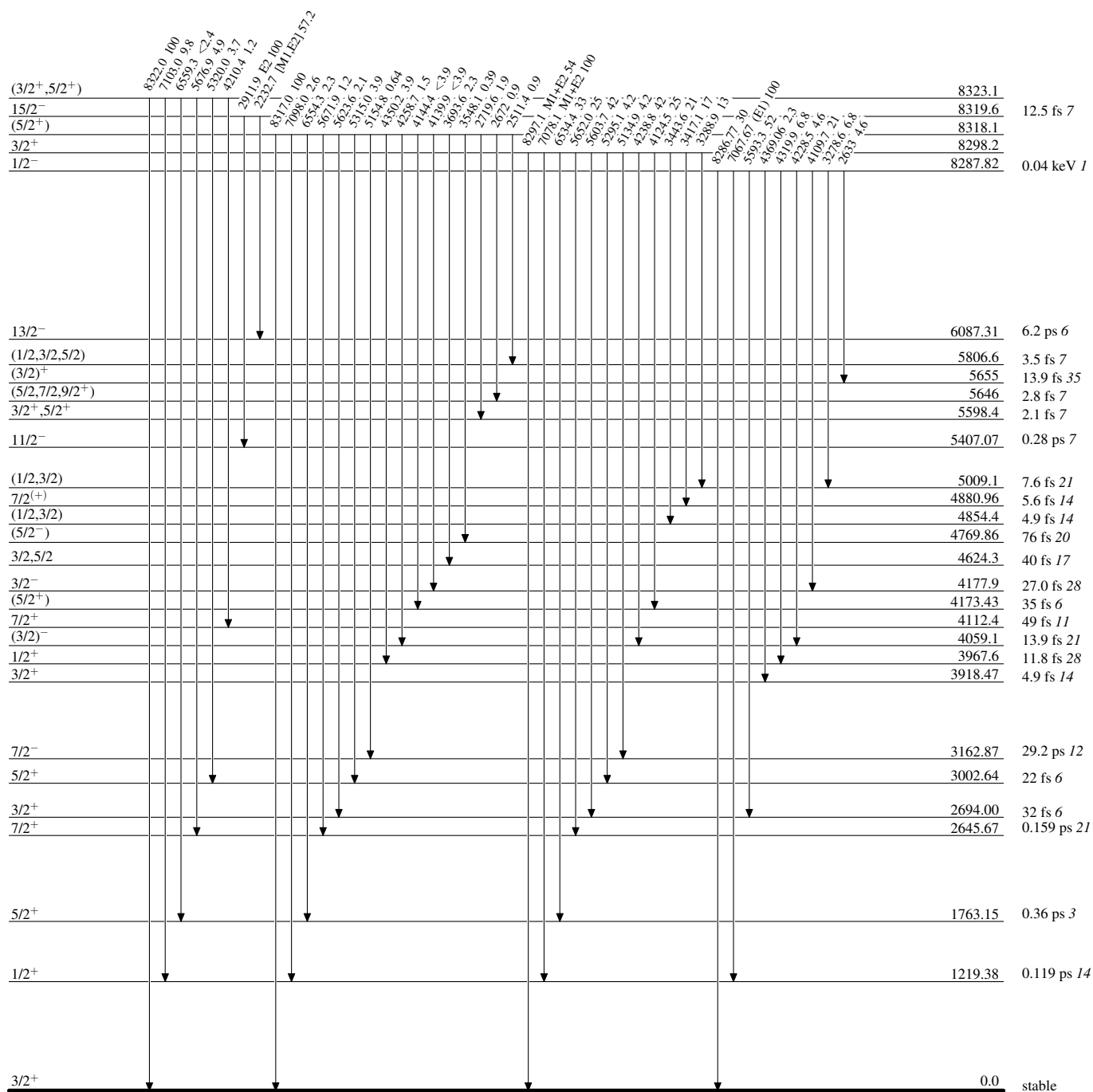
Intensities: Relative photon branching from each level



Adopted Levels, Gammas

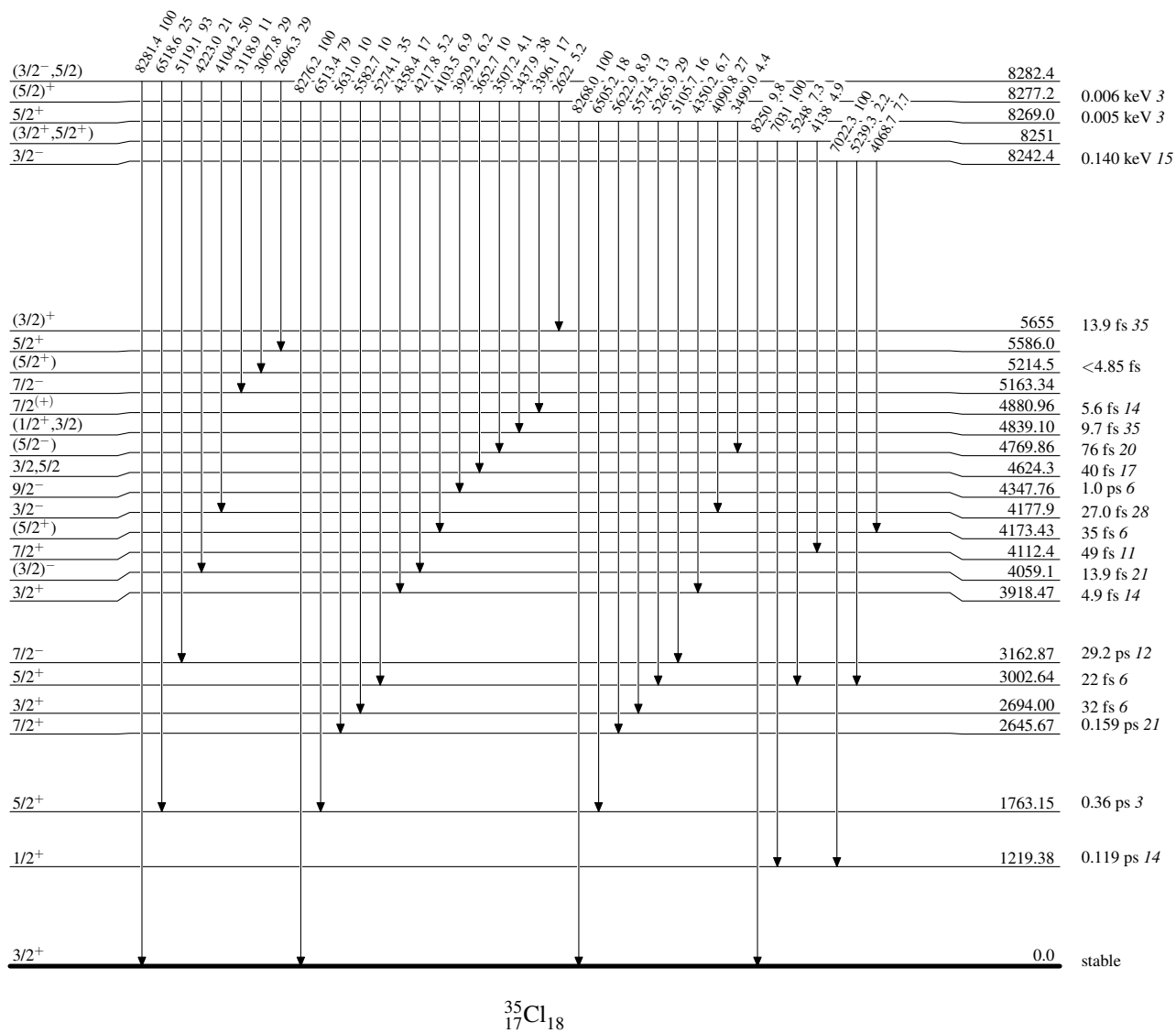
Level Scheme (continued)

Intensities: Relative photon branching from each level



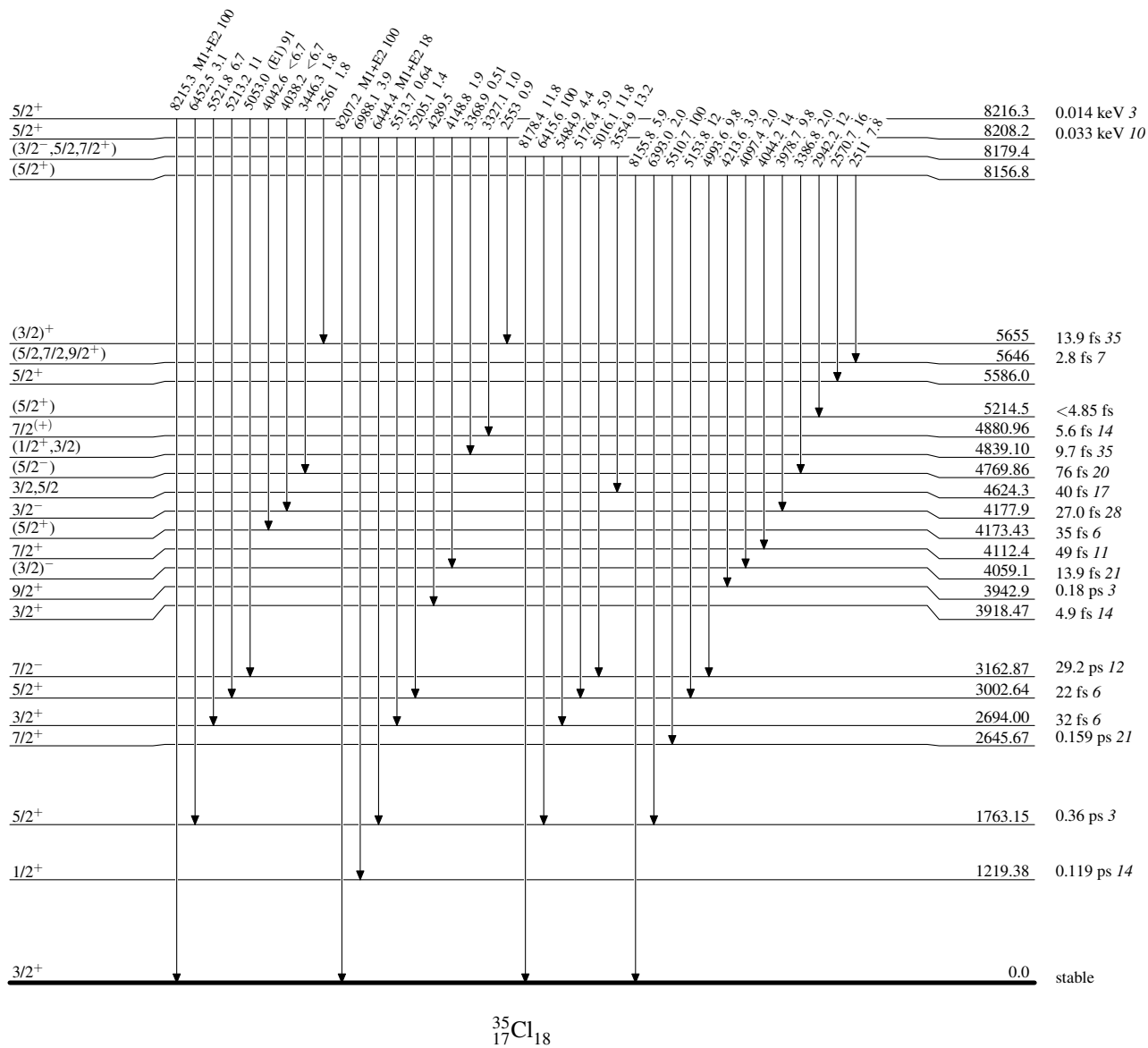
Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

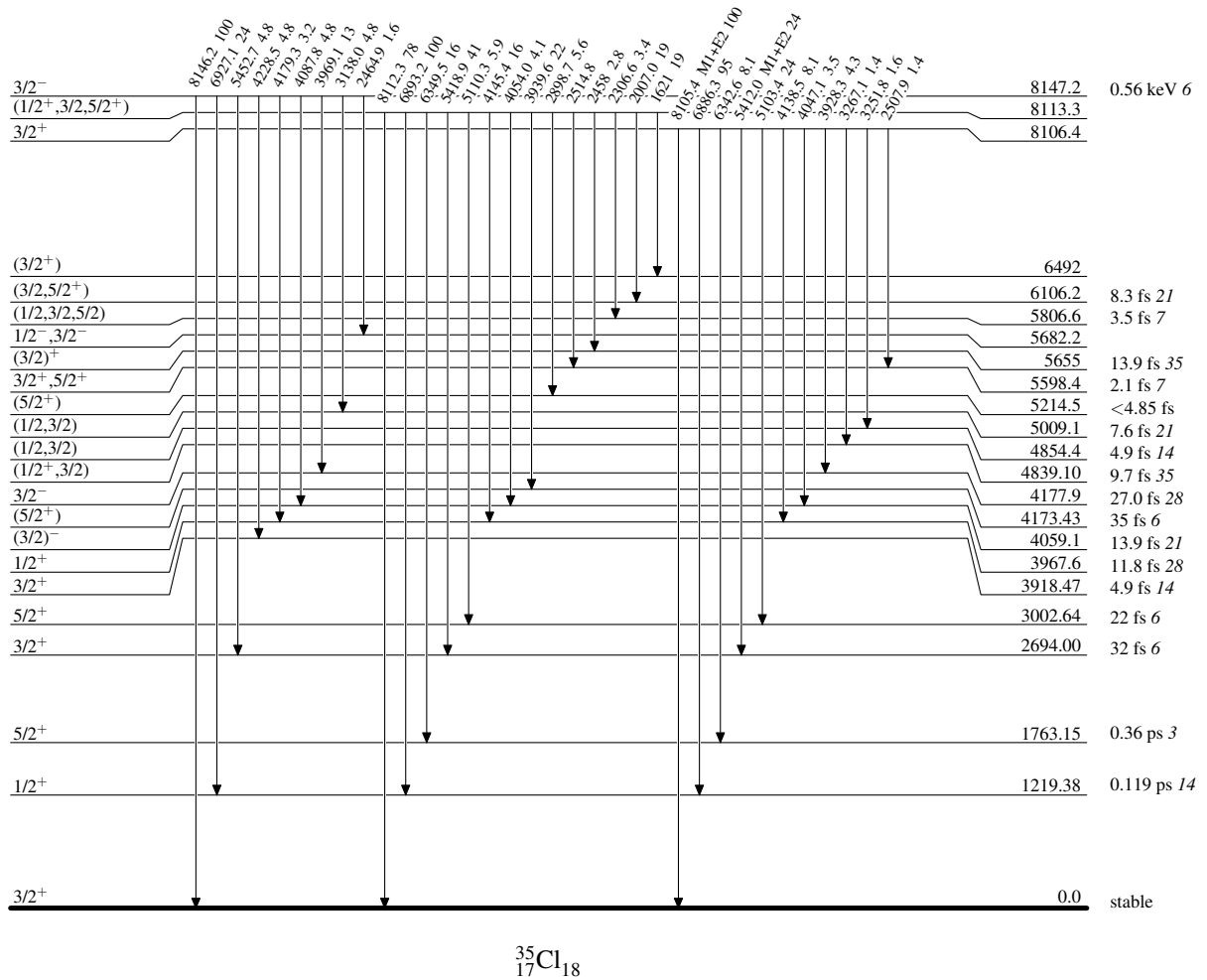
Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level



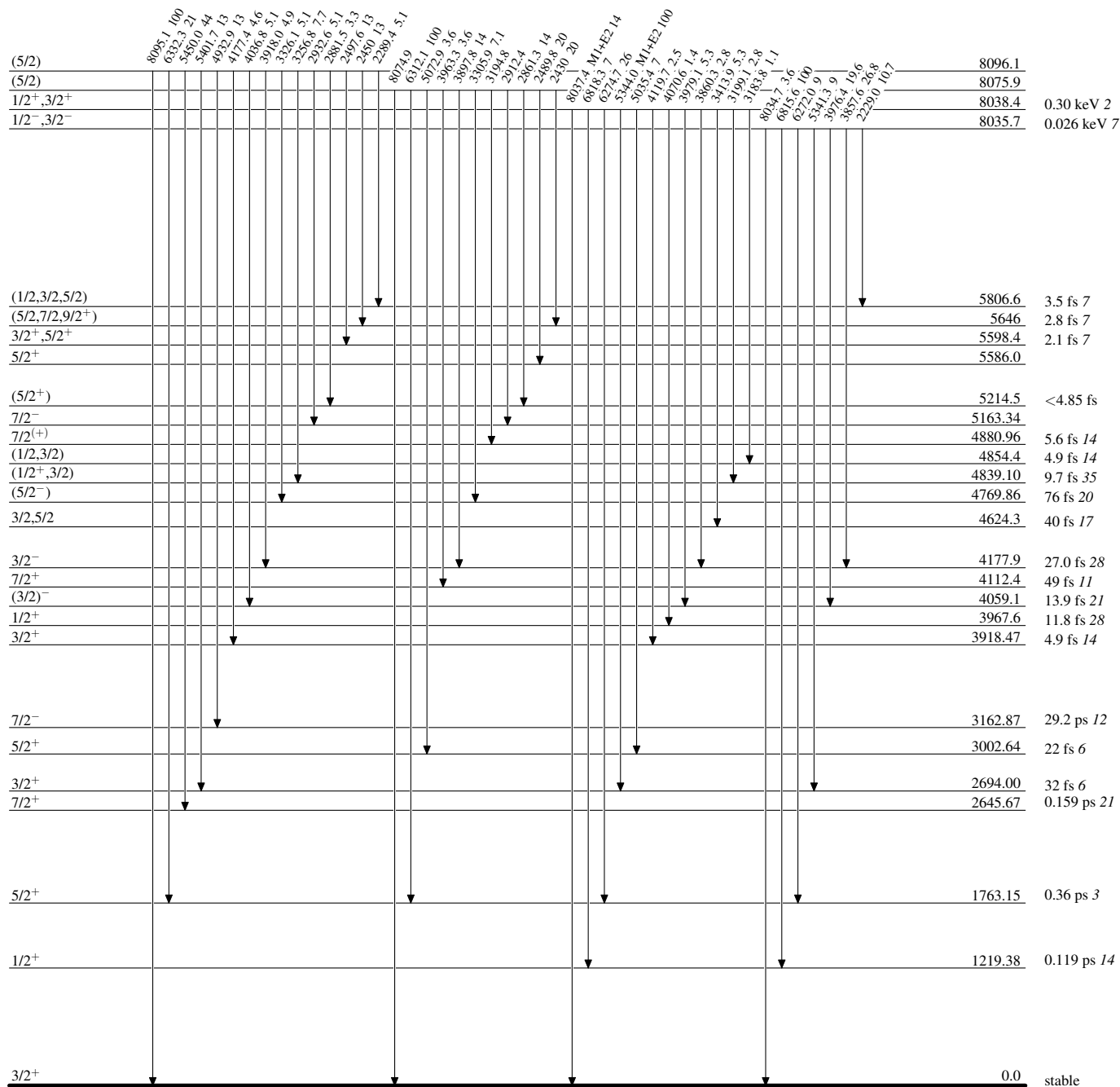
Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level



Adopted Levels, Gammas**Level Scheme (continued)**

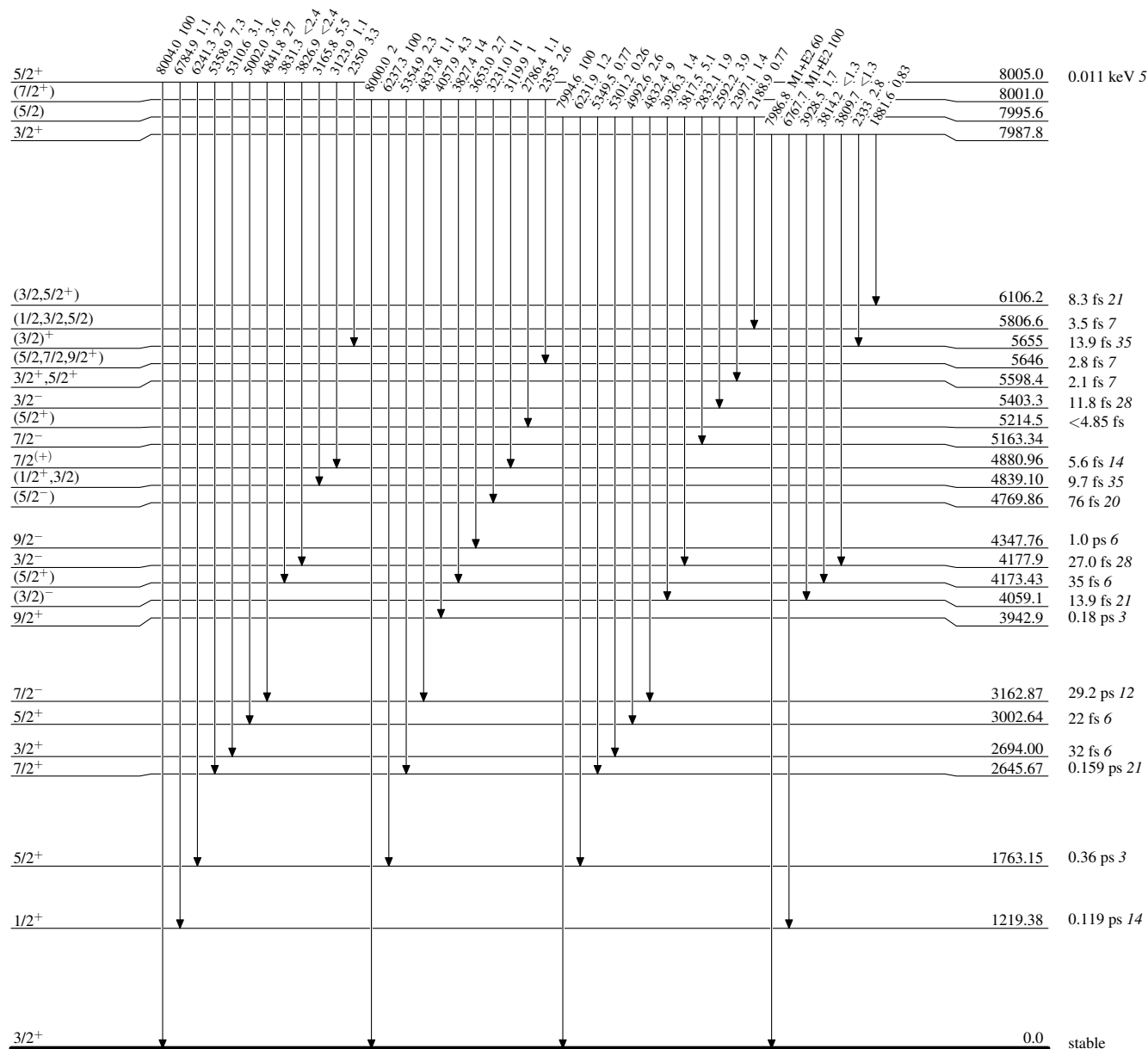
Intensities: Relative photon branching from each level



Adopted Levels, Gammas

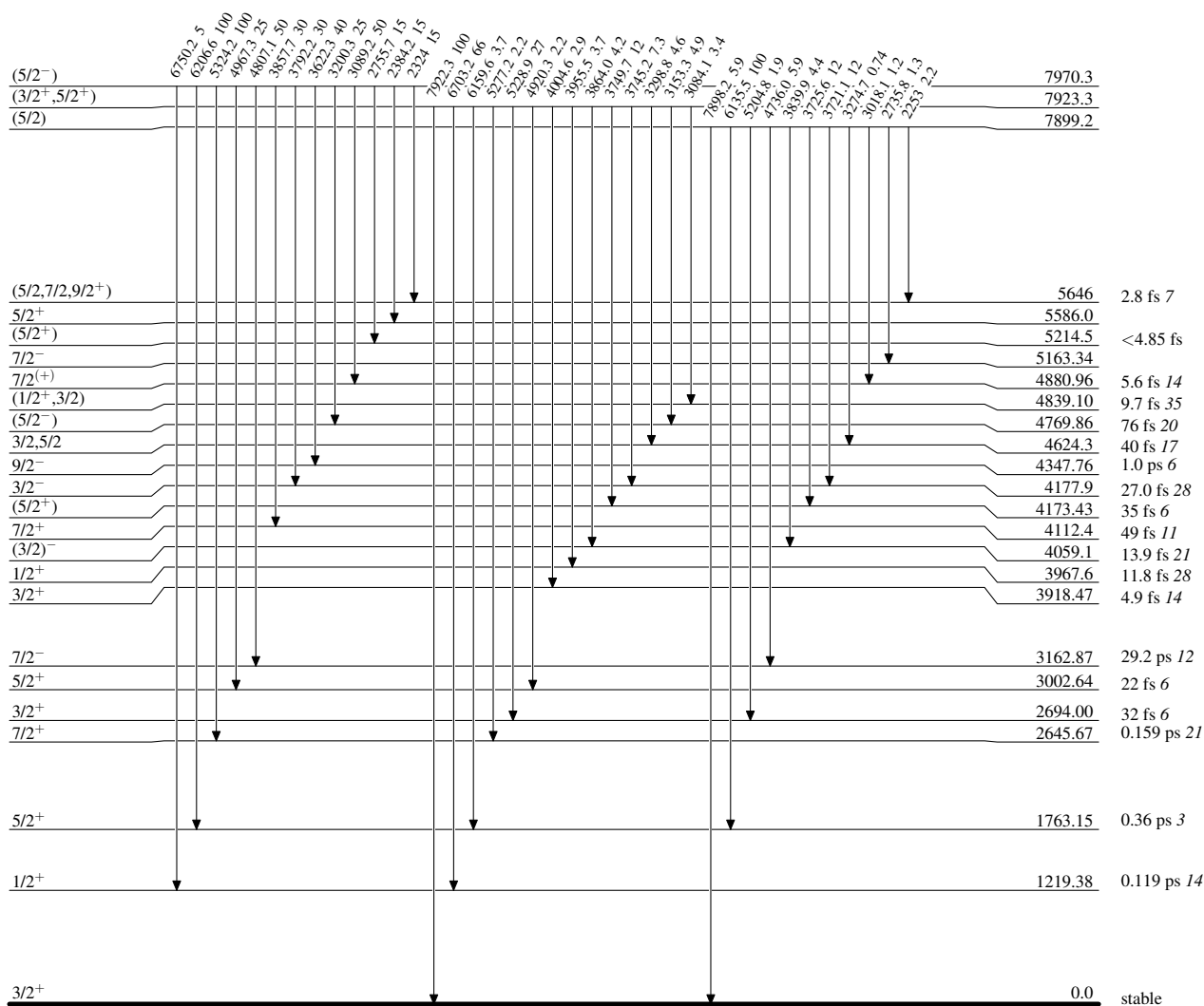
Level Scheme (continued)

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

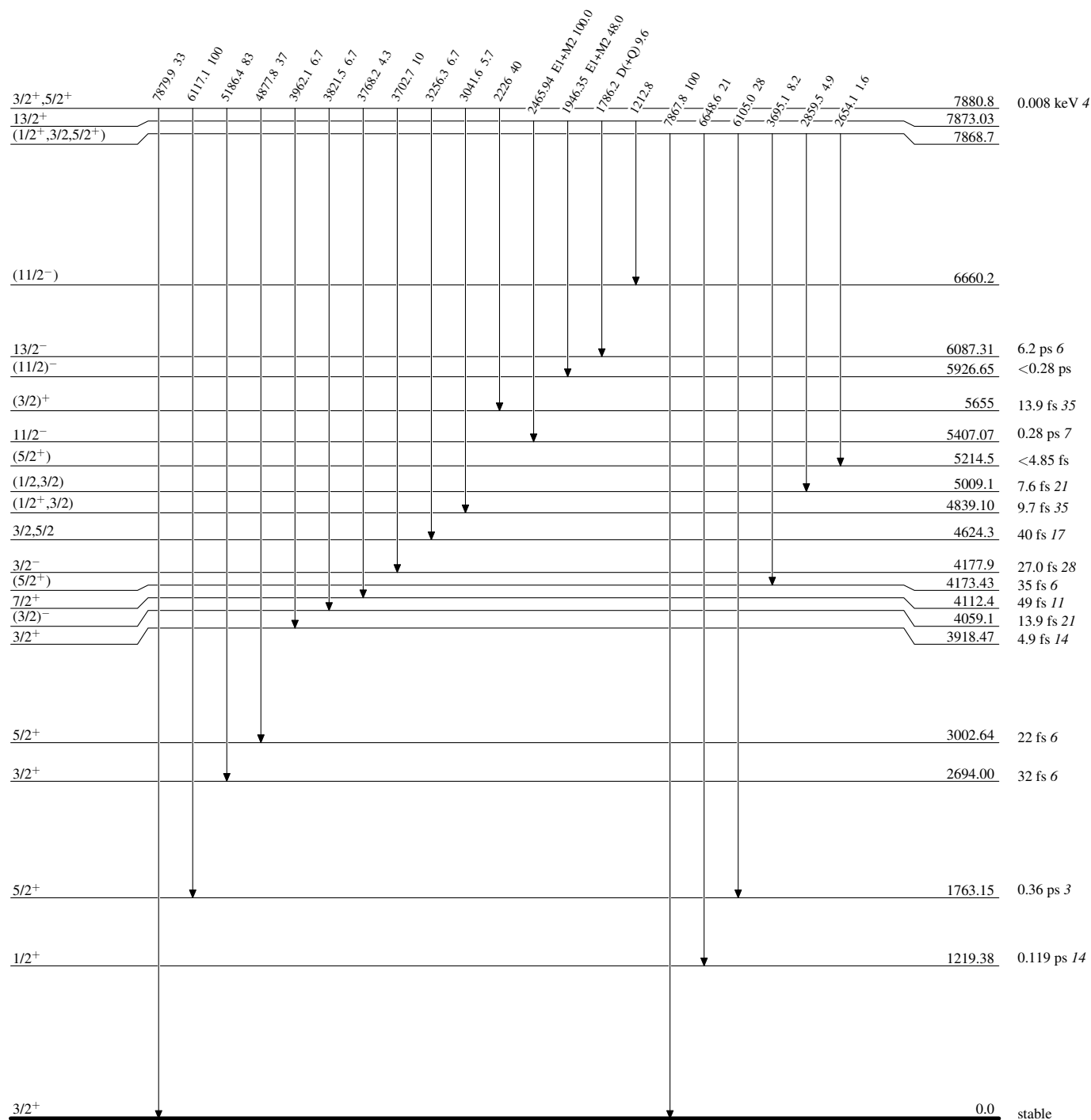
Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

Adopted Levels, Gammas
Level Scheme (continued)

Intensities: Relative photon branching from each level

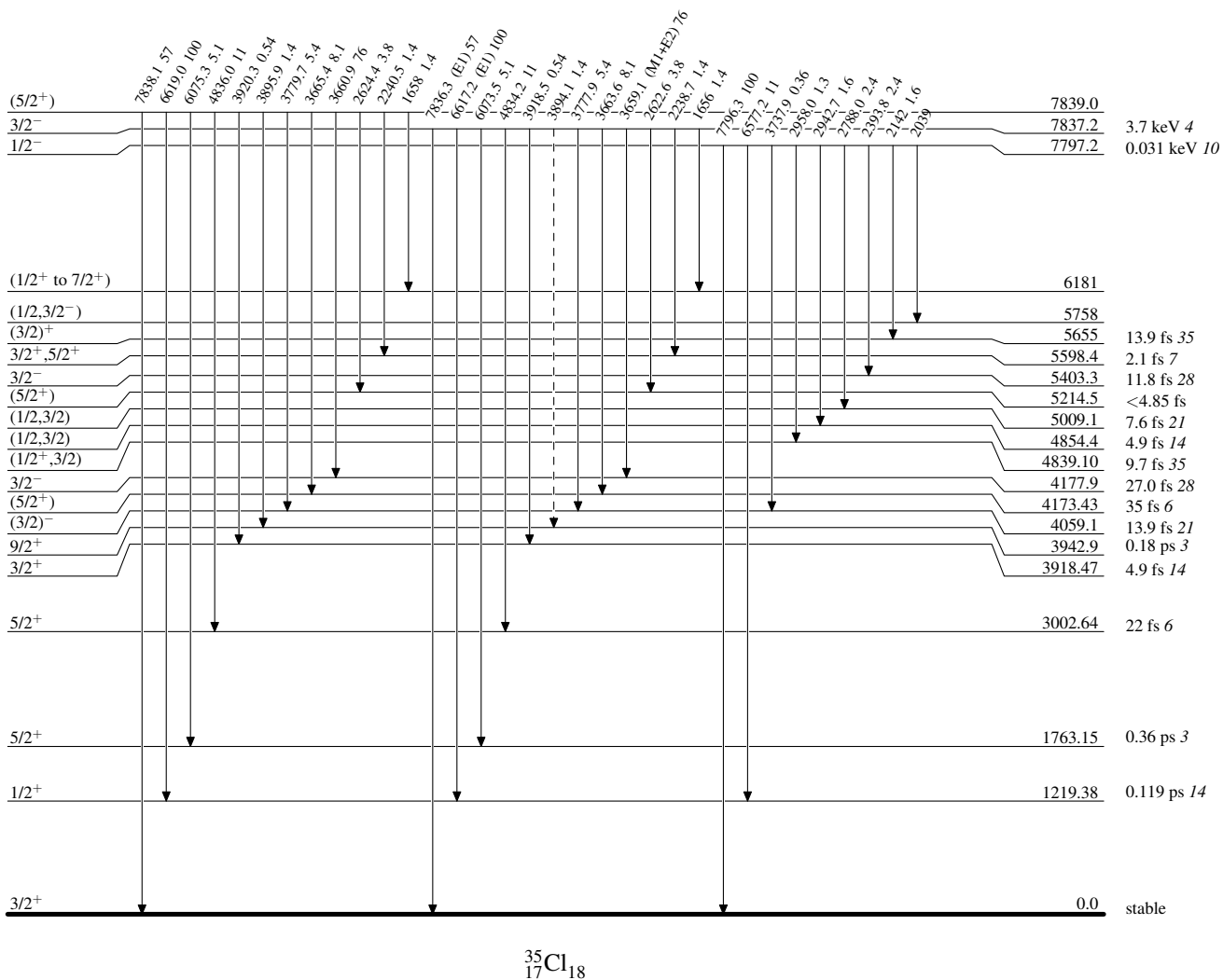


Adopted Levels, Gammas

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

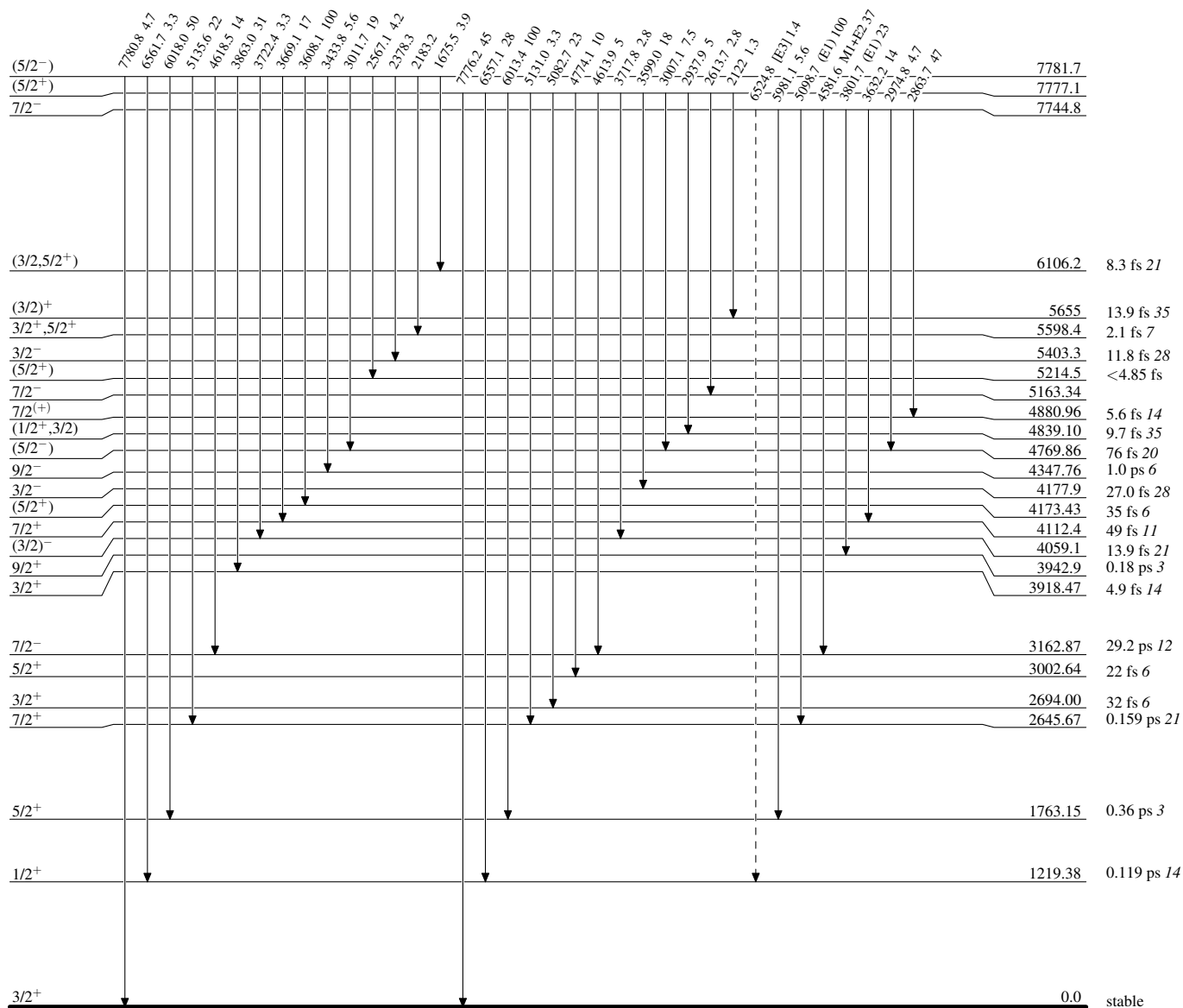
-----► γ Decay (Uncertain)

Adopted Levels, Gammas

Legend

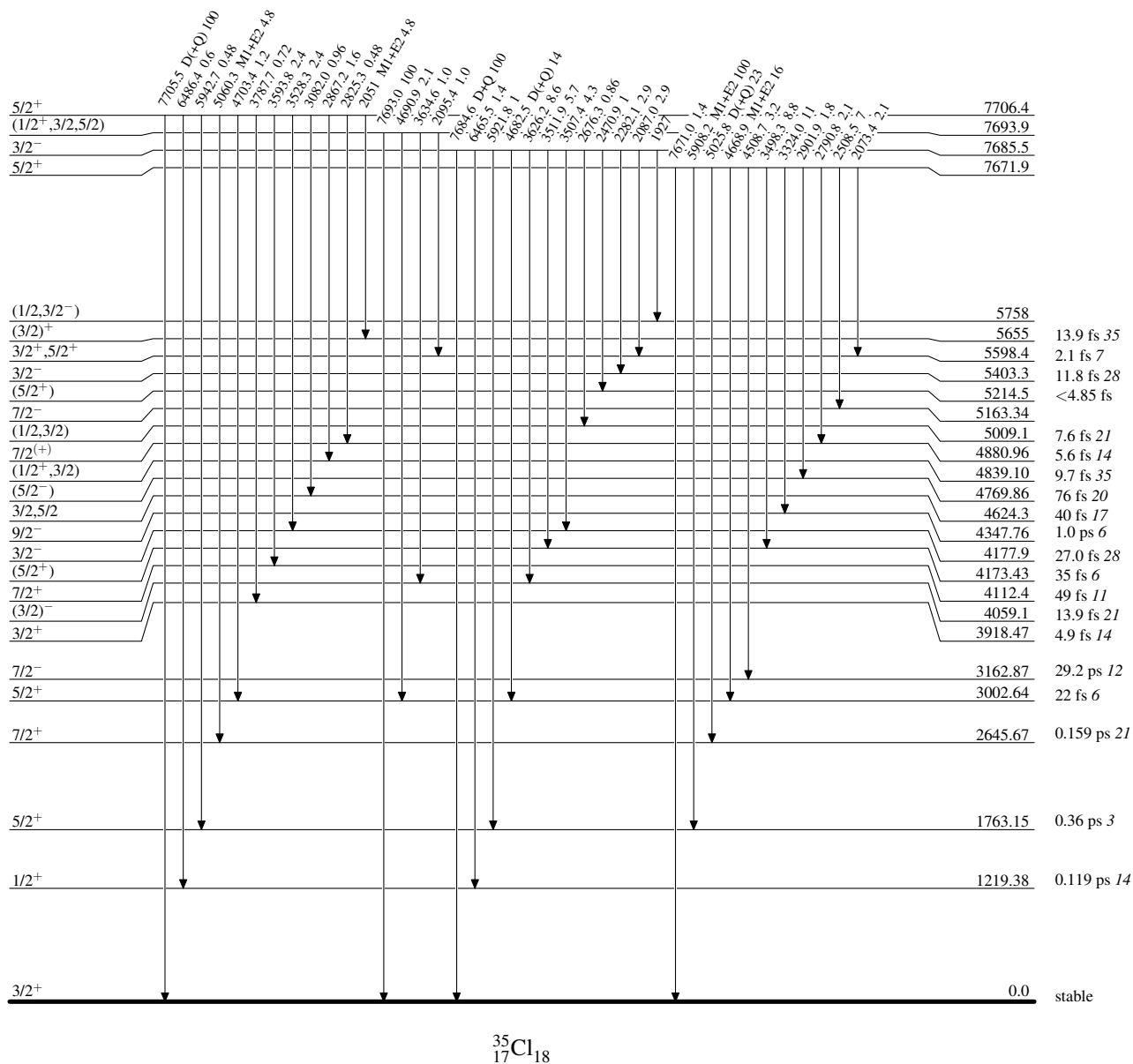
Level Scheme (continued)

Intensities: Relative photon branching from each level

-----► γ Decay (Uncertain) $^{35}_{17}\text{Cl}_{18}$

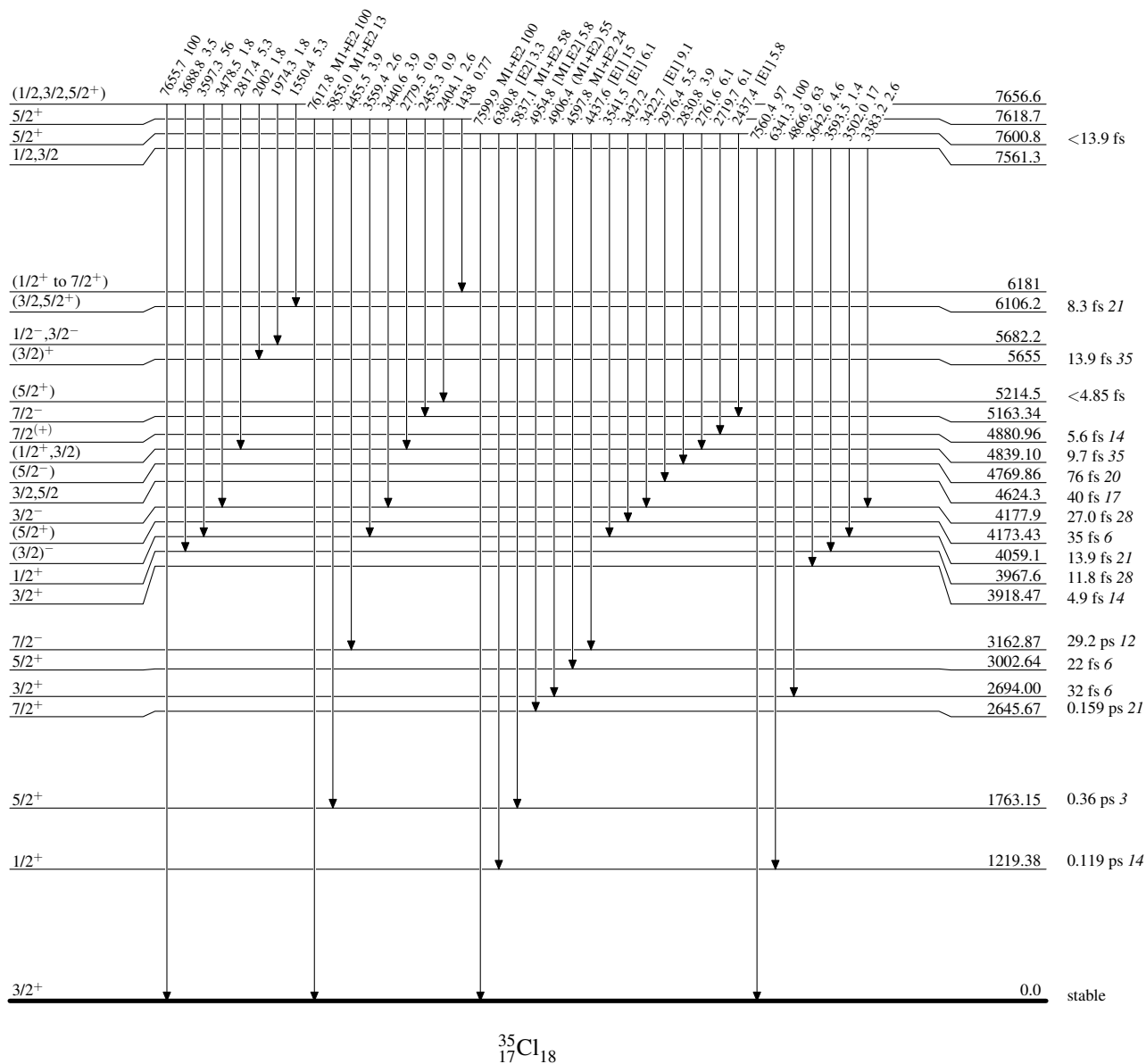
Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level



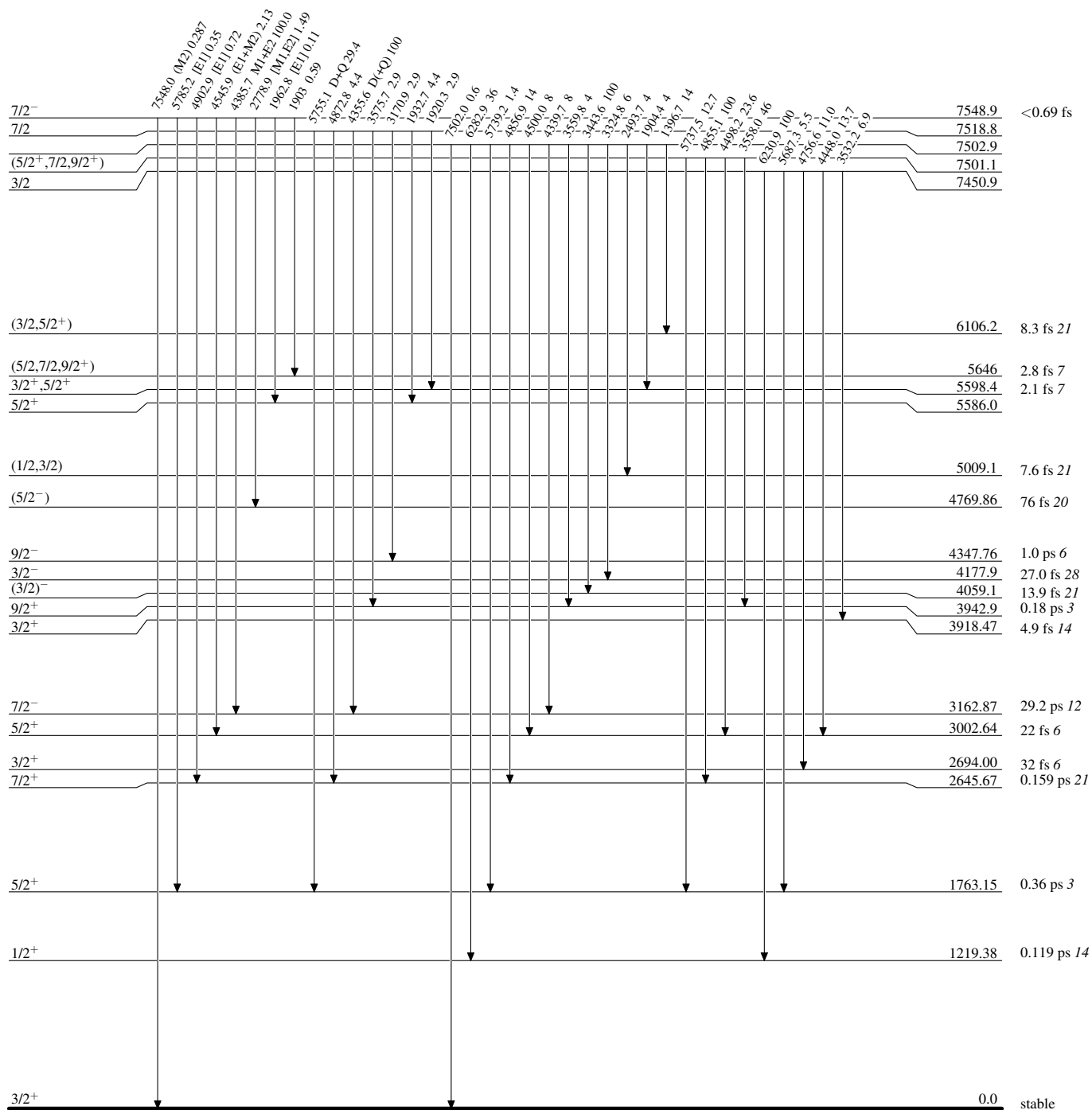
Adopted Levels, GammasLevel Scheme (continued)

Intensities: Relative photon branching from each level

 $^{35}_{17}\text{Cl}_{18}$

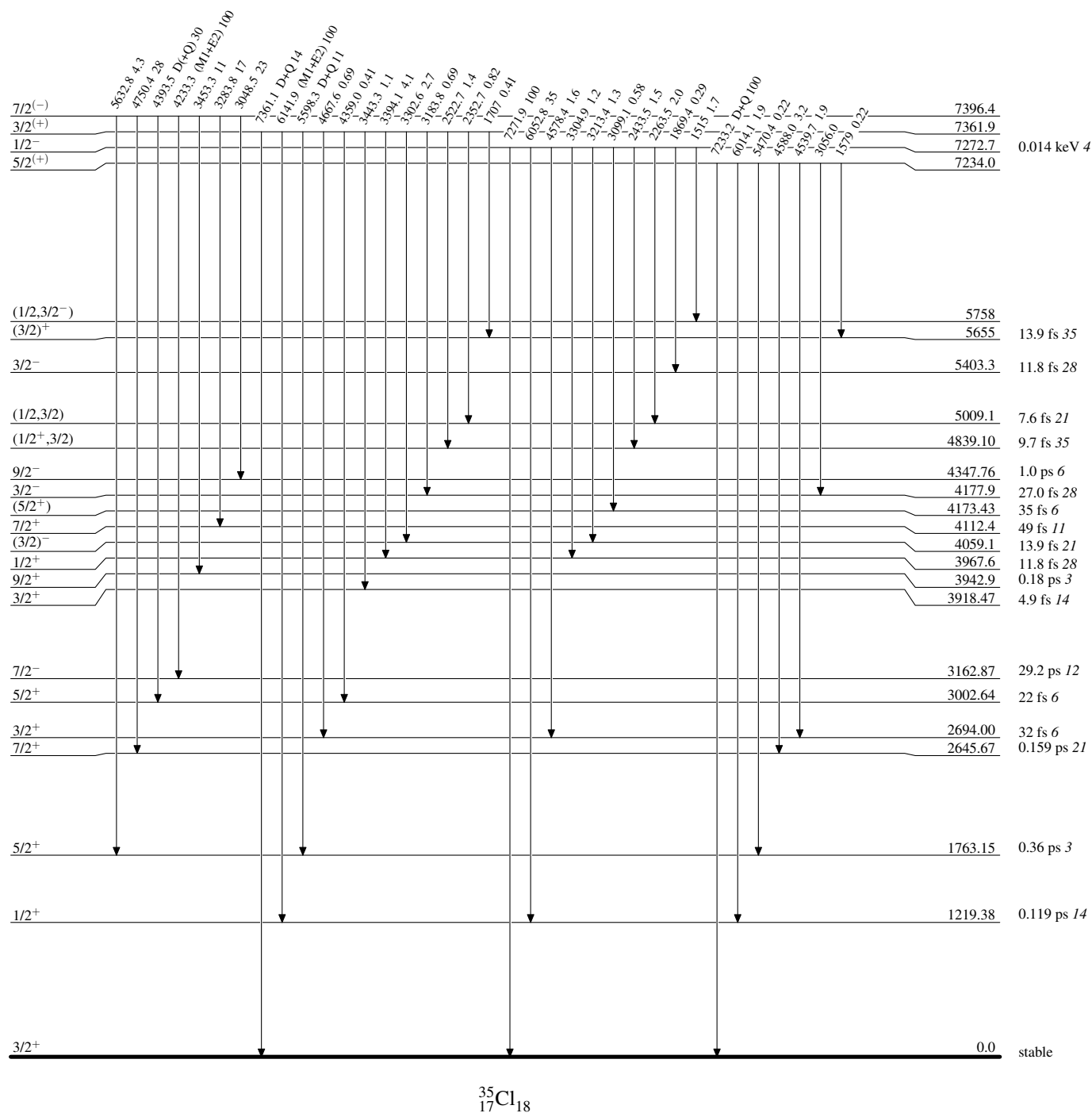
Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level



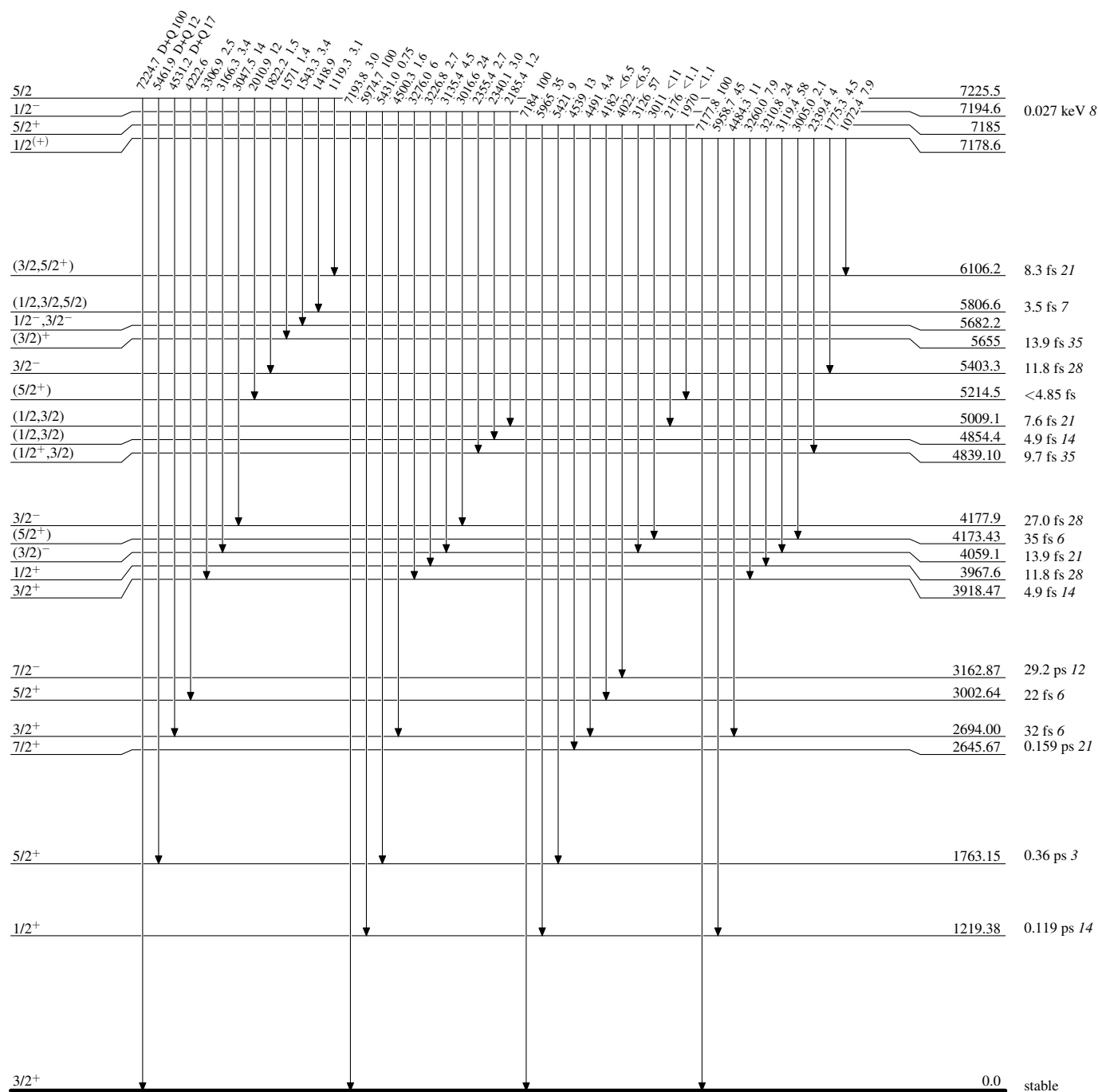
Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level



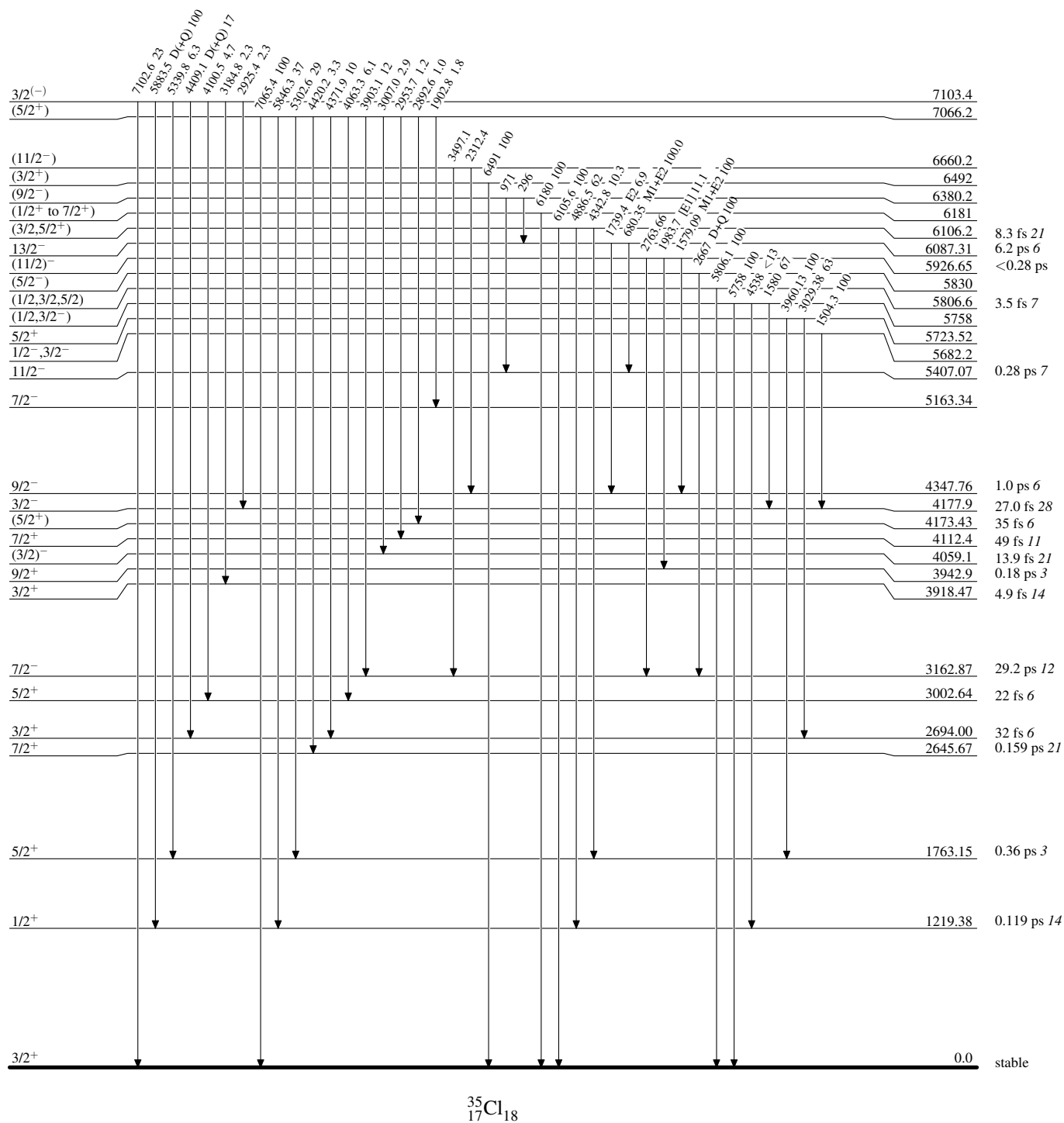
Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level



Adopted Levels, Gammas**Level Scheme (continued)**

Intensities: Relative photon branching from each level

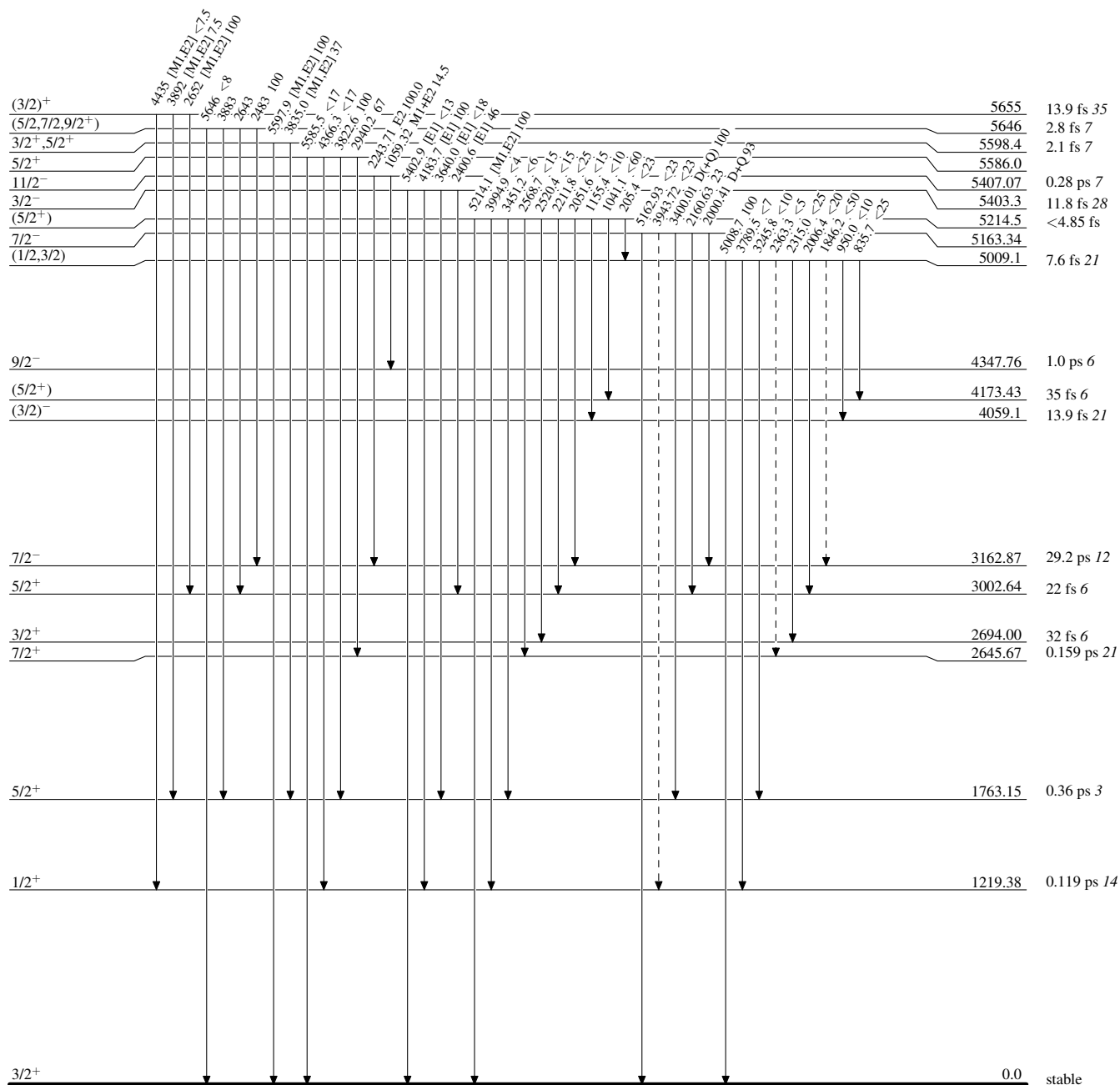


Adopted Levels, Gammas

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

-----► γ Decay (Uncertain)


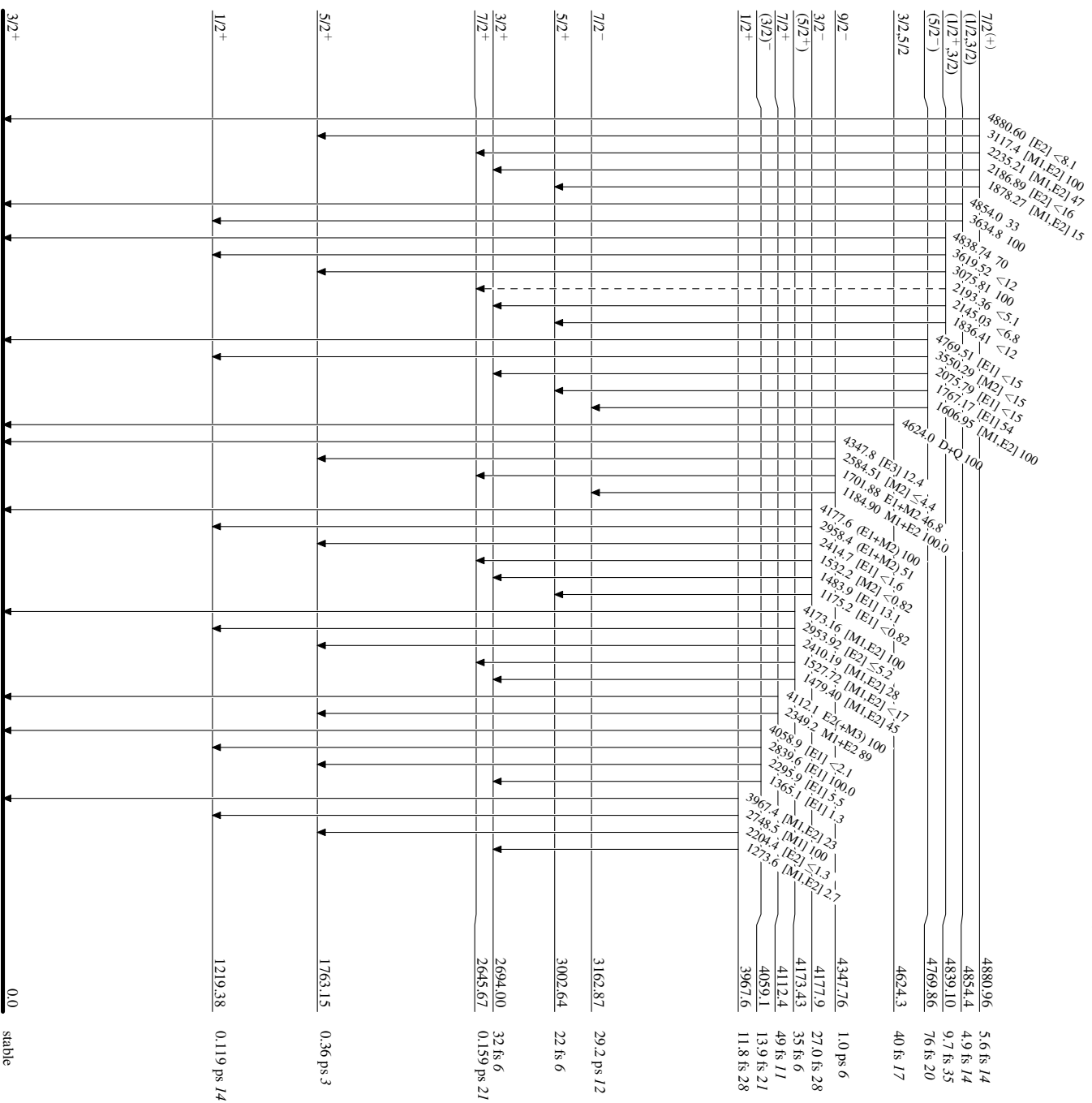
Adopted Levels, Gammas

Legend

Level Scheme (continued)

Intensities: Relative photon branching from each level

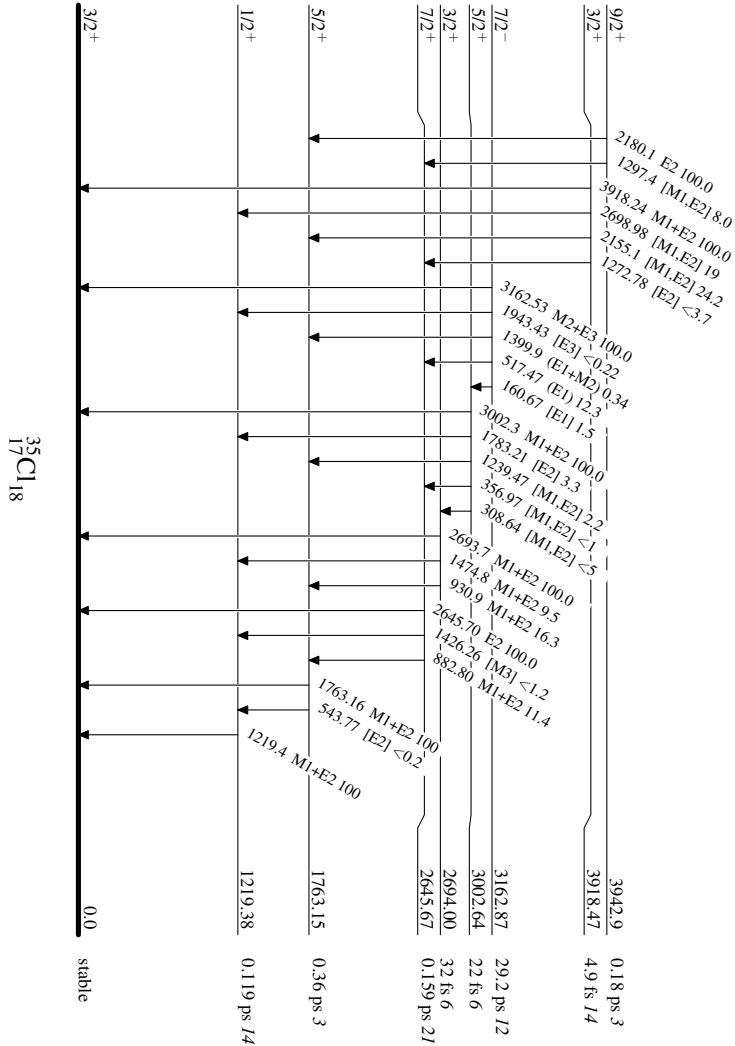
-----► γ Decay (Uncertain)



Adopted Levels, Gammas

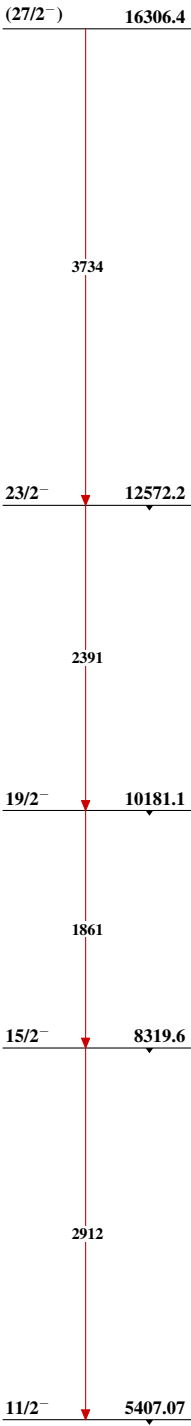
Level Scheme (continued)

Intensities: Relative photon branching from each level



Adopted Levels, Gammas

Band(A): Band based on $f_{7/2}$
orbital



$^{35}_{17}\text{Cl}_{18}$

Adopted Levels, Gammas (continued)Band(B): Band based on $13/2^+$ $21/2^+$ 11459.1 $19/2^+$ 10858.5

2615

2070

2015

 $17/2^+$ 8844.4
($15/2^+$) 8788.3

971

915

 $13/2^+$ 7873.03 $^{35}_{17}\text{Cl}_{18}$